Lecture notes. Lecture 6

Generative models and the physics of synthetic spin configurations

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1 New samples from data

1.1 Data as samples of a distribution

A Monte Carlo simulation of a magnet at temperature T returns configurations of spins, each drawn with the Boltzmann probability $e^{-H(\mathbf{s})/T}/Z$. A collection of handwritten digits is a set of images, each written by some person in some style. In both cases the data $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are independent draws from a probability distribution $p(\mathbf{x})$. For the magnet the distribution is known as an explicit formula and is still expensive to sample. A Markov chain has to be equilibrated and its successive configurations are correlated. For the digits nobody can write the distribution down.

A generative model learns an approximation $p_\theta(\mathbf{x})$ of $p(\mathbf{x})$ from the samples, where θ denotes all parameters of the model. It then draws new samples that are absent from the data set and could have been part of it. Learning turns a finite list of samples into a rule that assigns a probability to every possible $\mathbf{x}$, and sampling turns that rule into new $\mathbf{x}$. Both tasks are familiar from statistical physics. Fitting the parameters of a model to measured data is the first, and running a Monte Carlo simulation of a known Hamiltonian is the second. The new difficulty is the number of components. A digit has 784 grey values and a spin configuration of 32×32 sites has 1024 spins. A probability distribution over such objects cannot be tabulated, plotted or fitted like a histogram.

The variational autoencoder (VAE) is a generative model of this kind, which describes every sample by a few hidden numbers passed through a network.

1.2 Autoencoder codes as points

An autoencoder consists of two networks. The encoder E maps an image $\mathbf{x} \in [0,1]^{784}$ to a latent vector $\mathbf{z} = E(\mathbf{x}) \in \mathbb{R}^d$ with $d \ll 784$. The image holds the 784 grey values of a 28×28 digit of the MNIST collection (Modified National Institute of Standards and Technology) [1]. The decoder G maps it back, $\tilde{\mathbf{x}} = G(\mathbf{z})$. Both are trained together to minimise the mean squared error of the reconstruction, $\frac{1}{n}\sum_i \frac{1}{784}\|\mathbf{x}_i - G(E(\mathbf{x}_i))\|^2$, without labels. The decoder accepts any $\mathbf{z}$, which suggests using it as a generator by decoding latent vectors that no training image was mapped to.

The loss evaluates the decoder only at the n latent vectors $E(\mathbf{x}_i)$, so elsewhere its output is unconstrained. Figure 1 tests this for an autoencoder with $d = 2$, whose latent plane can be drawn. The codes of the digits form islands of different shapes and sizes, and the two coordinates have standard deviations of $3.8, 7.5$, values set by the training and by nothing else. Grid points inside an island decode to digits, and between two islands the decoded image morphs from one digit into the other. Grid points in the empty regions decode to blurred shapes that are no digit, because the training placed no requirement on latent vectors away from the codes.

With $d = 16$ the latent space cannot be drawn, so it is sampled. A random latent vector needs a distribution to be drawn from, and the autoencoder supplies none. Figure 2 tries two guesses, the standard normal distribution and a Gaussian fitted to the codes coordinate by coordinate. Both decode mostly to images that are no digits. The codes of the training images lie on a thin, curved set.

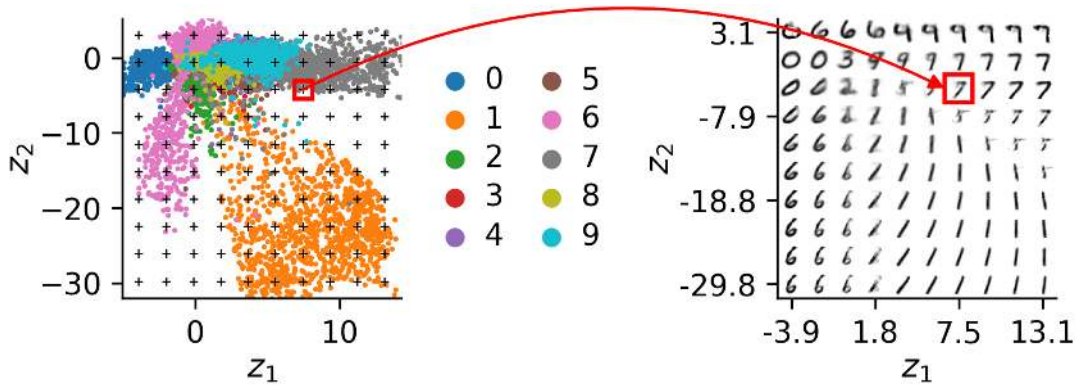


Figure 1: Autoencoder with a two-dimensional latent space. Left, the latent vectors $\mathbf{z} = E(\mathbf{x}) = (z_1, z_2)$ of the MNIST test digits, coloured by the true digit (labels used for colour only). A regular 10×10 grid of latent points (crosses) spans the 1st to the 99th percentile of the codes along each axis. Right, the decoded image $G(\mathbf{z})$ of every grid point, as grey levels from white (pixel value 0) to black (1), placed at the position of its grid point. The red square and arrow mark one grid point and its image.

A random point of $\mathbb{R}^{16}$, even one with the right mean and spread in every coordinate, almost never falls on it. Generation needs two ingredients that the autoencoder lacks, a known distribution of latent vectors from which to draw, and a decoder trained on every region that this distribution uses.



Figure 2: Images decoded by an autoencoder with $d = 16$ from random latent vectors. Top row, $\mathbf{z} \sim \mathcal{N}(0, I)$. Bottom row, every coordinate z_k drawn from a Gaussian with the mean and standard deviation of coordinate k over the codes of the training images. The panel labels these values m_k and s_k.

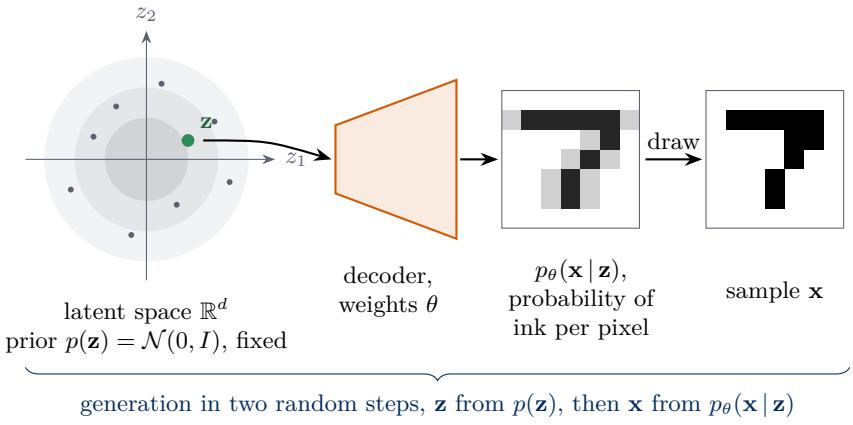


Figure 3: The variational autoencoder as a story of how one sample comes about, sketch on a 7×7 toy image. A latent vector $\mathbf{z}$ (green) is drawn from the fixed standard Gaussian (grey cloud, grey dots are other draws). The learned decoder (orange) turns it into a probability of ink for every pixel, shown as grey levels, and the image is drawn from these probabilities.

Figure 3 sketches how the VAE produces one sample.

2 Probability for generative models

A generative model is a probability distribution with adjustable parameters, and its training, evaluation and sampling are statements about probabilities. We introduce each notion on the smallest example that shows it, a coin, a pixel, a spin or a cloud of points.

2.1 Discrete outcomes, a coin, a pixel and a spin

A coin lands heads or tails. We write the outcome of one toss as $x = 1$ for heads and $x = 0$ for tails and call θ the probability of heads. The two probabilities $P(x = 1) = \theta$ and $P(x = 0) = 1 - \theta$ are written as one formula,

$$P(x \,|\, \theta) \coloneqq \theta^x (1-\theta)^{1-x}, \qquad x \in \{0, 1\}, \quad 0 \le \theta \le 1, \tag{2.1}$$

the Bernoulli distribution. For $x = 1$ the second factor equals one and leaves θ, and for $x = 0$ the first factor equals one and leaves $1 - \theta$. The vertical bar is read "given". It states that the probability of the outcome x is computed for a fixed value of the parameter θ. The symbol $\coloneqq$ marks a definition.

The Bernoulli distribution also describes the two objects of this chapter. A pixel of a black-and-white image is ink, $x_j = 1$, or background, $x_j = 0$. A decoder that outputs a number $\tilde{x}_j \in (0, 1)$ for this pixel uses it as the probability of ink, the role that θ plays for the coin. A spin $s_i = \pm 1$ of an Ising configuration becomes a Bernoulli variable through $x_i = (1 + s_i)/2$, and the probability that it points up plays the role of θ.

Outcomes are independent when the outcome of one leaves the probabilities of the others unchanged. The probability of a sequence of independent outcomes is then the product of the single probabilities. For n tosses with m heads,

$$P(x_1, \ldots, x_n \,|\, \theta) = \prod_{i=1}^{n} \theta^{x_i} (1-\theta)^{1-x_i} = \theta^m (1-\theta)^{n-m}, \qquad m = \sum_{i=1}^{n} x_i, \tag{2.2}$$

and because $\binom{n}{m}$ orders of the tosses give the same m, the number of heads has the binomial distribution $P(m \,|\, \theta) = \binom{n}{m} \theta^m (1-\theta)^{n-m}$. A model of an image whose 784 pixels are independent is a product of 784 Bernoulli factors of the form (2.2), each with its own θ_j.

2.2 Continuous quantities and densities

A grey value, a position or a latent coordinate can take any real value, and the probability of any single value is zero. Probability is then described by a density $p(x)$, a non-negative function whose area over an interval is the probability that the value falls in it,

$$P(a \le x \le b) = \int_a^b p(x)\, \mathrm{d}x, \qquad \int_{-\infty}^{\infty} p(x)\, \mathrm{d}x = 1. \tag{2.3}$$

A density is a probability per unit length, as a mass density is a mass per unit length. Its value at a point is no probability and can exceed one. The Gaussian, or normal, density with mean μ and standard deviation $\sigma > 0$ is

$$\mathcal{N}(x; \mu, \sigma^2) \coloneqq \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2} \Big). \tag{2.4}$$

The exponent is the squared distance of x from the centre μ in units of the width σ, and the prefactor makes the area equal to one. A fraction 0.683 of the area lies within one standard deviation of the mean, for every μ and σ. A narrow Gaussian must be tall to keep its area, and for $\sigma = 0.2$ its peak value is $1/\sqrt{2\pi \cdot 0.04} = 1.99$ (Figure 4). The standard Gaussian $\mathcal{N}(0, 1)$ has $\mu = 0$ and $\sigma = 1$. Its cumulative distribution function $\Phi(z) \coloneqq \int_{-\infty}^{z} \mathcal{N}(z'; 0, 1)\, \mathrm{d}z'$ is the probability of a value below z.

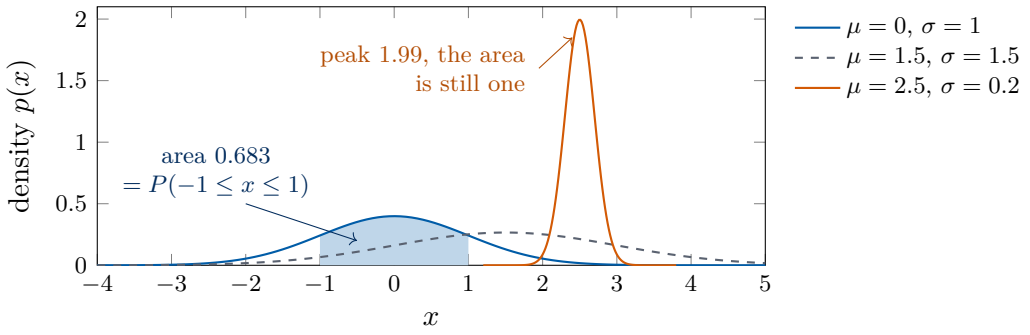


Figure 4: Gaussian densities (2.4). The mean μ moves the curve and the standard deviation σ sets its width. Probabilities are areas, as the shaded area between -1 and 1 under the standard Gaussian shows. The narrow orange curve has a peak value above one, which is allowed for a density.

2.3 Drawing a sample

The phrase "draw x from p", written $x \sim p$, names a random procedure whose outcomes, collected over many repetitions, fall into every interval $[a, b]$ with the frequency $P(a \le x \le b)$. A histogram of many draws, scaled to unit area, then approaches the density, and draws crowd where the density is high. One draw says little about the distribution, which only the collection of draws reflects. Monte Carlo simulation of the Ising model is such a procedure for the Boltzmann distribution. One configuration of the chain carries almost no information about the temperature, while a histogram of the magnetisation over many configurations shows the phase.

Every random procedure on a computer starts from the uniform distribution $\mathcal{U}(0, 1)$, whose draws u fall into any subinterval of $[0, 1]$ with a frequency equal to its length. A coin with bias θ is obtained by declaring heads when $u < \theta$. A continuous density with cumulative distribution $F(x) = P(X \le x)$ is obtained by the inverse transformation $x = F^{-1}(u)$. The event $x \le x_0$ happens if and only if $u \le F(x_0)$, which has the frequency $F(x_0)$. Values of u spread evenly over $[0, 1]$ land close together where F is steep, near the centre of a Gaussian, and far apart where it is flat, in the tails.

A Gaussian with any mean and width is obtained by shifting and stretching a standard one,

$$x = \mu + \sigma\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, 1) \quad \Longrightarrow \quad x \sim \mathcal{N}(\mu, \sigma^2). \tag{2.5}$$

The VAE uses this construction, a draw from a fixed simple distribution followed by a deterministic map, for its random latent vectors and, with a learned map, for its samples.

2.4 Expectations as averages over samples

The expectation of a function $f(x)$ under a density p is its average weighted by the density. It is estimated by an average over draws,

$$\mathbb{E}_{x\sim p}[f(x)] \coloneqq \int p(x)\, f(x)\, \mathrm{d}x \;\approx\; \frac{1}{n}\sum_{i=1}^{n} f(x_i), \qquad x_i \sim p. \tag{2.6}$$

The Monte Carlo estimate of a thermal average, $\langle O\rangle_T \approx \frac{1}{n}\sum_i O(\mathbf{s}_i)$, has this form. Its statistical error is $\sigma_f/\sqrt{n}$, with σ_f the standard deviation of $f(x)$ under p. Every loss of this chapter is an expectation over the data, and a minibatch of B samples estimates it in this way. The mean $\mu = \mathbb{E}[x]$ locates a distribution and the variance $\mathrm{Var}[x] \coloneqq \mathbb{E}[(x-\mu)^2] = \sigma^2$ measures its spread. For a vector $\mathbf{x}$ the covariance $\mathrm{Cov}[x_j, x_l] \coloneqq \mathbb{E}[(x_j - \mu_j)(x_l - \mu_l)]$ is positive when the two components tend to lie on the same side of their means. It is estimated by the sample covariance matrix, whose eigenvectors are the principal directions of principal component analysis.

2.5 Gaussians in several dimensions and independence

The Gaussian density of a vector $\mathbf{x} \in \mathbb{R}^d$ with mean $\boldsymbol{\mu}$ and covariance matrix Σ is

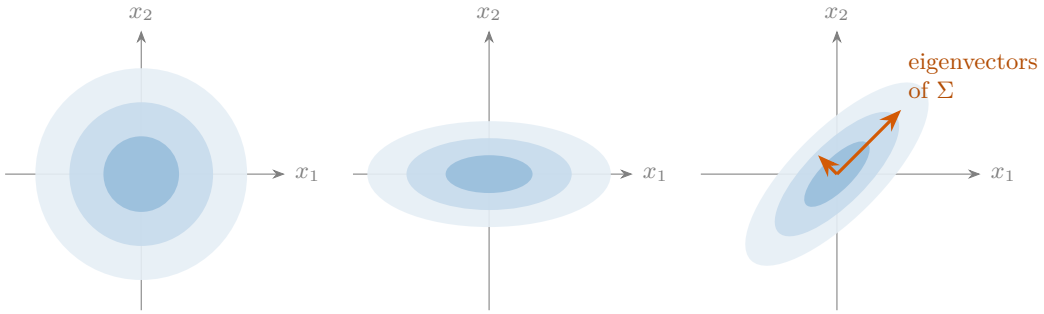
$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \Sigma) := \frac{1}{(2\pi)^{d/2}(\det \Sigma)^{1/2}} \exp\Big(-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big). \tag{2.7}$$

The quadratic form in the exponent is a squared distance from the centre in units of the spread. Points with equal density lie on ellipsoids whose axes are the eigenvectors of Σ, with half-lengths given by the square roots of its eigenvalues. Three cases recur in this chapter (Figure 5). The identity covariance $\Sigma = I$ gives circles. A diagonal covariance $\Sigma = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_d^2)$, written $\operatorname{diag}\boldsymbol{\sigma}^2$, gives ellipses aligned with the coordinate axes. A covariance with off-diagonal entries gives tilted ellipses.

When Σ is diagonal the quadratic form is a sum of squares and the exponential of a sum is a product,

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \operatorname{diag}\boldsymbol{\sigma}^2) = \prod_{j=1}^{d} \mathcal{N}(x_j; \mu_j, \sigma_j^2). \tag{2.8}$$

Two quantities are independent when their joint density is the product of their separate densities, so that knowing one gives no information about the other. The coordinates of a Gaussian with diagonal covariance are therefore independent. The prior of the VAE is the first case, $\mathcal{N}(0, I)$, and its d coordinates are d independent standard Gaussian numbers. The encoder of the VAE returns the second case, an ellipse aligned with the axes, for every image. The distribution of the latent vectors of a data set can look like the third case or like no Gaussian at all.



(a) $\Sigma = I$ (b) $\Sigma = \operatorname{diag}(\sigma_1^2, \sigma_2^2)$ (c) correlated

Figure 5: Contours of two-dimensional Gaussians with zero mean, sketch. (a) Identity covariance, circles, independent coordinates of equal spread. This is the prior of the VAE. (b) Diagonal covariance, ellipses aligned with the axes, independent coordinates of different spread. This is the shape of an encoder cloud. (c) A covariance with off-diagonal entries gives tilted ellipses whose axes are the eigenvectors of Σ (orange). A large x_1 then makes a large x_2 likely.

2.6 Joint, marginal and conditional distributions

A latent-variable model involves an observed quantity x and a hidden one z, and their joint density $p(x, z)$ assigns a density to every pair. The marginal density of x follows by summing over the hidden quantity. The conditional density of x given z is the joint density at fixed z, renormalised,

$$p(x) = \int p(x, z)\, \mathrm{d}z, \qquad p(x \mid z) = \frac{p(x, z)}{p(z)}, \qquad p(x, z) = p(x \mid z)\, p(z) = p(z \mid x)\, p(x). \tag{2.9}$$

The two factorisations of the joint density describe the same object in two orders. In the generative order z is drawn first and x is then drawn given z. In the inferential order x is observed, and we infer which z produced it. Equating the two factorisations gives Bayes' rule,

$$p(z \mid x) = \frac{p(x \mid z)\, p(z)}{p(x)}, \qquad p(x) = \int p(x \mid z')\, p(z')\, \mathrm{d}z'. \tag{2.10}$$

Statistical physics uses the same structure without naming it. The Boltzmann distribution of the Ising model at temperature T is a conditional distribution, $p(\mathbf{s}\,|\,T) = e^{-H(\mathbf{s})/T}/Z(T)$, the probability of a configuration given the temperature. A data set that mixes configurations from many temperatures, each chosen with probability $p(T)$, has the joint distribution $p(\mathbf{s}, T) = p(\mathbf{s}\,|\,T)\,p(T)$. Its marginal $p(\mathbf{s}) = \sum_T p(\mathbf{s}\,|\,T)\,p(T)$ is the distribution of a configuration whose temperature was not recorded. Bayes' rule then gives $p(T\,|\,\mathbf{s})$, the probabilities of the temperatures that could have produced a given snapshot. A supervised phase classifier learns an approximation of a closely related posterior, the probability $p(\text{phase}\,|\,\mathbf{s})$ that a snapshot belongs to the ordered phase. The VAE uses these four objects with the temperature replaced by a hidden vector $\mathbf{z}$ that nobody has measured.

2.7 Probability and likelihood

A model $p(x\,|\,\theta)$ gives the probability of data x when its parameters are θ. The same formula can be read in two ways. With θ fixed and x varied, it is a probability distribution over outcomes, and its sum over all outcomes is one. With the observed data x_{obs} fixed and θ varied, it is the likelihood of the parameters,

$$L(\theta) \coloneqq p(x_{\text{obs}}\,|\,\theta), \tag{2.11}$$

a function of θ that ranks parameter values by how probable they make the observed data. The likelihood is no probability distribution over θ, and its integral over θ can differ from one.

A coin with probability θ of heads tossed ten times gives k heads with the binomial probability $p(k\,|\,\theta) = \binom{10}{k}\theta^k(1-\theta)^{10-k}$. For a fair coin seven heads have the probability $p(7\,|\,0.5) = 0.117$ (Figure 6, left). If seven heads were observed, the likelihood $L(\theta) = 120\,\theta^7(1-\theta)^3$ is largest at $\theta = 0.7$, the observed fraction of heads. Its integral over $\theta \in [0, 1]$ is $120 \cdot 7!\,3!/11! = 1/11$ (Figure 6, right). Even at its maximum the likelihood is only $L(0.7) = 0.27$, because seven heads are one of eleven possible results and the best coin still produces the others.

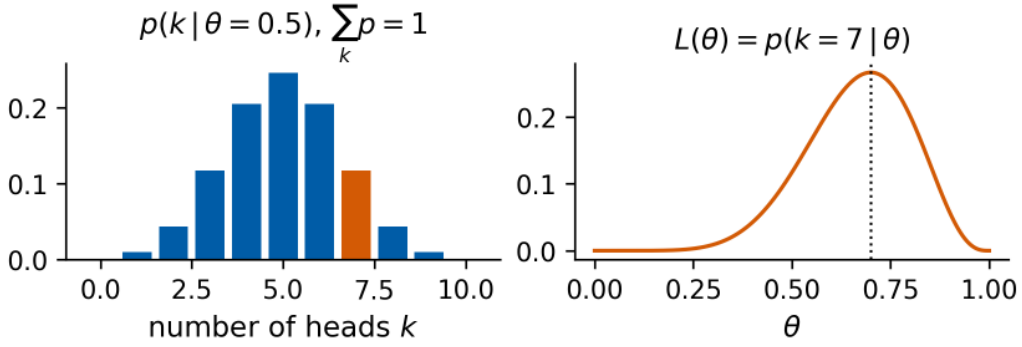


Figure 6: Probability and likelihood for a coin tossed ten times. Left, the probability $p(k\,|\,\theta = 0.5)$ of k heads against k, bars that add up to one, with $k = 7$ in orange. Right, the likelihood $L(\theta) = p(k = 7\,|\,\theta)$ against the probability of heads θ, with its maximum at $\theta = 0.7$ (dotted).

A probability distribution over θ requires a prior belief $p(\theta)$ and Bayes' rule (2.10) with z replaced by θ. If the coin is believed to be roughly fair, with $p(\theta) \propto \theta^4(1-\theta)^4$, the posterior after seven heads is $p(\theta\,|\,k{=}7) \propto \theta^7(1-\theta)^3 \cdot \theta^4(1-\theta)^4 = \theta^{11}(1-\theta)^7$. It peaks at $\theta = 11/18 = 0.61$, between the prior peak 0.5 and the likelihood peak 0.7. The denominator of Bayes' rule, $p(k{=}7) = \int p(7\,|\,\theta)\,p(\theta)\,\mathrm{d}\theta$, is the probability of the data averaged over all coins the prior allows. It is called the evidence or the marginal likelihood. Table 1 collects this vocabulary and its meaning in the VAE. The word likelihood has two uses in this chapter. In the table it is the probability of the data for a given value of the hidden quantity. This is $p(k\,|\,\theta)$ for the coin and $p_\theta(\mathbf{x}\,|\,\mathbf{z})$ as a function of $\mathbf{z}$ for the VAE. In training it is the probability of the data as a function of the model parameters. For the VAE this is the evidence $p_\theta(\mathbf{x})$ read as a function of the network weights θ. Training raises a lower bound on it.

Table 1: The vocabulary of Bayesian inference for the coin and for the latent vector of a variational autoencoder .

term	coin	meaning	variational autoencoder
prior	$p(\theta)$	belief before the data	$p(\mathbf{z}) = \mathcal{N}(0, I)$, fixed
likelihood	$p(k \mid \theta)$	probability of the observed data for given parameters, read as a function of the parameters	decoder $p_\theta(\mathbf{x} \mid \mathbf{z})$ read as a function of $\mathbf{z}$
posterior	$p(\theta \mid k)$	belief after the data	$p_\theta(\mathbf{z} \mid \mathbf{x})$, the latent vectors that could have produced $\mathbf{x}$
evidence	$p(k)$	probability of the data averaged over the prior	$p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x}\|\mathbf{z})p(\mathbf{z})\,\mathrm{d}\mathbf{z}$, the model of the data

3 Learning a probability distribution from samples

3.1 The modelling task

We are given a finite list of samples $\mathbf{x}_1, \ldots, \mathbf{x}_n$, assumed to be independent draws from an unknown distribution $p(\mathbf{x})$ that is itself never seen. For a coin, p is the Bernoulli distribution with the unknown bias of the real coin. For the digits, p is the distribution of all images that people produce when they write a digit, a distribution over $[0,1]^{784}$ that no formula describes. For the Ising model at temperature T, p is the Boltzmann distribution, known as an explicit formula and costly to sample near the critical point.

Learning p from the samples involves the three choices that a physicist makes when fitting a model to measurements. The first is a family of distributions $p_\theta(\mathbf{x})$, labelled by parameters θ, whose members can be evaluated at any $\mathbf{x}$ or sampled from. For a coin the family has one parameter, for a Gaussian a mean and a covariance. For a VAE it has the weights θ of a decoder network, while the encoder of the VAE is a training aid with its own weights ϕ. The second choice is a criterion of closeness between p_θ and p that can be computed from the samples alone, and the third an optimisation of θ under that criterion. A learned distribution always belongs to the chosen family, and training can at best reach the member closest to p. We judge the fitted model afterwards by the probability it assigns to samples held out from the training and by how closely new samples drawn from it resemble the data. Table 2 collects the cases of this chapter, with families that range from one number for a coin to several hundred thousand network weights for the digits.

Table 2: Learning a distribution, from a coin to images. In every row the parameters are fitted by maximum likelihood or, in the last row, by a lower bound on it.

data	family	parameters	drawing a new sample
coin tosses	Bernoulli (2.1)	θ, one number	heads when $u < \theta$, $u \sim \mathcal{U}(0,1)$
measurements	Gaussian (2.4)	μ, σ^2	$x = \mu + \sigma\varepsilon$, $\varepsilon \sim \mathcal{N}(0,1)$
three groups of points	mixture (3.9)	weights, means, covariances	a hidden label k, then its Gaussian
digits, spins	latent-variable model (4.1)	network weights θ	$\mathbf{z} \sim \mathcal{N}(0, I)$, then $p_\theta(\mathbf{x} \mid \mathbf{z})$

3.2 Maximum likelihood

Every member p_θ of the family is a candidate description of the data. As in Section 1, θ denotes the parameters of the model, for the VAE the weights of the decoder network. The number $p_\theta(\mathbf{x}_i)$ is the probability of the observed sample $\mathbf{x}_i$ if the data had been produced by this candidate, or its probability density for continuous data. For independent samples the probability of the data set under the candidate is the product $L(\theta) = \prod_i p_\theta(\mathbf{x}_i)$, the likelihood of Section 2.7. Maximum likelihood

chooses the candidate under which the observed data are most probable,

$$\theta^{*} \coloneqq \arg\max_{\theta} \ell(\theta), \qquad \ell(\theta) \coloneqq \sum_{i=1}^{n} \ln p_{\theta}(\mathbf{x}_i). \tag{3.1}$$

Being an increasing function, the logarithm leaves the position of the maximum unchanged. It turns the product into a sum that a minibatch can estimate and keeps the numbers representable. A thousand factors of 0.01 give 10^{-2000}, far below the smallest normal double-precision number of about 10^{-308}, while their logarithm -4605 is an ordinary number. The negative log-likelihood divided by n is the loss to be minimised, and its value does not depend on the size of the data set.

A digit is one of an astronomical number of possible images, and even the true distribution gives each of them a tiny probability. The absolute value of the likelihood therefore carries no meaning. A model that inks every pixel with probability one half gives every binary image of 28×28 pixels the probability 2^{-784}, a negative log-likelihood of $784 \ln 2 = 543$ nats per image. The nat is the unit of a logarithm to the base e, 1 nat $= 1/\ln 2 = 1.44$ bits. These 543 nats serve as a reference for the trained models of this chapter. Only differences of ℓ between candidates, or between models evaluated on the same held-out data, are meaningful.

For a Gaussian family the maximum can be found analytically, and the result is familiar. With $p_{\theta}(x) = \mathcal{N}(x; \mu, \sigma^2)$ the negative log-likelihood per sample is

$$-\frac{1}{n}\ell(\mu, \sigma^2) = \frac{1}{2n\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2 + \frac{1}{2} \ln\left(2\pi\sigma^2\right), \tag{3.2}$$

and setting its derivatives with respect to μ and σ^2 to zero,

$$-\frac{1}{n\sigma^2} \sum_{i} (x_i - \mu) = 0, \qquad -\frac{1}{2n\sigma^4} \sum_{i} (x_i - \mu)^2 + \frac{1}{2\sigma^2} = 0, \tag{3.3}$$

gives the sample mean $\bar{\mu} = \frac{1}{n}\sum_i x_i$ and the sample variance $\tilde{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \bar{\mu})^2$. Computing the mean and the spread of a set of measurements is maximum likelihood for a Gaussian model. For a coin the same calculation gives $\tilde{\theta} = m/n$, the fraction of heads, as Figure 6 shows for $m = 7$ and $n = 10$. For a network no formula exists, and the maximum is found by gradient ascent on ℓ, the standard procedure for training neural networks.

3.3 Maximum likelihood and the Kullback–Leibler divergence

For $n \to \infty$ the average $\frac{1}{n}\ell$ tends to an expectation over the data distribution, equation (2.6). Adding and subtracting $\mathbb{E}_p[\ln p]$ splits it into two parts,

$$\frac{1}{n}\ell(\theta) \;\longrightarrow\; \mathbb{E}_{\mathbf{x}\sim p}\big[\ln p_{\theta}(\mathbf{x})\big] = \underbrace{\mathbb{E}_p\big[\ln p(\mathbf{x})\big]}_{-S[p]} - \underbrace{\mathbb{E}_p\Big[\ln \frac{p(\mathbf{x})}{p_{\theta}(\mathbf{x})}\Big]}_{\mathrm{KL}(p \,\|\, p_{\theta})}. \tag{3.4}$$

The first part is minus the entropy $S[p] \coloneqq -\mathbb{E}_p[\ln p]$ of the data distribution, which is independent of θ. The quantity $-\ln p(\mathbf{x})$ is the surprise of observing $\mathbf{x}$, large for a rare outcome, and the entropy is the average surprise. A fair coin has $S = \ln 2 = 0.693$ nats, one bit per toss, and a coin with $\theta = 0.9$, whose outcome is easier to guess, has 0.325 nats. In statistical physics the same quantity, multiplied by Boltzmann's constant, is the Gibbs entropy of the Boltzmann distribution. The second part is the Kullback–Leibler (KL) divergence

$$\mathrm{KL}(p \,\|\, q) \coloneqq \mathbb{E}_{\mathbf{x}\sim p}\Big[\ln \frac{p(\mathbf{x})}{q(\mathbf{x})}\Big] = \int p(\mathbf{x}) \ln \frac{p(\mathbf{x})}{q(\mathbf{x})} \, \mathrm{d}\mathbf{x}, \tag{3.5}$$

the average extra surprise when data drawn from p are described by q. Maximum likelihood therefore minimises the divergence of the model from the data distribution. The best negative log-likelihood per

sample that any model can reach on new data is the entropy $S[p]$. For discrete data such as binary images the entropy is positive, so even a model equal to the data distribution has a nonzero loss.

A coin shows what the divergence measures. A fair coin, $p = (0.5, 0.5)$, described by a model of a bent coin, $q = (0.9, 0.1)$, gives $\mathrm{KL}(p\|q) = 0.5 \ln \frac{0.5}{0.9} + 0.5 \ln \frac{0.5}{0.1} = -0.294 + 0.805 = 0.511$ nats. Tails, which the model considers rare, occur half of the time, and their extra surprise dominates. The opposite order gives $\mathrm{KL}(q\|p) = 0.368$ nats. The divergence is asymmetric, and the order of its arguments determines what it penalises.

The divergence is never negative. The proof rests on the inequality $\ln u \le u - 1$ for every $u > 0$, with equality only at $u = 1$. The straight line $u - 1$ touches the curve $\ln u$ at $u = 1$ and lies above it everywhere else, since the logarithm bends downwards. With $u = q(\mathbf{x})/p(\mathbf{x})$ and the average over $\mathbf{x} \sim p$,

$$-\mathrm{KL}(p \,\|\, q) = \mathbb{E}_p\Big[\ln \frac{q}{p}\Big] \le \mathbb{E}_p\Big[\frac{q}{p} - 1\Big] = \int q(\mathbf{x})\,\mathrm{d}\mathbf{x} - 1 = 0, \tag{3.6}$$

with equality only for $q = p$. The cross-entropy loss of classification applies this construction to labels, as the negative log-likelihood of the observed labels under the probabilities that the network assigns to them.

3.4 Reconstruction losses as negative log-likelihoods

A network that describes data returns numbers, for example 784 values between 0 and 1. These numbers are the parameters of a distribution, for example the probability of ink of every pixel. A sample drawn from that distribution is a third object, for example an image in which every pixel is ink or background. The loss compares the data with the distribution, whereas a displayed generated image usually shows the numbers without drawing a sample.

If the components x_j of a sample are real numbers with Gaussian noise of fixed width σ_x around the outputs $\tilde{x}_j$, then

$$-\ln p(\mathbf{x} \,|\, \tilde{\mathbf{x}}) = \frac{1}{2\sigma_x^2} \sum_j (x_j - \tilde{x}_j)^2 + \text{const}, \tag{3.7}$$

the squared reconstruction error of an autoencoder divided by $2\sigma_x^2$. If every component is a bit, ink $x_j = 1$ or background $x_j = 0$, and $\tilde{x}_j \in (0, 1)$ is the probability of ink, every pixel is a Bernoulli variable (2.1) with $\theta = \tilde{x}_j$. For independent pixels

$$-\ln p(\mathbf{x} \,|\, \tilde{\mathbf{x}}) = \mathrm{BCE}(\mathbf{x}, \tilde{\mathbf{x}}) \coloneqq -\sum_j \Big[x_j \ln \tilde{x}_j + (1 - x_j) \ln(1 - \tilde{x}_j)\Big], \tag{3.8}$$

the binary cross-entropy (BCE). A pixel predicted with certainty and correctly costs nothing, and an undecided prediction $\tilde{x}_j = \frac{1}{2}$ costs $\ln 2 = 0.69$ nats. The cost of a confident wrong prediction grows without bound. For the grey values of MNIST, which lie between 0 and 1, the BCE is still used as a loss. It is then smallest at $\tilde{x}_j = x_j$ and no longer a normalised likelihood, so numbers in nats computed with it are statements about this loss. Choosing a reconstruction loss is choosing a likelihood. The squared error assumes Gaussian noise, and the BCE assumes binary pixels that are independent given the network output. Spins are binary without approximation, while their independence given the latent vector is a mean-field approximation.

3.5 Mixtures as two-step draws

The three groups of points in Figure 7 show the procedure on data that can be drawn. For 1500 points in the plane, a single Gaussian fitted by maximum likelihood has the sample mean and the sample covariance as parameters. Its mean log-likelihood per point is $\ell/n = -4.64$. Its density has one peak, placed between the groups where there are no data, and points drawn from it fill that empty space. A mixture of three Gaussians,

$$p_\theta(\mathbf{x}) = \sum_{k=1}^{3} \pi_k \,\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \Sigma_k), \qquad \pi_k \ge 0, \quad \sum_k \pi_k = 1, \tag{3.9}$$

is again a normalised density, because each component has area one and the weights add up to one. Its maximum has no explicit formula, and it is fitted by gradient ascent on ℓ. It reaches $\ell/n = -3.68$, a higher log-likelihood of the same data, and points drawn from it look like the data.

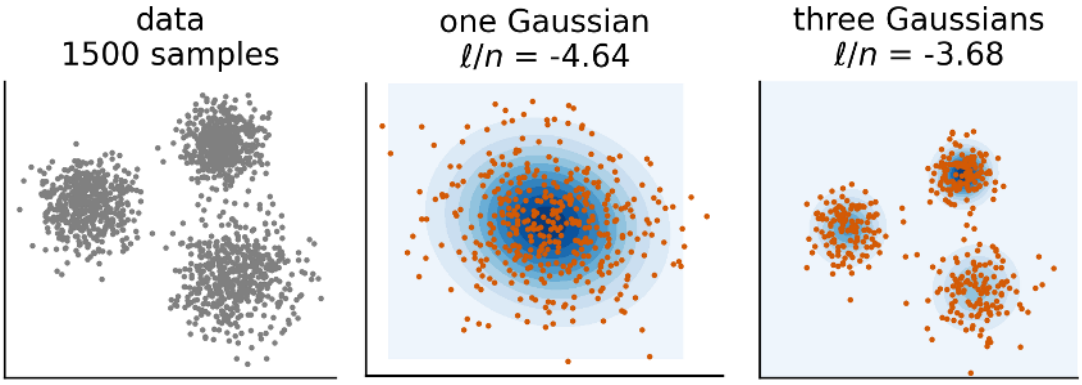


Figure 7: Maximum likelihood in the plane. Left, 1500 data points from three groups. Middle, one Gaussian fitted by maximum likelihood (blue, its density) and 400 points drawn from it (orange). Right, the same for a mixture of three Gaussians. The titles give the mean log-likelihood per point ℓ/n.

The mixture is the first latent-variable model of the chapter, and the vocabulary of Section 2.6 describes its structure. Drawing from it takes two steps. A hidden label k is drawn with probability $p(k) = \pi_k$, and then the point is drawn from the Gaussian of that component, $p_\theta(\mathbf{x} \mid k) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \Sigma_k)$. Only the point is recorded. Its probability is the marginal, a sum over all labels that could have produced it, $p_\theta(\mathbf{x}) = \sum_k p_\theta(\mathbf{x} \mid k)\, p(k)$, as in equation (3.9). Bayes' rule gives the posterior $p_\theta(k \mid \mathbf{x}) = \pi_k \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \Sigma_k)/p_\theta(\mathbf{x})$, the probability that a given point came from group k. For a point deep inside one group the posterior is close to one for that group, and for a point between two groups it is shared. The VAE keeps this structure and replaces the discrete hidden label by a continuous hidden vector. A choice among a few prototypes describes a handwritten digit poorly, since slant, thickness, size and style vary continuously.

3.6 Direct densities for images

In one or two dimensions a density can be written down directly, as a histogram, a Gaussian or a mixture. For images all three fail. A binary image of 28×28 pixels can take $2^{784} \approx 10^{236}$ values, so a histogram with one bin per image cannot even be stored. The $6 \cdot 10^4$ training images would leave all bins but a negligible fraction empty. A single Gaussian over the 784 grey values has the mean image as its centre, a grey blur in which all digits overlap. Its samples are this blur plus correlated noise.

A model of independent pixels is the first family that can be fitted to images (Figure 8). Every pixel of a binarised digit, ink where the grey value exceeds 0.5, is treated as an independent coin with its own probability of ink ϖ_j. The model is then $p_{\boldsymbol{\varpi}}(\mathbf{x}) = \prod_{j=1}^{784} \varpi_j^{x_j} (1 - \varpi_j)^{1 - x_j}$, a product of 784 Bernoulli factors of the form (2.2). Maximum likelihood works pixel by pixel, as for one coin, and gives $\tilde{\varpi}_j$ equal to the fraction of training digits with ink at pixel j, the mean image. At 108 pixels near the border no training digit has ink, and $\tilde{\varpi}_j = 0$ would give a test digit with ink there the probability zero. The probabilities are therefore kept between 10^{-3} and $1 - 10^{-3}$. The fitted model has a negative log-likelihood of 206.2 nats per digit on the training digits and 206.0 on the test digits. Both lie far below the 543 nats of coin flips with probability one half, and they agree because 784 parameters cannot memorise $6 \cdot 10^4$ digits. Digits drawn from the model have ink in roughly the right region and no strokes. In a real digit ink at one pixel makes ink at its neighbours likely, because they belong to one stroke, and a product of independent factors cannot express this. The decoder of the VAE keeps the product form, with probabilities computed from a latent vector, and lets the latent vector carry what the pixels share.

The digits occupy a vanishing and highly structured part of the space of all images. Almost every binary image looks like noise. An image of a digit can be changed into another acceptable digit only

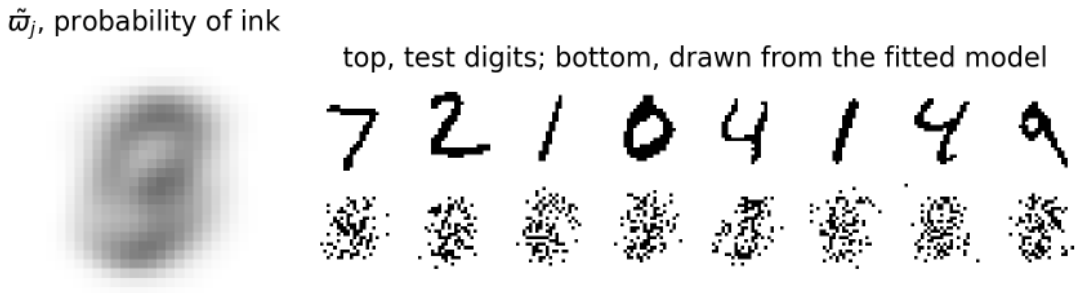


Figure 8: The model of independent pixels fitted to binarised MNIST digits by maximum likelihood. Left, the fitted probabilities of ink $\tilde{\varpi}_j$, the mean binarised image, white for 0 and black for 1. Right, top row, eight binarised test digits. Bottom row, eight digits drawn from the fitted model, every pixel inked with its own probability.

along a few directions, such as thickness, slant or size. According to the manifold hypothesis, the data lie close to a thin, curved set of low dimension inside $\mathbb{R}^{784}$. The same holds for spin configurations at a given temperature. Their relevant directions are the collective variables, the magnetisation, the energy and the positions of domain walls, and the remaining randomness is local. A model of such data has to put this structure into the family itself.

3.7 Gaussian random numbers passed through a learned map

The structure can be built in by drawing a few random numbers from a simple known distribution. A flexible function then places them on the data manifold. In one dimension we can show that starting from a Gaussian costs no generality. Let $z \sim \mathcal{N}(0,1)$ and $x = g(z)$ with an increasing function g. A small interval $\mathrm{d}z$ is mapped to an interval of length $g'(z)\,\mathrm{d}z$, and the probability $p(z)\,\mathrm{d}z$ it carries is conserved, so

$$p_x(x) = \left.\frac{p(z)}{g'(z)}\right|_{z=g^{-1}(x)} . \tag{3.10}$$

Where g is steep the probability is spread thin and p_x is low, and where g is flat it is compressed and p_x is high. Figure 9 uses a function that is steep near $z = 0$, and the single peak of the Gaussian becomes two peaks. Any target distribution with cumulative distribution F is reached by $g(z) = F^{-1}(\Phi(z))$. The number $\Phi(z)$ is uniform on $(0,1)$, and F^{-1} turns a uniform number into a draw from F, as in Section 2.3. A flexible enough function can therefore turn a standard Gaussian into any distribution. The same equation describes a change of variables in statistical mechanics, for example from the speed of a particle to its kinetic energy.

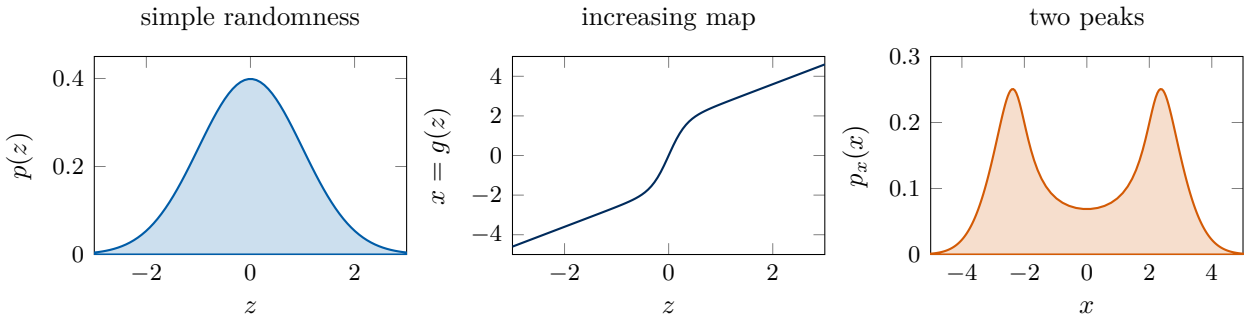


Figure 9: Change of variables, equation (3.10). A standard Gaussian z (left) is passed through the increasing function $g(z) = z + 1.6\tanh(3z)$ (middle), which is steep near $z = 0$. The steep part spreads the central probability thinly, and the density of $x = g(z)$ (right) has two peaks. The function is chosen for illustration.

In many dimensions a latent vector $\mathbf{z} \in \mathbb{R}^d$ is drawn from $\mathcal{N}(0, I)$ and a network maps it to a point of the data space. These points form a d-dimensional surface in $\mathbb{R}^{784}$, which the network bends onto the data manifold. A surface has zero volume, and a distribution concentrated on it would give

zero probability to every real image lying slightly off it. The network output is therefore taken as the centre of a narrow distribution. This distribution is a Gaussian of width σ_x or a product of Bernoulli pixels whose probabilities of ink are the outputs. The conditional distribution $p_\theta(\mathbf{x} \mid \mathbf{z})$ gives every image a nonzero probability, and the model of the data is its marginal over $\mathbf{z}$, a latent-variable model.

4 Latent-variable models

4.1 A hidden cause and an observation

A latent-variable model replaces the discrete label of the mixture by a latent vector $\mathbf{z} \in \mathbb{R}^d$ and the sum by an integral,

$$p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x} \mid \mathbf{z})\, p(\mathbf{z})\, \mathrm{d}\mathbf{z}. \tag{4.1}$$

Four distributions appear in it, and only one of them is learned directly (Table 3). The prior $p(\mathbf{z})$ is the distribution of the hidden cause before any data are seen. It is fixed before training as the standard normal distribution $\mathcal{N}(0, I)$, so that drawing a latent vector means drawing d independent Gaussian numbers. The decoder $p_\theta(\mathbf{x} \mid \mathbf{z})$ is the distribution of the observation once the cause is fixed. A network with parameters θ computes its parameters from $\mathbf{z}$, for digits the probability of ink of every pixel as in equation (3.8). The evidence, or marginal likelihood, $p_\theta(\mathbf{x})$ is the model of the data. It has no formula of its own, follows from the prior and the decoder, and is sampled in two steps, $\mathbf{z}$ from the prior and then $\mathbf{x}$ from the decoder. The posterior $p_\theta(\mathbf{z} \mid \mathbf{x})$ describes the inverse problem, the latent vectors that could have produced an observed $\mathbf{x}$. Bayes' rule connects the four,

$$p_\theta(\mathbf{z} \mid \mathbf{x}) = \frac{p_\theta(\mathbf{x} \mid \mathbf{z})\, p(\mathbf{z})}{p_\theta(\mathbf{x})}. \tag{4.2}$$

Table 3: The pieces of a latent-variable model and of the variational autoencoder.

piece	what it is	learned or fixed	used for
prior $p(\mathbf{z})$	$\mathcal{N}(0, I)$	fixed before training	generation
decoder $p_\theta(\mathbf{x} \mid \mathbf{z})$	network output sets the probabilities of ink, or a Gaussian mean	learned, weights θ	generation and training
evidence $p_\theta(\mathbf{x})$	$\int p_\theta(\mathbf{x}\|\mathbf{z})\, p(\mathbf{z})\, \mathrm{d}\mathbf{z}$	implied by prior and decoder	the model of the data
posterior $p_\theta(\mathbf{z} \mid \mathbf{x})$	latent vectors that could have produced $\mathbf{x}$, Bayes' rule	implied, cannot be computed	interpreting a sample
encoder $q_\phi(\mathbf{z} \mid \mathbf{x})$	Gaussian cloud that approximates the posterior	learned, weights ϕ	training, encoding real samples

Fixing the prior costs no generality, since by the change of variables of Section 3.7 a flexible decoder can bend the standard Gaussian into whatever shape the data require. A learned prior would only shift work from the decoder to the prior.

As an analogy from statistical mechanics, $\mathbf{z}$ resembles a macrostate such as the magnetisation of a ferromagnet and $\mathbf{x}$ a microstate, a configuration of all spins. The posterior then reads the macrostate off a microstate. The magnetisation is a known function of the spins, while the latent vector is hidden and its meaning is set by the training.

4.2 A continuous mixture

Equation (4.1) is the mixture (3.9) with a continuous label. Replacing the integral by a sum over a fine grid of latent values $\mathbf{z}_r$, each representing a small cell of volume $\Delta\mathbf{z}$, gives

$$p_\theta(\mathbf{x}) \approx \sum_r \underbrace{p(\mathbf{z}_r)\,\Delta\mathbf{z}}_{\text{weight } \pi_r}\; p_\theta(\mathbf{x}\,|\,\mathbf{z}_r), \tag{4.3}$$

a mixture of very many narrow components, one per grid cell. Each component is centred on the decoded image of its latent vector, and its weight is the prior probability of its cell. Because the decoder is a continuous function, neighbouring cells decode to similar images and the components form a continuous family. The mixture of Section 3.5 had three unrelated components, whereas a latent-variable model has infinitely many, ordered by $\mathbf{z}$.

4.3 The intractable integral

Training should maximise $\ln p_\theta(\mathbf{x}_i)$ for every training sample, and this needs the integral (4.1). For one latent dimension the grid sum (4.3) suffices. For $d = 16$ a grid of ten values per coordinate has 10^{16} points, each requiring a pass through the decoder. The remaining option is an estimate of the integral as an average over the prior, equation (2.6),

$$p_\theta(\mathbf{x}) = \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\big[p_\theta(\mathbf{x}\,|\,\mathbf{z})\big] \approx \frac{1}{K}\sum_{k=1}^{K} p_\theta(\mathbf{x}\,|\,\mathbf{z}_k), \qquad \mathbf{z}_k \sim \mathcal{N}(0, I), \tag{4.4}$$

and this estimate fails for a different reason (Figure 10). A given digit is produced only by latent vectors in a small region, the region where its posterior lives. Elsewhere the decoder produces other digits, and $p_\theta(\mathbf{x}\,|\,\mathbf{z})$, a product of 784 pixel probabilities, is smaller by many orders of magnitude. Almost every draw from the prior misses the region, and the rare draw that hits it dominates the average. If the region holds a fraction 10^{-6} of the prior probability, an illustrative value, about a million draws are needed for a single useful term. This cost arises for every image and every training step. The posterior (4.2) is out of reach for the same reason, since it divides by $p_\theta(\mathbf{x})$. Training therefore needs a guess, for every sample, of where its relevant latent vectors lie.

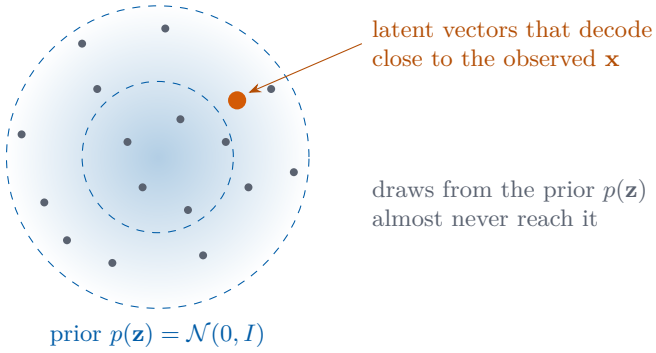


Figure 10: Sampling the prior to estimate $p_\theta(\mathbf{x})$, sketch. The prior (blue shading, circles of radius 1 and 2) spreads over the latent plane, while only a small region (orange) decodes close to the observed sample. Grey dots are draws from the prior, and none of them contributes to the average (4.4).

5 The variational autoencoder

5.1 Gaussian clouds of latent vectors

The variational autoencoder [3, 4] supplies the guess with an encoder network and changes the autoencoder of Section 1 in two ways. The first change concerns the encoder, which returns for an

input $\mathbf{x}$ a Gaussian cloud of latent vectors in place of a single point,

$$q_\phi(\mathbf{z}\,|\,\mathbf{x}) := \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_\phi(\mathbf{x}), \operatorname{diag} \boldsymbol{\sigma}^2_\phi(\mathbf{x})), \tag{5.1}$$

whose centre $\boldsymbol{\mu}_\phi(\mathbf{x})$ and widths $\sigma_{\phi,k}(\mathbf{x})$, d numbers each, are outputs of a network with parameters ϕ. The covariance is diagonal, case (b) of Figure 5, so the cloud is an ellipse aligned with the latent axes and is described by $2d$ numbers. During training the decoder receives a random point of this cloud, a different one at every step. Every latent vector near $\boldsymbol{\mu}_\phi(\mathbf{x})$ must therefore decode to an image close to $\mathbf{x}$. The decoder is thus trained on neighbourhoods of the codes, and where the clouds of different images overlap it learns to produce intermediate images.

Clouds alone give no distribution to draw from yet, since with the reconstruction term alone they would shrink back to scattered points (Figure 13). The second change adds a penalty on the difference between each cloud and the prior, measured by the divergence $\mathrm{KL}\,(q_\phi(\mathbf{z}\,|\,\mathbf{x})\,\|\,\mathcal{N}(0, I))$. It grows when a centre moves away from the origin and when a width departs from one, most steeply when it shrinks. Under the two pressures the clouds of all training images overlap and fill the region where $\mathcal{N}(0, I)$ puts its probability. A latent vector drawn from the prior then lands where the decoder was trained (Figure 11). The sum of the reconstruction loss and this penalty is the negative of a lower bound on $\ln p_\theta(\mathbf{x})$, so training the VAE is approximate maximum likelihood.

The quantity that training should raise, $\ln p_\theta(\mathbf{x})$, is an integral over all latent vectors that cannot be computed (Section 4). Bayes' rule, averaged over the encoder cloud, splits it without approximation into two parts. The first, the evidence lower bound (ELBO), can be computed from a few latent vectors drawn from the cloud. The second is the divergence of the cloud from the true posterior. This divergence cannot be computed and is never negative, so the ELBO lies below $\ln p_\theta(\mathbf{x})$, and raising the ELBO improves the decoder and the encoder together. Written out, the ELBO is a reconstruction term minus the KL penalty, and the loss of the VAE is its negative.

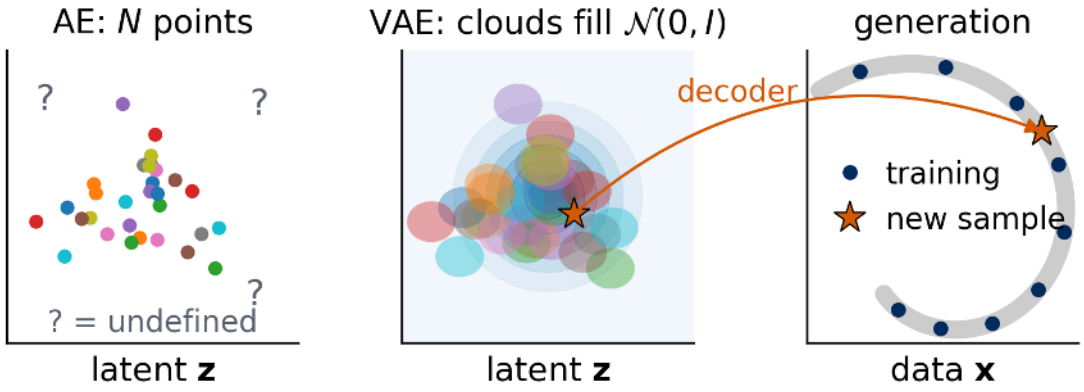


Figure 11: Points and clouds, sketch. Left, an autoencoder maps n images (N in the panel title) to n latent points (colours are classes), and nothing is defined between them (question marks). Middle, a VAE maps every image to a Gaussian cloud $q_\phi(\mathbf{z}\,|\,\mathbf{x})$, and the clouds fill the prior $\mathcal{N}(0, I)$ (blue shading). Right, a new latent vector (star) is decoded into a new sample in data space, here between training points (dots) on a curve (grey).

5.2 The bound from Bayes' rule

Bayes' rule (4.2), written for one image $\mathbf{x}$, holds for every latent vector $\mathbf{z}$. Its logarithm reads

$$\ln p_\theta(\mathbf{x}) = \ln p_\theta(\mathbf{x}\,|\,\mathbf{z}) + \ln p(\mathbf{z}) - \ln p_\theta(\mathbf{z}\,|\,\mathbf{x}).$$

The left-hand side is independent of $\mathbf{z}$, so averaging both sides over latent vectors drawn from the encoder cloud, $\mathbf{z} \sim q_\phi(\mathbf{z}\,|\,\mathbf{x})$, leaves it unchanged. On the right-hand side we add and subtract

$\ln q_\phi(\mathbf{z}\,|\,\mathbf{x})$ inside the average and group the terms,

$$\begin{aligned}\ln p_\theta(\mathbf{x}) &= \mathbb{E}_{q_\phi}\Big[\ln p_\theta(\mathbf{x}\,|\,\mathbf{z}) + \ln\frac{p(\mathbf{z})}{q_\phi(\mathbf{z}\,|\,\mathbf{x})}\Big] + \mathbb{E}_{q_\phi}\Big[\ln\frac{q_\phi(\mathbf{z}\,|\,\mathbf{x})}{p_\theta(\mathbf{z}\,|\,\mathbf{x})}\Big] \\ &= \underbrace{\mathbb{E}_{q_\phi}\big[\ln p_\theta(\mathbf{x}\,|\,\mathbf{z})\big] - \mathrm{KL}\left(q_\phi(\mathbf{z}\,|\,\mathbf{x})\,\|\,p(\mathbf{z})\right)}_{\mathrm{ELBO}(\mathbf{x})} + \underbrace{\mathrm{KL}\left(q_\phi(\mathbf{z}\,|\,\mathbf{x})\,\|\,p_\theta(\mathbf{z}\,|\,\mathbf{x})\right)}_{\text{gap}}.\end{aligned} \tag{5.2}$$

The averages of $\ln[p/q_\phi]$ and of $\ln[q_\phi/p_\theta(\mathbf{z}|\mathbf{x})]$ are recognised as minus and plus a divergence (3.5). The first part is the evidence lower bound,

$$\mathrm{ELBO}(\mathbf{x}) \coloneqq \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\ln p_\theta(\mathbf{x}\,|\,\mathbf{z})\big]}_{\text{reconstruction}} - \underbrace{\mathrm{KL}\left(q_\phi(\mathbf{z}\,|\,\mathbf{x})\,\|\,p(\mathbf{z})\right)}_{\text{distance from the prior}} \;\le\; \ln p_\theta(\mathbf{x}), \tag{5.3}$$

named after the evidence $p_\theta(\mathbf{x})$ of Table 1. Both of its terms can be computed. The reconstruction term is estimated from a few latent vectors drawn from the cloud. The divergence of a Gaussian cloud from $\mathcal{N}(0, I)$ is an explicit function of its centre and widths. The second part of equation (5.2), the gap, is the divergence of the cloud from the true posterior. It cannot be computed, because the posterior is out of reach. A divergence is never negative, equation (3.6), so the ELBO never exceeds $\ln p_\theta(\mathbf{x})$, and it equals $\ln p_\theta(\mathbf{x})$ when the cloud is the posterior.

5.3 The size of the gap

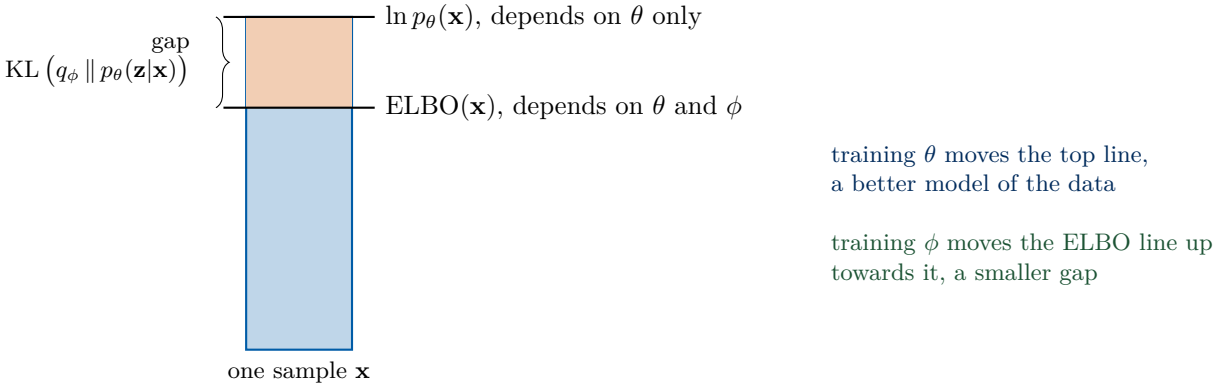


Figure 12: Equation (5.2) as a bar. The log-likelihood of a sample (top line) is the ELBO plus a gap equal to the divergence of the encoder cloud from the true posterior. Raising the ELBO with respect to the encoder parameters ϕ leaves the top line in place and shrinks the gap. Raising it with respect to the decoder parameters θ improves the model of the data as long as the gap stays small.

Drawn as a bar in Figure 12, the identity shows how one objective trains both networks. Maximising the ELBO with respect to ϕ leaves $\ln p_\theta(\mathbf{x})$ unchanged and can only shrink the gap, which makes the encoder a better approximation of the posterior. Maximising it with respect to θ raises the bound and, as long as the gap stays small, the log-likelihood. The word variational refers to the choice of the best member of a family of approximating distributions, here the family of Gaussian clouds. The variational principle of quantum mechanics has the same structure and serves as an analogy. For any normalised trial state the expectation value of the Hamiltonian bounds the ground-state energy from above, $\langle\psi|\hat{H}|\psi\rangle \ge E_0$. The best member of a family of trial states gives the tightest bound, and the difference measures how far the trial state is from the ground state. In the VAE the bound lies below $\ln p_\theta(\mathbf{x})$ and is maximised, and the gap is the divergence of the trial cloud from the posterior.

5.4 A numerical example with two latent values

With two latent values every quantity of the last two subsections becomes a number that can be checked directly. Let z take the values 1 and 2 with prior probabilities $p(z) = \frac{1}{2}$ each, and let the

decoder give an observed image $\mathbf{x}$ the probabilities $p(\mathbf{x}\,|\,z{=}1) = 0.8$ and $p(\mathbf{x}\,|\,z{=}2) = 0.1$. The integral (4.1) becomes a sum of two terms,

$$p(\mathbf{x}) = 0.8 \cdot 0.5 + 0.1 \cdot 0.5 = 0.45, \qquad \ln p(\mathbf{x}) = -0.799,$$

and Bayes' rule gives the posterior $p(z{=}1\,|\,\mathbf{x}) = 0.40/0.45 = 0.889$ and $p(z{=}2\,|\,\mathbf{x}) = 0.111$. Table 4 evaluates the bound for three guesses $q(z)$. For a guess $q(z)$ the ELBO is the sum $\sum_z q(z)\big[\ln p(\mathbf{x}\,|\,z) + \ln p(z) - \ln q(z)\big]$, and it lies below $\ln p(\mathbf{x})$ by the divergence of the guess from the posterior. The closer a guess is to the posterior, the smaller the gap, and for the posterior itself the bound becomes an equality. The first two columns split the same bound into reconstruction and divergence from the prior. The posterior pays 0.34 nats of divergence from the prior and gains 0.81 nats in reconstruction compared with the prior as a guess.

Table 4: The ELBO of a model with two latent values for three guesses $q(z)$ of the posterior. The evidence is $\ln p(\mathbf{x}) = -0.799$ and the gap is $\ln p(\mathbf{x}) - \text{ELBO}$.

guess $q(1)$, $q(2)$	reconstruction $\mathbb{E}_q[\ln p(\mathbf{x}\|z)]$	$\mathrm{KL}(q \,\|\, p(z))$	ELBO	gap
0.5, 0.5 (prior)	-1.263	0.000	-1.263	0.464
0.7, 0.3	-0.847	0.082	-0.929	0.131
0.889, 0.111 (posterior)	-0.454	0.344	-0.799	0

5.5 The two terms of the bound

The two terms of equation (5.3) act as opposite forces on the clouds (Figure 13). The reconstruction term pushes the clouds of different images apart and shrinks them, since then a latent vector identifies its image. Acting alone it returns an autoencoder with points scattered anywhere and holes between them. The divergence term pulls every centre towards the origin and every width towards one. Acting alone it makes every cloud equal to the prior, and the latent vector carries no information about $\mathbf{x}$. The decoder, which appears only in the reconstruction term, then has nothing to learn from. The trained clouds are as narrow as the reconstruction needs and packed together inside the prior. A related failure occurs with both terms present. When the decoder can rebuild the data well without the latent vector, the divergence pushes the clouds onto the prior and the decoder learns to ignore $\mathbf{z}$. This failure, called posterior collapse [6], can affect individual latent coordinates of a trained model.

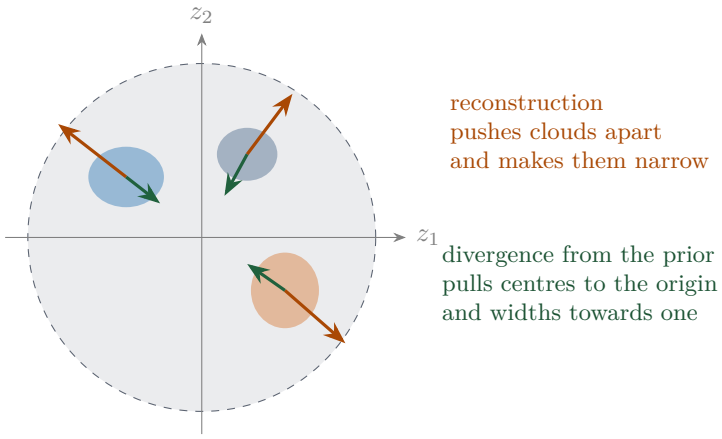


Figure 13: The two terms of the ELBO as forces on the encoder clouds of three images, sketch. The maximum of the bound balances the two, with clouds packed together inside the prior (grey disc).

5.6 The divergence of a Gaussian cloud from the prior

For one coordinate with $q = \mathcal{N}(\mu, \sigma^2)$ and $p = \mathcal{N}(0, 1)$, the logarithms of the densities are $\ln q(z) = -\frac{1}{2}\ln(2\pi\sigma^2) - (z-\mu)^2/2\sigma^2$ and $\ln p(z) = -\frac{1}{2}\ln(2\pi) - z^2/2$. The averages under q of the two squares

are $\mathbb{E}_q[(z-\mu)^2] = \sigma^2$ and $\mathbb{E}_q[z^2] = \mu^2 + \sigma^2$, so

$$\mathrm{KL}(q \,\|\, p) = \mathbb{E}_q[\ln q - \ln p] = \underbrace{-\tfrac{1}{2}\ln\sigma^2 - \tfrac{1}{2}}_{\mathbb{E}_q[\ln q] + \frac{1}{2}\ln 2\pi} + \underbrace{\tfrac{1}{2}(\mu^2+\sigma^2)}_{-\mathbb{E}_q[\ln p] - \frac{1}{2}\ln 2\pi} = \tfrac{1}{2}(\mu^2 + \sigma^2 - \ln\sigma^2 - 1). \tag{5.4}$$

For a diagonal Gaussian both densities factorise over the coordinates, equation (2.8), the logarithm of a product is a sum, and the divergence is a sum over coordinates,

$$\mathrm{KL}\left(q_\phi(\mathbf{z} \,|\, \mathbf{x}) \,\|\, \mathcal{N}(0, I)\right) = \sum_{k=1}^{d} \mathrm{KL}_k, \qquad \mathrm{KL}_k \coloneqq \tfrac{1}{2}(\mu_k^2 + \sigma_k^2 - \ln\sigma_k^2 - 1). \tag{5.5}$$

Each coordinate contributes separately. The term μ_k^2 is the cost of moving the centre, and moving it by one prior standard deviation costs half a nat. The term $\sigma_k^2 - \ln\sigma_k^2 - 1$ is the cost of a narrow cloud and vanishes at $\sigma_k = 1$. Narrowing a cloud from width 1 to $1/e$ costs $(e^{-2}+1)/2 = 0.57$ nats, and every further factor e costs about one nat more. A VAE with $d = 2$ trained on the digits has clouds of median width $0.04, 0.04$ along its two axes. By equation (5.4) with $\mu_k = 0$ this width costs about 2.8 nats per coordinate, plus about half a nat for centres spread like the prior. The divergence is the price, in nats, of the information that the latent vector carries about the image. Averaged over the data it is an upper bound on that information. A narrow cloud pins down the image and costs much, and a wide cloud that overlaps with the clouds of other images carries little information and costs little.

5.7 The reparametrisation trick

The reconstruction term is estimated with one latent vector drawn from the cloud per image and training step. Its gradient with respect to the decoder parameters θ is ordinary. Its gradient with respect to the encoder parameters ϕ poses a problem, because ϕ enters only through the distribution from which $\mathbf{z}$ is drawn. A sampling routine gives no information about how its output would move if $\boldsymbol{\mu}$ or $\boldsymbol{\sigma}$ changed slightly. The remedy is the shift-and-stretch rule (2.5), in which a parameter-free noise is drawn and then shifted and stretched,

$$\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}) + \boldsymbol{\sigma}_\phi(\mathbf{x}) \odot \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, I), \qquad \frac{\partial z_k}{\partial \mu_k} = 1, \quad \frac{\partial z_k}{\partial \sigma_k} = \varepsilon_k, \tag{5.6}$$

where $\odot$ denotes the product component by component. This $\mathbf{z}$ has the distribution (5.1). An average over q_ϕ becomes an average, over the fixed distribution of $\boldsymbol{\varepsilon}$, of a differentiable function of ϕ, $\mathbb{E}_{q_\phi}[f(\mathbf{z})] = \mathbb{E}_{\boldsymbol{\varepsilon}}[f(\boldsymbol{\mu}_\phi + \boldsymbol{\sigma}_\phi \odot \boldsymbol{\varepsilon})]$. The randomness enters as an external input, and the gradient passes from the loss through the decoder and $\mathbf{z}$ into the encoder (Figure 14). The encoder outputs $\ln \boldsymbol{\sigma}^2$, which can take any real value while $\sigma_k^2 = e^{\ln\sigma_k^2}$ stays positive.

5.8 The loss, the network and the training

The loss of the VAE is the negative ELBO averaged over a minibatch of B images, with one latent vector per image,

$$\mathcal{L}(\phi, \theta) = \frac{1}{B}\sum_{b=1}^{B}\Big[\underbrace{\mathrm{BCE}\left(\mathbf{x}_b, \tilde{\mathbf{x}}_b\right)}_{-\ln p_\theta(\mathbf{x}_b|\mathbf{z}_b)} + \underbrace{\tfrac{1}{2}\sum_{k=1}^{d}\left(\mu_{bk}^2 + \sigma_{bk}^2 - \ln\sigma_{bk}^2 - 1\right)}_{\mathrm{KL}(q_\phi(\mathbf{z}|\mathbf{x}_b)\,\|\,\mathcal{N}(0,I))}\Big], \qquad \tilde{\mathbf{x}}_b = \mathrm{sigm}\left(G_\theta(\mathbf{z}_b)\right), \tag{5.7}$$

where G_θ is the decoder network and $\mathbf{z}_b$ is drawn as in equation (5.6). The 784 outputs of the decoder network, the logits, are turned into probabilities of ink by the logistic function $\mathrm{sigm}(u) = 1/(1+e^{-u})$ applied to every component. The network for the digits is dense (Figure 15). The encoder narrows the 784 pixels through layers of 400 and 200 units with the rectified linear unit $\mathrm{ReLU}(u) = \max(0, u)$

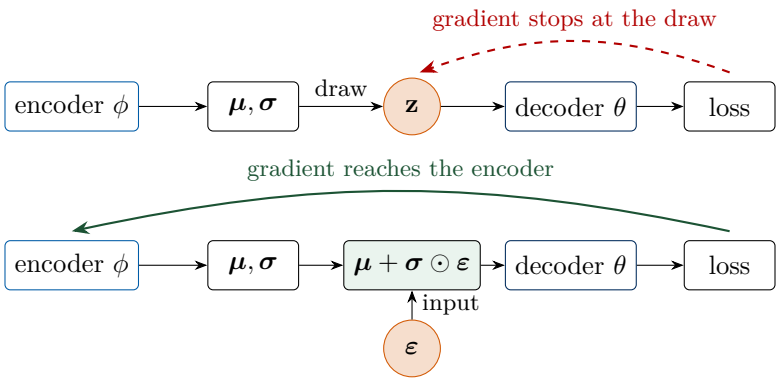


Figure 14: The reparametrisation trick as a computation graph, sketch. Top, a latent vector drawn inside the network (orange node) blocks the gradient of the loss on its way to the encoder. Bottom, the noise $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, I)$ enters as an external input, $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}$ is a differentiable function of the encoder outputs, and the gradient reaches the encoder parameters ϕ.

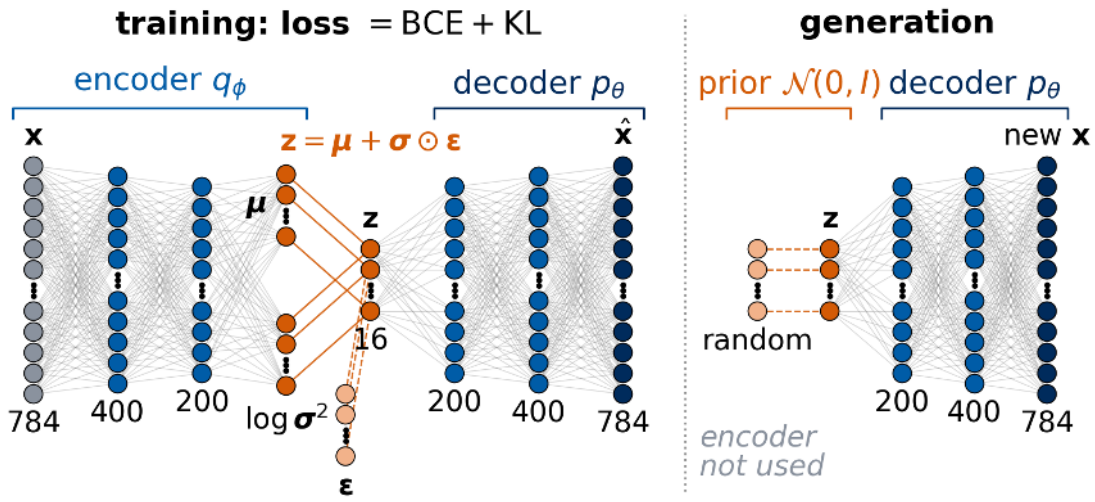


Figure 15: The VAE for MNIST with $d = 16$ in its two uses. Left, training. The encoder (layer widths below the columns) ends in two heads, $\boldsymbol{\mu}$ and $\log \boldsymbol{\sigma}^2$ (orange). The noise $\boldsymbol{\varepsilon}$ (light orange) enters as a separate input, $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}$, and the decoder returns the probabilities of ink $\tilde{\mathbf{x}}$. The loss is the BCE plus the KL term (5.7). Right, generation. A random $\mathbf{z}$ drawn from the prior replaces the encoder, and the decoder alone produces a new image.

to two linear heads, $\boldsymbol{\mu}$ and $\ln \boldsymbol{\sigma}^2$. The decoder mirrors it. One encoder serves every input, old and new, so no parameters are stored for individual images, a sharing called amortised inference.

Neither network is ever given the correct latent vector of an image. At the start both have random weights. The encoder sends a minibatch of images to clouds at arbitrary places, and the decoder turns the latent vectors drawn from these clouds into nearly uniform grey images. One backward pass through the reparametrised latent vectors gives the gradient of the loss (5.7) with respect to all weights. The decoder weights move so that the latent vectors it receives rebuild their images better. The encoder weights move so that each image gets a latent vector from which the current decoder rebuilds it better and a cloud closer to the prior. After many steps the encoder places similar images at nearby latent vectors, because the decoder can then serve them with similar outputs. The decoder has by then learned what the latent vectors mean.

5.9 A VAE on points near a spiral

Points $\mathbf{x} \in \mathbb{R}^2$ near a spiral are generated by the kind of process a VAE assumes, and every part of the model can be seen on them. A hidden number $z_{\text{true}} \sim \mathcal{N}(0, 1)$ chooses a position along the spiral, and the point is placed near that position with Gaussian scatter of width $\sigma_x = 0.08$. The data set has 4000 such points, and the hidden positions are kept for evaluation only. We train on the points a

VAE with one latent number and a Gaussian decoder of width σ_x around a decoded point $\mathbf{g}_\theta(z)$. The loss is equation (5.7) with the squared error (3.7) in place of the BCE, and z_{true} and the curve stay hidden from the training.

In Figure 16 the decoder curve $\mathbf{g}_\theta(z)$, traced for z from -3 to 3, follows the spiral. It is the learned map of Section 3.7, which takes the one-dimensional Gaussian prior and lays it along the curve. The model density $p_\theta(\mathbf{x})$ is this curve thickened by the decoder noise σ_x. Points generated in two steps, $z \sim \mathcal{N}(0,1)$ and then $\mathbf{x}$ from the decoder, fall on the data. The encoder has learned the inverse map. It encodes every point into a narrow cloud on the latent line, median width 0.04, at the position where it sits along the curve. The cloud centres correlate with the hidden position with $|\rho| = 0.993$. Here $\rho = \mathrm{Cov}(a,b)/\sqrt{\mathrm{Var}(a)\,\mathrm{Var}(b)}$ is the Pearson correlation coefficient of two quantities a and b. It equals ± 1 for a linear relation without scatter and 0 in the absence of a linear relation. Without being told that a hidden cause exists, the latent variable has recovered it up to its sign and a smooth change of scale. The average of all clouds,

$$q_\phi(\mathbf{z}) \coloneqq \frac{1}{n}\sum_{i=1}^{n} q_\phi(\mathbf{z}\,|\,\mathbf{x}_i), \tag{5.8}$$

is called the aggregate posterior, the distribution of latent values that the encoded data occupy. Here its mean is -0.02 and its standard deviation 1.04, close to the prior, so points drawn from the prior decode onto the data. The mean KL term per point is 3.4 nats. Equation (5.4) gives about 3.2 nats for a cloud of width 0.04 whose centre has unit variance over the data. Almost all of the KL term is thus the price of locating every point on the spiral to that precision.

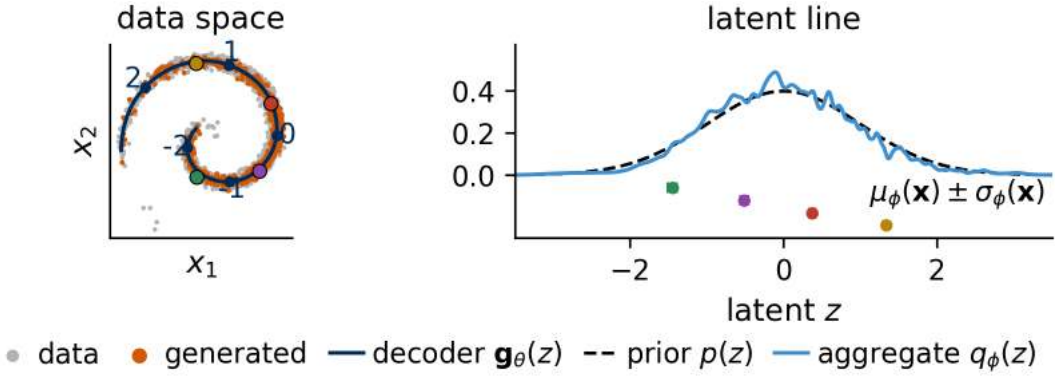


Figure 16: A VAE with one latent number on points near a spiral. Left, data space with coordinates (x_1, x_2), the training points (grey), the decoder curve $\mathbf{g}_\theta(z)$ for $z \in [-3,3]$ (navy, labelled at $z = -2,\ldots,2$) and 500 points generated from $z \sim \mathcal{N}(0,1)$ (orange). Four training points are marked in colour. Right, the latent line, with the prior density $p(z)$ (dashed), the aggregate posterior $q_\phi(z)$ (blue) and the clouds $\mu_\phi(\mathbf{x}) \pm \sigma_\phi(\mathbf{x})$ of the four marked points (coloured bars, drawn below the axis).

5.10 Tightness of the bound on digits

Two VAEs are trained on the 60 000 MNIST training digits without labels, one with $d = 2$, whose latent plane can be drawn, and one with $d = 16$. On the test digits the $d = 16$ model reaches $-\,\mathrm{ELBO} = 101.0$ nats per image, of which 22.7 nats are the KL term, against 141.9 nats for $d = 2$. A model that inks every pixel with probability one half would cost $784 \ln 2 = 543$ nats (Section 3.2). The KL term is the price of the information the latent vector carries about the image. The model with $d = 16$ buys more of it than the one with $d = 2$ and reconstructs better in return.

To compare the bound with the log-likelihood itself, for a subset of 200 test images we estimate $\ln p_\theta(\mathbf{x})$ with $K = 1000$ draws in two ways, both computed as logarithms of sums. The first is the average (4.4) over the prior. The second is the importance-sampling average over the encoder cloud, $p_\theta(\mathbf{x}) \approx \frac{1}{K}\sum_k p_\theta(\mathbf{x}\,|\,\mathbf{z}_k)\,p(\mathbf{z}_k)/q_\phi(\mathbf{z}_k\,|\,\mathbf{x})$ with $\mathbf{z}_k \sim q_\phi$. It rewrites the integral (4.1) as an average over the cloud, with the ratio p/q_ϕ correcting for the change of the sampling distribution. Its draws

therefore land where the image is explained. The mean ELBO of this subset is -98.1 nats and the importance-sampling estimate is -93.3 nats. The gap of equation (5.2) thus averages about 4.8 nats, and the bound is close to tight. Sampling the prior gives -154.8 nats for the same quantity, far too low, because almost none of the 1000 prior draws lands where $p_\theta(\mathbf{x} \mid \mathbf{z})$ is large. This measures the failure sketched in Figure 10. The logarithm of an importance-sampling average is itself a lower bound that tightens as K grows [5].

5.11 The four quantities on one image

Table 5 collects the quantities of this section for the digits model with $d = 16$. No formula gives the log-likelihood $\ln p_\theta(\mathbf{x})$, the target of maximum likelihood, and the ELBO that the network computes lies a few nats below it. Their gap measures how far the Gaussian clouds of the encoder are from the true posteriors and is small for this model. The loss that the training prints is minus the ELBO, the BCE plus the KL term. Its KL part is the price of the information that the latent vector carries.

Table 5: The ELBO and its relatives for the VAE with $d = 16$ on MNIST, in nats per image. The first three rows are means over the subset of 200 test images used for the importance-sampling estimate, the last two rows means over all test images.

quantity	role	how it is obtained	value
$\ln p_\theta(\mathbf{x})$	what maximum likelihood wants to raise	estimated with $K = 1000$ draws from the cloud	-93.3
$\mathrm{ELBO}(\mathbf{x})$	what training raises, never above $\ln p_\theta(\mathbf{x})$	reconstruction minus KL, equation (5.3)	-98.1
gap	divergence of the cloud from the true posterior	difference of the two rows above, equation (5.2)	4.8
loss $-$ ELBO	what the training prints	BCE plus KL, equation (5.7)	101.0
KL part of the loss	price of the information in $\mathbf{z}$	closed form (5.5)	22.7

6 Content of a trained VAE

A trained VAE consists of a decoder and an encoder. The decoder is a map from the latent space to distributions of images, and the encoder is an approximate posterior that places every image in the latent space. Together with the fixed prior they define the model distribution $p_\theta(\mathbf{x})$, which can be sampled and has no explicit formula. We examine each of them on the digits and compare them with the autoencoder of Section 1.

6.1 Points and clouds

The latent plane of the $d = 2$ models shows the effect of the divergence term (Figure 17). The autoencoder places one point per image, with coordinate spreads of $3.8, 7.5$ chosen by the training, and leaves empty regions between the islands of codes. The VAE centres have standard deviations $1.03, 1.09$, close to the prior, and fill the disc where $\mathcal{N}(0, I)$ puts its probability. Each cloud is small, with a median width of $0.04, 0.04$ along the two axes, because a code of two numbers must locate an image accurately. The 60 000 training clouds nevertheless lie densely side by side. Images of the same digit still sit together, although no label was used, and the regions of different digits now border on each other.

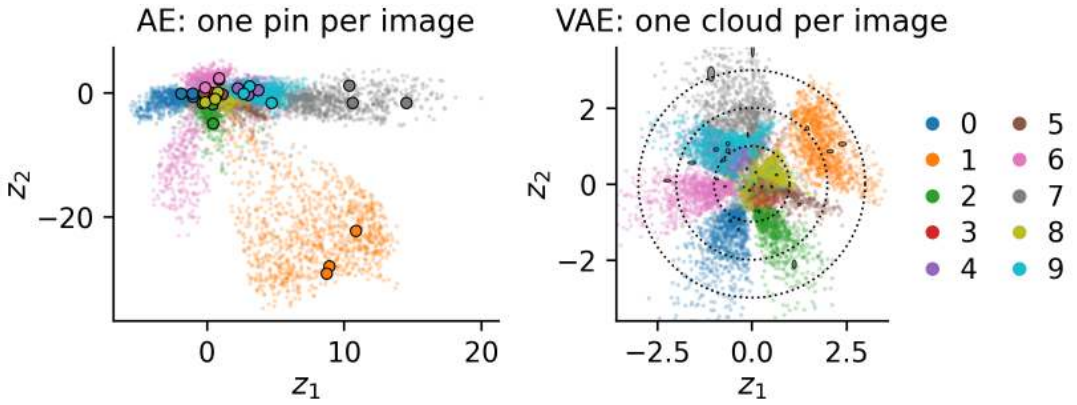


Figure 17: Latent planes with $d = 2$, coordinates (z_1, z_2). Left, the autoencoder codes $E(\mathbf{x})$ of all test digits (small dots) and of three digits per class (large dots). Right, the VAE centres $\boldsymbol{\mu}_\phi(\mathbf{x})$ of all test digits (small dots), the clouds $\boldsymbol{\mu}_\phi \pm \boldsymbol{\sigma}_\phi$ of three digits per class (ellipses), and the circles $|\mathbf{z}| = 1, 2, 3$ (dotted). Colours show the true digit. The axes of the two panels differ.

The decoded grid of the VAE uses the prior to place the grid points. Along each axis the values are the quantiles $z^{(j)} = \Phi^{-1}((j + \frac{1}{2})/10)$, $j = 0, \ldots, 9$, of the standard normal distribution, obtained by the inverse transformation of Section 2.3. Every cell then carries prior probability 1/100, and the grid pictures what sampling the prior produces. Every cell decodes to a digit-like image, including the cells between two digits (Figure 18), whereas the autoencoder grid of Figure 1 had cells that decode to no digit. The images are blurred, since a decoded image is the mean of $p_\theta(\mathbf{x} \mid \mathbf{z})$, the average of all pixel patterns compatible with $\mathbf{z}$.

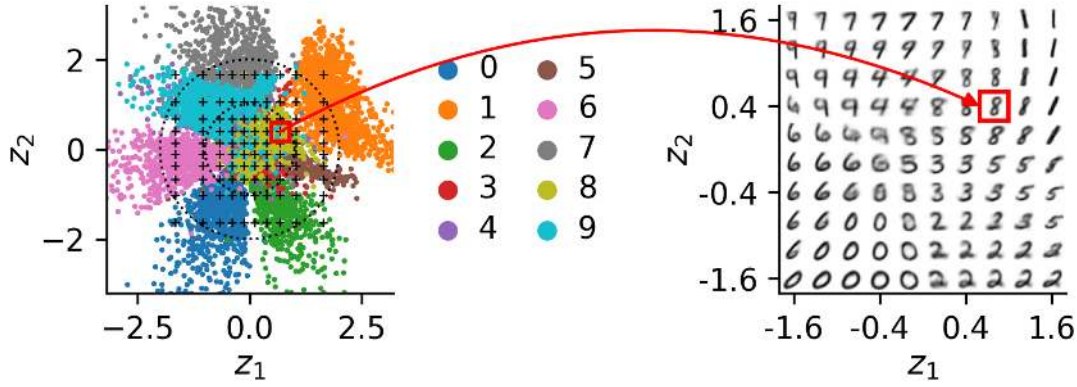


Figure 18: VAE with $d = 2$, layout as in Figure 1. Left, the centres $\boldsymbol{\mu}_\phi(\mathbf{x})$ of the test digits and a 10×10 grid of latent points at the quantiles of $\mathcal{N}(0, 1)$, from -1.64 to 1.64 along each axis. Right, the decoder mean $\text{sigm}(G_\theta(\mathbf{z}))$, the probability of ink of every pixel as a grey level, at every grid point.

6.2 Reconstruction, generation and samples

A trained VAE serves for reconstruction and for generation. Reconstruction encodes an existing image and decodes a point of its cloud, $\mathbf{x} \to \mathbf{z} \sim q_\phi(\mathbf{z} \mid \mathbf{x}) \to p_\theta(\mathbf{x} \mid \mathbf{z})$. Generation draws $\mathbf{z}$ from the prior and decodes it, which samples $p_\theta(\mathbf{x})$ in two steps without the encoder. For generation the encoder is only a training device. It told the decoder which latent vectors belong to which images, and the divergence term made these vectors fill the prior, so the prior can replace the encoder afterwards. For $d = 16$ (Figure 19) the reconstructions keep the identity and most of the style of their inputs, while the generated digits belong to no input. Most are clear, and some lie between two digits.

The decoder returns a distribution of images, $p_\theta(\mathbf{x} \mid \mathbf{z}) = \prod_j \tilde{x}_j^{x_j}(1 - \tilde{x}_j)^{1-x_j}$, and usually its mean $\tilde{\mathbf{x}}$ is shown. A sample of the model draws every pixel independently, black with probability $\tilde{x}_j$, the coin of Section 2.1 tossed once per pixel (Figure 20). Inside a stroke, where $\tilde{x}_j$ is close to one, nearly every pixel comes out black. At the edges, where $\tilde{x}_j$ is near one half, the pixels are black or white at

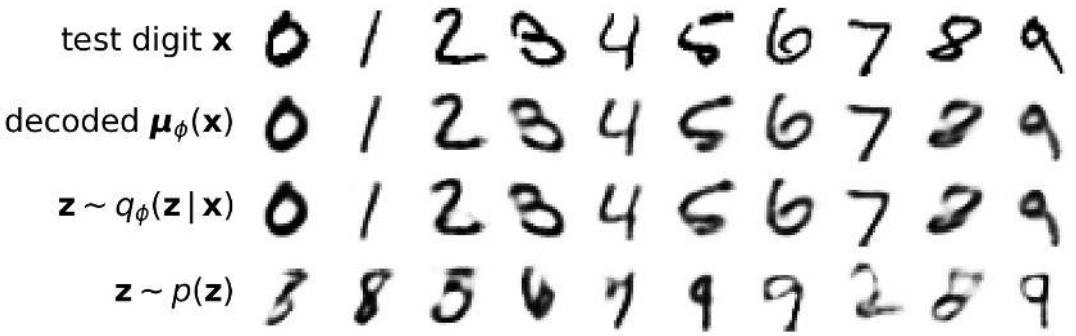


Figure 19: VAE with $d = 16$. First row, one test digit $\mathbf{x}$ per class. Second row, the decoder mean at the centre of its cloud, $\boldsymbol{\mu}_\phi(\mathbf{x})$. Third row, at a random point of the cloud. Fourth row, at ten latent vectors drawn from the prior, unrelated to the images above them. Grey levels are probabilities of ink.

random, and two samples of the same mean differ there. The speckle is the correct output of a model whose pixels are independent given $\mathbf{z}$. Every correlation between pixels, such as the continuity of a stroke, must be carried by $\mathbf{z}$, and independent coin flips fill in everything that $\mathbf{z}$ leaves open. A spin configuration must consist of spins ± 1, so for spins a sample has to be shown.

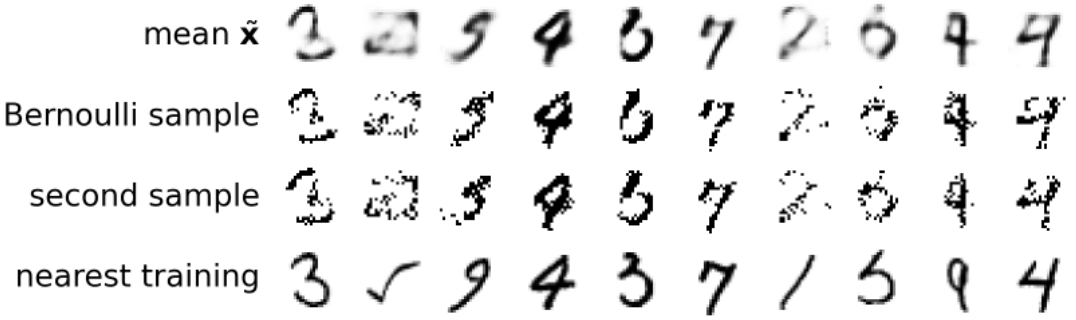


Figure 20: Decoder mean and decoder samples, $d = 16$. First row, ten decoder means $\tilde{\mathbf{x}}$ at latent vectors drawn from the prior. Second and third rows, two independent samples of each, every pixel black with probability $\tilde{x}_j$. Fourth row, the training image closest to each mean in Euclidean distance.

The generated digits differ from the training images. For 500 generated means the Euclidean distance to the nearest of the 60 000 training images is on average 4.48. Real test digits, which the network never saw, lie at 4.19. A sample is a new combination of features seen in the data, a slant from here and a loop from there.

6.3 Three distributions in the latent space

Three distributions describe the latent vectors of a trained VAE (Table 6). The prior $\mathcal{N}(0, I)$ is fixed and supplies the latent vectors for generation. Each image has its own encoder cloud, a Gaussian with diagonal covariance that approximates the true posterior $p_\theta(\mathbf{z} \mid \mathbf{x})$. The aggregate posterior (5.8), the average of all clouds, describes where the encoded data lie. It is a mixture of many ellipses and in general no Gaussian. The divergence term pulls each cloud towards the prior, while the clouds must stay distinguishable to reconstruct their images, so $q_\phi(\mathbf{z})$ equals $p(\mathbf{z})$ only approximately. Regions that the prior visits and no training image was encoded into are called prior holes. Latent vectors drawn there decode to less clear images.

For $d = 16$ the per-coordinate standard deviations of $q_\phi(\mathbf{z})$ lie between 0.97 and 1.04, and its projection on two coordinates looks like the prior (Figure 21). In sixteen dimensions most of the probability of $\mathcal{N}(0, I)$ lies on a thin shell of radius near $\sqrt{16} = 4$. The squared length $|\mathbf{z}|^2$ is a sum of sixteen squares of standard Gaussian numbers, each with mean one. The mean length is 4.01 for the aggregate posterior and 3.93 for the prior. The two distributions nevertheless differ in shape. The median distance from a prior sample to the nearest encoded test image is 2.61, against 1.55 between

Table 6: Distributions in the latent space of a variational autoencoder.

distribution	form	Gaussian	covariance
prior $p(\mathbf{z})$	$\mathcal{N}(0, I)$, chosen before training	yes	identity
encoder cloud $q_\phi(\mathbf{z}\|\mathbf{x})$	$\mathcal{N}\big(\boldsymbol{\mu}_\phi(\mathbf{x}), \operatorname{diag}\boldsymbol{\sigma}^2_\phi(\mathbf{x})\big)$, one per image	yes, by construction	diagonal
true posterior $p_\theta(\mathbf{z}\|\mathbf{x})$	Bayes' rule with the decoder	in general no	in general not diagonal
aggregate posterior $q_\phi(\mathbf{z})$	$\frac{1}{n}\sum_i q_\phi(\mathbf{z}\|\mathbf{x}_i)$	no, a mixture	any

encoded images. Part of what the prior proposes lies off the encoded data, consistent with generated digits being on average less clear than reconstructions.

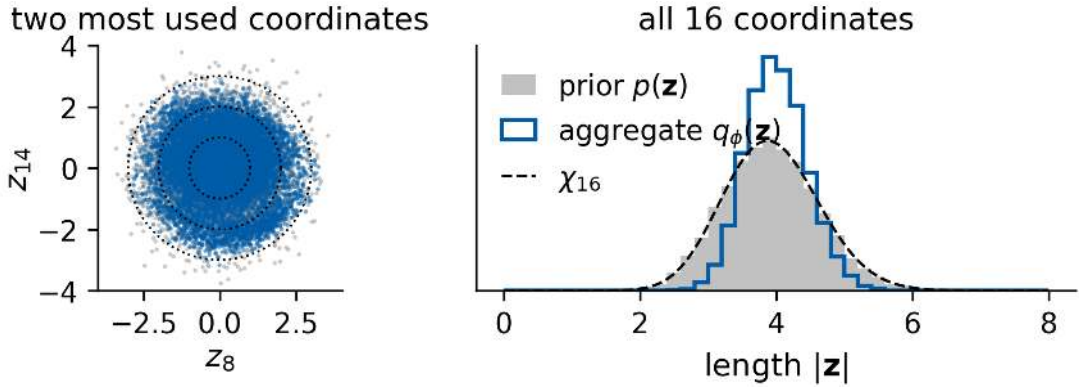


Figure 21: Aggregate posterior and prior, $d = 16$. Left, 10^4 samples of $q_\phi(\mathbf{z})$ (blue, obtained by encoding a test image and drawing a point of its cloud) and of $p(\mathbf{z})$ (grey). They are projected on the two coordinates with the largest KL term, z_8 and z_{14}, with circles of radius 1, 2, 3 (dotted). Right, histograms of the length $|\mathbf{z}|$ for both distributions and the exact chi distribution with 16 degrees of freedom, χ_{16} (dashed), which describes $|\mathbf{z}|$ under the prior.

6.4 Lines and coordinates of the latent space

Decoding the straight line $\mathbf{z}(t) = (1-t)\,\boldsymbol{\mu}_\phi(\mathbf{x}_a) + t\,\boldsymbol{\mu}_\phi(\mathbf{x}_b)$ between the centres of two test digits gives plausible digits at every $t \in [0, 1]$. A 1 bends into a 7, or a 4 closes into a 9. The pixel cross-fade $(1-t)\mathbf{x}_a + t\mathbf{x}_b$ passes through two superposed digits (Figure 22). The straight line in the latent space is a curved path through the space of images that stays near the data manifold. The decoder of the spiral behaved in the same way in one dimension.

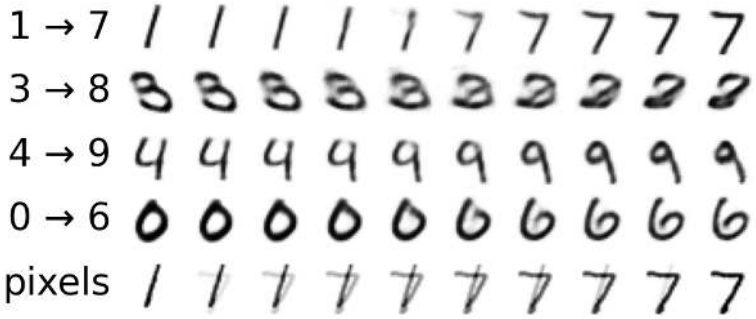


Figure 22: Interpolation, $d = 16$. Each of the first rows decodes $\mathbf{z}(t) = (1-t)\boldsymbol{\mu}_\phi(\mathbf{x}_a) + t\boldsymbol{\mu}_\phi(\mathbf{x}_b)$ for t from 0 (left) to 1 (right) between two test digits. The last row shows the pixel cross-fade $(1-t)\mathbf{x}_a + t\mathbf{x}_b$ of the first pair.

The use of a coordinate is measured by its KL term averaged over the test images,

$$\overline{\mathrm{KL}}_k := \left\langle \tfrac{1}{2}(\mu_k^2 + \sigma_k^2 - \ln \sigma_k^2 - 1) \right\rangle_{\mathbf{x}}, \tag{6.1}$$

an upper bound, in nats, on the information that coordinate k carries about the image. A coordinate with $\overline{\mathrm{KL}}_k$ above 0.1 nats is called active. One with $\overline{\mathrm{KL}}_k \approx 0$ has the prior as its cloud for every image and is ignored by the decoder, a collapsed coordinate in the sense of Figure 13. In the $d = 16$ model all sixteen coordinates are active, with $\overline{\mathrm{KL}}_k$ from 0.65 to 2.23 nats. The role of a coordinate is seen in a latent traversal, which moves one coordinate of the code of one image and keeps the others fixed (Figure 23). Each of the four most used coordinates changes several properties at once, slant together with width, thickness together with size, and often the digit. We measure this on 300 test digits. There 7 of the 16 coordinates change both the slant and the amount of ink by more than 0.3 standard deviations of these quantities over real digits.

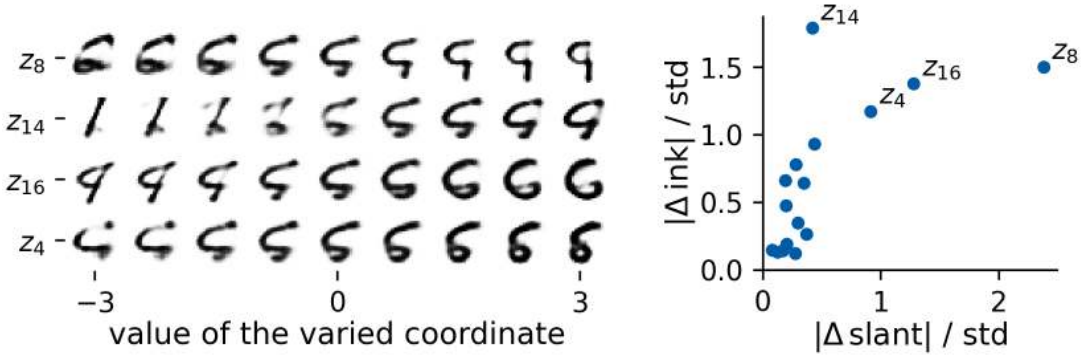


Figure 23: Latent traversals, $d = 16$. Left, the code $\boldsymbol{\mu}_\phi(\mathbf{x})$ of one test 5 with one coordinate (rows, the four with the largest $\overline{\mathrm{KL}}_k$) set to values from -3 to 3. Right, for each of the 16 coordinates (dots), the change of the slant and of the ink when the coordinate goes from -2 to 2. The slant is $-\mathrm{Cov}(u, v)/\mathrm{Var}(v)$ of the ink distribution over the pixel column u and row v, and the ink is the mean pixel value. Both changes are averaged over 300 test digits and divided by the standard deviation of the property over real test digits.

The mixing has its origin in the loss. The prior $\mathcal{N}(0, I)$ is unchanged by any rotation $\mathbf{z} \to Q\mathbf{z}$ of the latent space, with Q an orthogonal matrix, and so is the divergence of a Gaussian cloud from it. The decoder can absorb the rotation, $G'(\mathbf{z}) = G(Q^\top \mathbf{z})$. Apart from the restriction of the clouds to diagonal covariance, the ELBO therefore prefers no particular set of axes. A rotated latent space, whose every axis mixes slant, thickness and identity, scores as well as one in which they are separated. The VAE learns a good density and a smooth map, and the individual coordinates of its latent space carry no guaranteed meaning.

7 A classifier as a referee

Generated images have so far been judged by eye. A measurement needs a referee, a network trained on real data whose accuracy on real test data is known, applied unchanged to synthetic images. The referee for the digits is a small convolutional neural network (CNN). Two 3×3 convolutions with 16 and 32 channels and ReLU, each followed by 2×2 max-pooling, reduce the image from 28×28 to 7×7. A dense layer of 128 units then gives ten outputs $f_y(\mathbf{x})$. The probability of digit y is $p(y \,|\, \mathbf{x}) = e^{f_y} / \sum_{y'} e^{f_{y'}}$. The network is trained with the cross-entropy $-\frac{1}{n} \sum_i \ln p(y_i \,|\, \mathbf{x}_i)$ over labelled training digits $(\mathbf{x}_i, y_i)$, and $\max_y p(y \,|\, \mathbf{x})$ is its confidence. On the 10 000 real test digits it reaches the accuracy 0.9854.

Every test of synthetic data is first run on real data, which gives the baseline. The classifier is confident, $\max_y p(y \,|\, \mathbf{x}) > 0.9$, for 96% of the real test digits and for 92% of the VAE reconstructions with $d = 16$. It is confident for 54% of the VAE samples decoded from the prior. For the images that the autoencoder of Figure 2 decodes from a Gaussian fitted to its codes, the fraction is 37%.

The ordering is consistent with the prior holes, since latent vectors from the aggregate posterior give clearer digits than those from the prior. The autoencoder has no prior at all.

A confusion matrix makes the test of reconstructions precise. Its entry

$$C_{ab} \coloneqq \frac{\#\{\text{images of digit } a \text{ assigned to digit } b\}}{\#\{\text{images of digit } a\}}, \qquad \sum_b C_{ab} = 1, \tag{7.1}$$

is computed here with a the label of a real test digit and b the classifier label of its reconstruction from one $\mathbf{z} \sim q_\phi(\mathbf{z} \,|\, \mathbf{x})$. The mean diagonal $\frac{1}{10} \operatorname{tr} C$ is 0.985 for the real images themselves, 0.953 for the reconstructions with $d = 16$ and 0.783 with $d = 2$ (Figure 24). In two dimensions digits that look alike share a region of the plane, and the worst case is a 4 read as a 9 in 35% of the reconstructions.

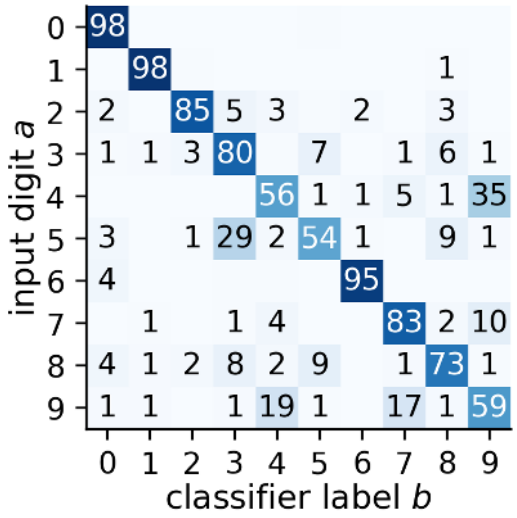


Figure 24: Confusion matrix of VAE reconstructions, $d = 2$. Row a, the digit of the real test input. Column b, the classifier label of its reconstruction. Entries C_{ab} in per cent of row a.

8 The conditional variational autoencoder

8.1 Generation on request

A VAE trained on the digits draws digits of every class in the proportions of the data, and the user cannot request a 7. The class of a generated digit is set by where the random latent vector happens to land (Figure 18). In physics this limitation is more serious. A generator of Ising configurations is useful only if it produces configurations at a requested temperature, a known, labelled factor that a simulation records for every configuration.

The object to be learned is then a conditional distribution, $p(\mathbf{x} \,|\, \mathbf{c})$, the distribution of samples with a given condition $\mathbf{c}$. For digits $\mathbf{c}$ encodes the label y, for spins the temperature, or the point (g, T) of a phase diagram. It is the conditional distribution of Section 2.6, and the Boltzmann distribution $p(\mathbf{s} \,|\, T)$ is an example. The conditional VAE (cVAE) [7] learns it with one change to the VAE, the condition given as an additional input to both networks (Figure 25).

The encoder becomes $q_\phi(\mathbf{z} \,|\, \mathbf{x}, \mathbf{c})$ with input $[\mathbf{x}, \mathbf{c}]$, where $[\cdot, \cdot]$ joins two vectors into one. The decoder becomes $p_\theta(\mathbf{x} \,|\, \mathbf{z}, \mathbf{c})$ with input $[\mathbf{z}, \mathbf{c}]$, and the model is

$$p_\theta(\mathbf{x} \,|\, \mathbf{c}) = \int p_\theta(\mathbf{x} \,|\, \mathbf{z}, \mathbf{c})\, p(\mathbf{z})\, \mathrm{d}\mathbf{z}, \tag{8.1}$$

with the same prior $\mathcal{N}(0, I)$ for every condition. For digits the condition is the one-hot vector, $c_j = \delta_{j,y}$ for the label y, ten numbers equal to zero except a one at position y. A single number y would tell the network that 3 lies between 2 and 4, an order that the digits lack. A temperature has an order and a scale and enters as a number.

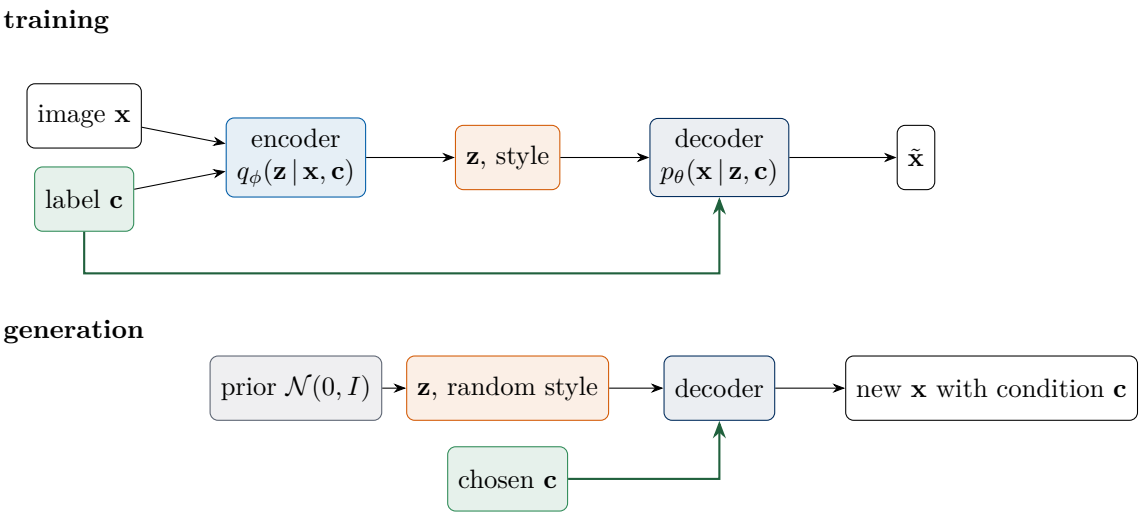


Figure 25: The conditional VAE, sketch. Top, training. The encoder receives the image and its condition, the decoder receives a latent vector drawn from the cloud and, a second time, the condition (green). Bottom, generation without the encoder. The condition is chosen, the latent vector is drawn from the prior, and the decoder produces a new sample with the chosen condition.

8.2 The conditional bound and the content of the latent vector

The derivation of Section 5 carries over with $\mathbf{c}$ held fixed in every distribution. Bayes' rule at fixed $\mathbf{c}$ and the average over $q_\phi(\mathbf{z}\,|\,\mathbf{x},\mathbf{c})$ give

$$\begin{aligned} \ln p_\theta(\mathbf{x}\,|\,\mathbf{c}) &= \mathrm{ELBO}(\mathbf{x},\mathbf{c}) + \mathrm{KL}\left(q_\phi(\mathbf{z}\,|\,\mathbf{x},\mathbf{c})\,\|\,p_\theta(\mathbf{z}\,|\,\mathbf{x},\mathbf{c})\right), \\ \mathrm{ELBO}(\mathbf{x},\mathbf{c}) &= \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x},\mathbf{c})}\left[\ln p_\theta(\mathbf{x}\,|\,\mathbf{z},\mathbf{c})\right]}_{\text{rebuild } \mathbf{x} \text{ from } \mathbf{z} \text{ and the condition}} - \underbrace{\mathrm{KL}\left(q_\phi(\mathbf{z}\,|\,\mathbf{x},\mathbf{c})\,\|\,\mathcal{N}(0,I)\right)}_{\text{price of the information in } \mathbf{z}}. \end{aligned} \tag{8.2}$$

The loss is again the BCE plus the explicit divergence (5.5), and the training data are pairs $(\mathbf{x}_i,\mathbf{c}_i)$.

The condition changes what the latent vector carries. The KL term charges every nat that $\mathbf{z}$ stores about the image, while the information in $\mathbf{c}$ reaches the decoder free of charge. A plain VAE must pay in $\mathbf{z}$ for the identity of a digit, the largest single factor of variation of an image, about $\ln 10 = 2.3$ nats for ten equally frequent digits. In a cVAE storing the digit in $\mathbf{z}$ would duplicate free information at a cost. The loss therefore favours a latent vector that holds the information missing from the label, the style of the handwriting, its slant, thickness, width and size. The cVAE thus splits a sample into a chosen part, the condition, and a random part, the latent vector.

8.3 Two recipes

Generation at a chosen condition fixes $\mathbf{c}$, draws a latent vector from the prior and decodes,

$$\mathbf{c} \text{ chosen}, \qquad \mathbf{z} \sim \mathcal{N}(0,I), \qquad \tilde{\mathbf{x}} = \mathrm{sigm}\left(G_\theta([\mathbf{z},\mathbf{c}])\right). \tag{8.3}$$

Many $\mathbf{z}$ with one $\mathbf{c}$ give many ways of writing one digit, and one $\mathbf{z}$ with many $\mathbf{c}$ gives one hand writing every digit. Style transfer starts from a real image $\mathbf{x}$ with condition $\mathbf{c}$ and encodes it with its own condition to the centre $\boldsymbol{\mu}_\phi(\mathbf{x},\mathbf{c})$ of its cloud. This centre is then decoded with another condition $\mathbf{c}'$. The result is the digit $\mathbf{c}'$ written in the hand of $\mathbf{x}$, provided that the latent vector holds the style and leaves the identity to the condition.

The cVAE for the digits has the dense layers of the VAE. The ten numbers of the one-hot condition are appended to the 784 pixels at the encoder input and to the latent vector at the decoder input (Figure 26). We train a cVAE with $d = 8$ on the training digits and their labels. Decoded with the ten conditions, one $\mathbf{z}$ keeps its slant, stroke width and size while the digit changes. The style transfer of real test digits carries their handwriting to the other digits (Figure 27).

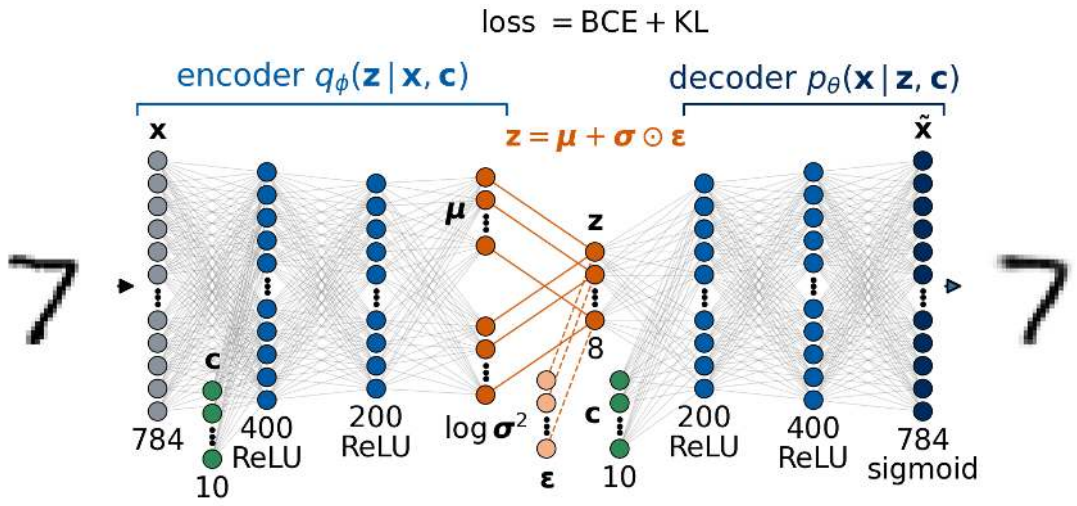


Figure 26: The cVAE for MNIST with $d = 8$. The one-hot condition $\mathbf{c}$ (green) enters the first layer of the encoder together with the image and the first layer of the decoder together with $\mathbf{z}$. The rest is drawn as in Figure 15.

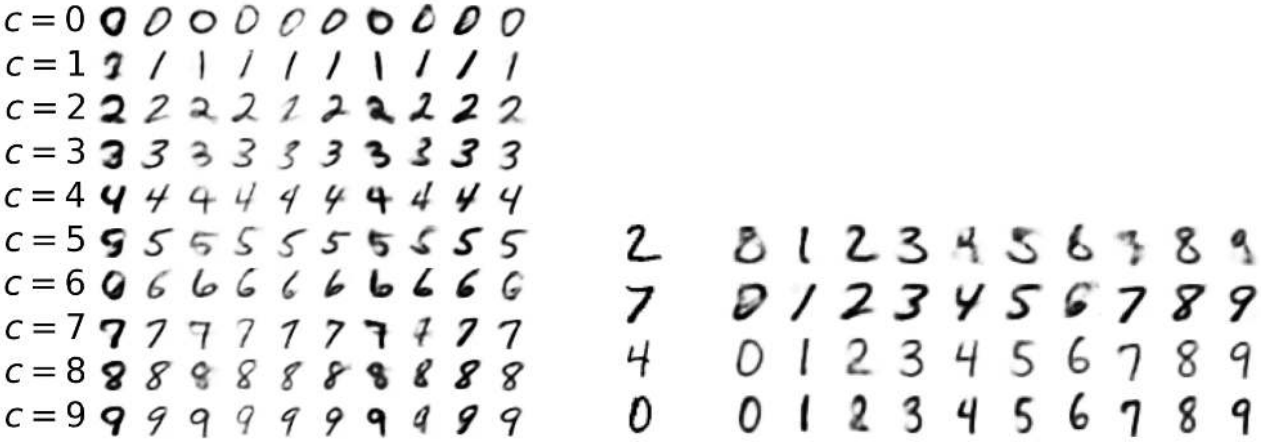


Figure 27: The cVAE with $d = 8$. Left, decoder means $\mathrm{sigm}(G_\theta(\mathbf{z}, \mathbf{c}))$. Row, the condition $\mathbf{c}$ for the digits 0 to 9. Column, one latent vector $\mathbf{z} \sim \mathcal{N}(0, I)$, the same in every row. Right, style transfer. Each row starts with a real test digit $\mathbf{x}$, followed by $\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}, \mathbf{c}_y)$ decoded with the conditions of the digits 0 to 9.

8.4 The digit outside the latent space

If $\mathbf{z}$ carries only the style, the digit should be invisible in the latent plane, and a cVAE with $d = 2$ confirms this. In the VAE the centres of different digits occupy different regions (Figure 17), and in the cVAE the centres of all digits share one disc (Figure 28). We quantify this by assigning each test digit to the digit whose mean latent centre is nearest. This rule is correct for a fraction 0.73 of the test digits in the VAE and 0.18 in the cVAE, against 0.1 by chance. The latent vector of the cVAE has given up almost all information about the identity. Decoding the grid with the condition of the digit 7 gives a 7 in every cell, and moving across the grid changes only how it is written.

8.5 Conditions between the training values

The condition is an input like any other, and the decoder also accepts values never seen in training. Along the straight line $\mathbf{c}(t) = (1 - t)\,\mathbf{e}_4 + t\,\mathbf{e}_9$ between the one-hot vectors of 4 and 9, with $\mathbf{e}_k$ the k-th unit vector, the decoded images change continuously. They keep the style of each $\mathbf{z}$. The referee switches from 4 to 9 near the middle of the line (Figure 29). The decoder interpolates between the conditions it was trained on. No digit lies between 4 and 9, so these intermediate images have no meaning, whereas a temperature or a point of a phase diagram has intermediate values. The decoder

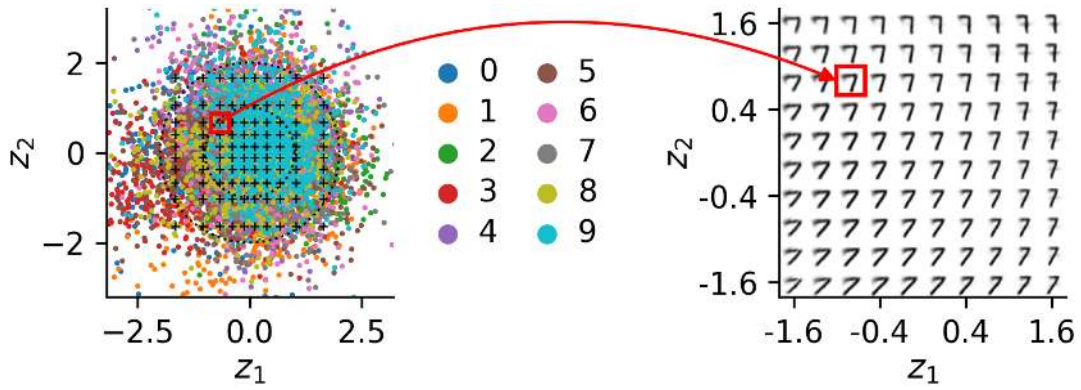


Figure 28: The cVAE with $d = 2$, layout as in Figure 18. Left, the centres $\boldsymbol{\mu}_\phi(\mathbf{x}, \mathbf{c})$ of the test digits, each encoded with its own label and coloured by it, and the quantile grid. Right, the decoder mean at every grid point with the condition fixed to the digit 7.

can then generate configurations at a temperature that was never simulated, and the result can be checked against a simulation at that temperature.

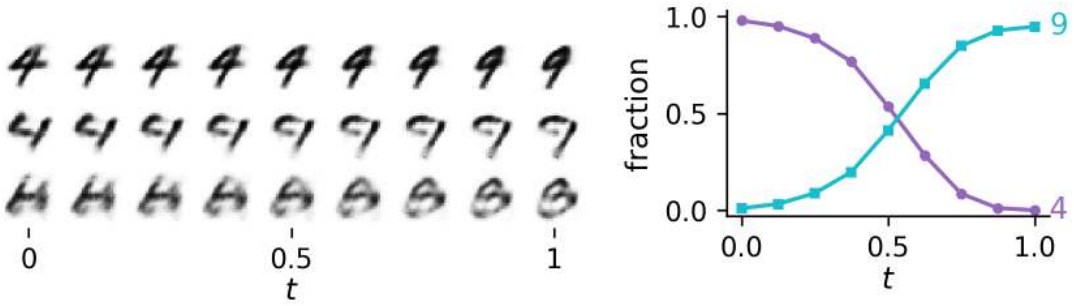


Figure 29: Conditions between two digits, $d = 8$. Left, three fixed latent vectors (rows) decoded with $\mathbf{c}(t) = (1-t)\mathbf{e}_4 + t\mathbf{e}_9$ for t from 0 to 1. Right, the fraction of the images decoded from 1000 latent vectors $\mathbf{z} \sim \mathcal{N}(0, I)$ that the referee assigns to 4 (purple) and to 9 (cyan), against t.

8.6 Requested digits judged by the referee

We measure with the referee of Section 7 how often the cVAE draws the requested digit. For each requested digit a, 1000 images are generated with $\mathbf{z} \sim \mathcal{N}(0, I)$, and C_{ab} is now the fraction of them assigned to digit b. The diagonal has the mean 0.950, a little below the values of the referee on real test digits (Figure 30). The largest confusion is a requested 2 read as a 0 in 3.2% of the samples.

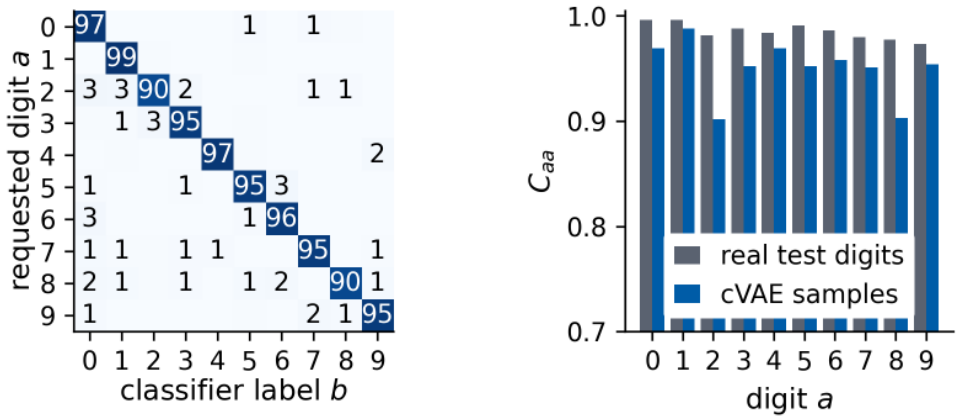


Figure 30: Requested digits. Left, confusion matrix C_{ab}, in per cent, of 1000 cVAE samples per requested digit a (rows) against the referee label b (columns). Right, the diagonal C_{aa} of the cVAE samples (blue) next to the diagonal of the referee on real test digits (grey), for every digit a.

In physics a generator is conditioned on the temperature T of an Ising model or on the point (g, T) of a phase diagram. The referees are then networks trained on real Monte Carlo configurations. A spin model adds a stricter referee, because its distribution is known, $p(\mathbf{s}\,|\,T) = e^{-H(\mathbf{s})/T}/Z$, and every observable of statistical mechanics can be computed on the synthetic configurations.

9 Spin models and their observables

9.1 Two Hamiltonians

Spins $s_i = \pm 1$ sit on the $N = L^2$ sites of an $L \times L$ square lattice with periodic boundaries. In the Ising model each spin interacts with its four nearest neighbours. In the J_1–J_2 model it also interacts with its four diagonal neighbours,

$$H_{\text{Ising}}(\mathbf{s}) = -J \sum_{\langle ij \rangle} s_i s_j, \qquad H_{J_1 J_2}(\mathbf{s}) = -J_1 \sum_{\langle ij \rangle} s_i s_j + J_2 \sum_{\langle\langle ij \rangle\rangle} s_i s_j, \qquad g \coloneqq \frac{J_2}{J_1}, \tag{9.1}$$

where $\langle ij \rangle$ runs over the $2N$ nearest-neighbour bonds and $\langle\langle ij \rangle\rangle$ over the $2N$ diagonal bonds, each counted once. The couplings $J = J_1 = 1$ and the Boltzmann constant $k_B = 1$ set the units (Figure 31). Both energies are unchanged by the global flip $\mathbf{s} \to -\mathbf{s}$. In thermal equilibrium at temperature T a configuration occurs with the Boltzmann probability

$$p(\mathbf{s}\,|\,T) = \frac{e^{-H(\mathbf{s})/T}}{Z}, \qquad Z \coloneqq \sum_{\mathbf{s}} e^{-H(\mathbf{s})/T}. \tag{9.2}$$

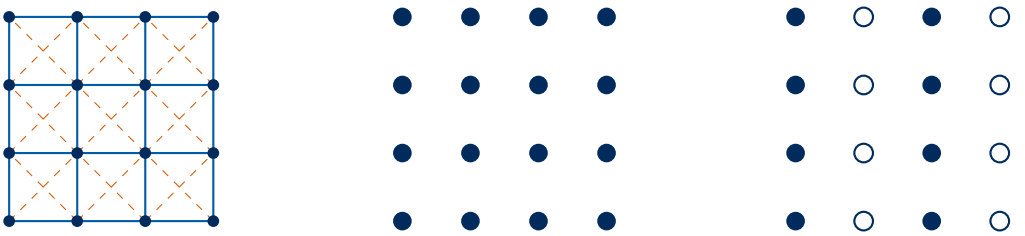


Figure 31: The J_1–J_2 model, sketch. Left, ferromagnetic nearest-neighbour bonds J_1 (blue) and antiferromagnetic diagonal bonds J_2 (orange) of a 4×4 piece of the lattice. Middle and right, the two ground states, the ferromagnet (all spins up, filled dots) and vertical stripes (columns of alternating sign), with their energies per spin e. The Ising model is the case $g = 0$.

The two-dimensional Ising model has a continuous transition at Onsager's temperature $T_c = 2/\ln(1+\sqrt{2}) = 2.269$ [8]. Below it most spins point the same way and the magnetisation per spin $m \coloneqq \frac{1}{N}\sum_i s_i$ is close to ± 1. Above it the spins form small random domains and m is close to zero. In the J_1–J_2 model the diagonal bonds compete with the nearest-neighbour bonds. Counting bonds gives the energy per spin $e = -2 + 2g$ of the ferromagnet and $e = -2g$ of the stripe pattern, columns or rows of alternating sign. The ground state is ferromagnetic for $g < \frac{1}{2}$ and striped for $g > \frac{1}{2}$. The transition temperature falls to zero at $g = \frac{1}{2}$, where the two orders compete [9, 10, 11]. The ferromagnetic transition is continuous and belongs to the Ising universality class, while the stripe transition close to $g = \frac{1}{2}$ shows Ashkin–Teller criticality and pseudo-first-order behaviour [11]. Stripes have $m = 0$ and are measured by the Fourier components of the spins at the wave vectors $(\pi, 0)$ and $(0, \pi)$,

$$m_x \coloneqq \frac{1}{N}\sum_{x,y} (-1)^x s_{x,y}, \qquad m_y \coloneqq \frac{1}{N}\sum_{x,y} (-1)^y s_{x,y}, \qquad m_{\text{str}} \coloneqq \sqrt{m_x^2 + m_y^2}, \tag{9.3}$$

where (x, y) labels the sites. Vertical stripes have $m_x = \pm 1$, horizontal stripes $m_y = \pm 1$, and $m_{\text{str}} = 1$ for ideal stripes of either orientation.

9.2 Observables and their estimators

Every observable is an average over the Boltzmann distribution, $\langle O\rangle \coloneqq \frac{1}{Z}\sum_{\mathbf{s}} O(\mathbf{s})\, e^{-H(\mathbf{s})/T}$, and the partition function also generates the thermodynamics. The energy per spin is $\langle e\rangle = \langle H\rangle/N$. The specific heat per spin is $\frac{1}{NT^2}\,\partial^2 \ln Z/\partial(1/T)^2 = (\langle H^2\rangle - \langle H\rangle^2)/(NT^2)$, the variance of the energy divided by NT^2. The sum over 2^N configurations is out of reach for $N = 900$. Monte Carlo therefore draws n configurations $\mathbf{s}^{(1)}, \ldots, \mathbf{s}^{(n)}$ with probability $p(\mathbf{s}\,|\,T)$ and estimates every average by a sample mean, $\overline{O} \coloneqq \frac{1}{n}\sum_k O(\mathbf{s}^{(k)})$. A generator is judged by the same sample means over its own configurations. With $e^{(k)}$ and $m^{(k)}$ the energy and magnetisation per spin of configuration k, the estimators are

$$\begin{gathered}\overline{e} = \frac{1}{n}\sum_k e^{(k)}, \qquad \overline{|m|} = \frac{1}{n}\sum_k |m^{(k)}|, \qquad c = \frac{N}{T^2}\bigl(\overline{e^2} - \overline{e}^{\,2}\bigr), \qquad \chi = \frac{N}{T}\bigl(\overline{m^2} - \overline{|m|}^{\,2}\bigr), \\ U_4 = 1 - \frac{\overline{m^4}}{3\,\overline{m^2}^{\,2}}, \qquad G(r) = \frac{1}{2nN}\sum_k \sum_{\mathbf{r}} \sum_{\mu=1,2} s^{(k)}_{\mathbf{r}}\, s^{(k)}_{\mathbf{r}+r\hat{\mathbf{e}}_\mu}.\end{gathered} \tag{9.4}$$

The mean $|m|$ is used because on a finite lattice without a field both signs of m are equally likely and $\langle m\rangle = 0$. The specific heat c, an italic c distinct from the bold condition vector $\mathbf{c}$ of the cVAE, measures the fluctuations of the energy. The susceptibility χ measures those of the order parameter, and both peak at a transition. The Binder cumulant U_4 measures the shape of the distribution $P(m)$ of the magnetisation, $2/3$ for two narrow peaks at $\pm|m|$ and zero for a Gaussian [12]. The correlation function $G(r)$ is the average product of two spins r lattice spacings apart along a lattice direction $\hat{\mathbf{e}}_\mu$, with $\mathbf{r} = (x, y)$ the position of a site. Its nearest-neighbour value $G(1) = -\langle e\rangle/2$ is minus half the Ising energy per spin. The histograms of $e^{(k)}$ and $m^{(k)}$ estimate the full distributions $P(e)$ and $P(m)$, and the J_1–J_2 energy includes the diagonal bonds.

9.3 Monte Carlo data and a baseline

The Metropolis algorithm [13] proposes to flip one spin and accepts the flip with probability $\min(1, e^{-\Delta H/T})$, where ΔH is the change of energy. Its stationary distribution is the Boltzmann distribution (9.2), with all the correlations between neighbouring spins that H implies. The chain must be equilibrated and its successive configurations are correlated, which costs computing time. The Swendsen–Wang algorithm [14] flips whole clusters of spins and decorrelates faster near T_c. The Ising data are 256 Swendsen–Wang configurations of the 30×30 lattice at each of 37 temperatures between 1.5 and 3.5, denser near T_c. At every temperature 128 configurations train the networks, and the remaining ones form two independent held-out sets, A and B, of 64 configurations each. The J_1–J_2 data are Metropolis configurations of the 32×32 lattice from chains cooled from high temperature. They cover a grid of 81 values of $g \in [0, 1]$ and 79 temperatures $T \in [0.1, 4]$, 100 per grid point, split into 60 for training and 20 each for sets A and B.

A difference between a synthetic and a real estimate is meaningful only relative to the difference between two real sets. For an observable $O(T)$ we therefore express the comparison in noise units,

$$R_O \coloneqq \frac{\mathrm{mean}_T\ \bigl|O_{\mathrm{syn}}(T) - O_A(T)\bigr|}{\mathrm{mean}_T\ \bigl|O_B(T) - O_A(T)\bigr|}, \tag{9.5}$$

where the means run over the temperatures. A value $R_O \approx 1$ means that the synthetic set is as close to set A as an independent Monte Carlo set of the same size, and $R_O \gg 1$ signals a real discrepancy. For the Ising energy the denominator is 0.011 per spin. Averages can agree while distributions differ. We therefore also compare distributions by the Wasserstein-1 distance of two samples x_k and y_k of equal size n, $\mathcal{W}_1 \coloneqq \frac{1}{n}\sum_k |x_{(k)} - y_{(k)}|$, with $x_{(k)}$ the k-th smallest value. This mean absolute difference of the sorted values is the average distance by which probability has to be moved to turn one histogram into the other.

10 A conditional VAE for spins

10.1 A product of Bernoulli factors

The generator is the cVAE with a spin configuration in place of an image and the temperature in place of the digit, and a configuration is stored as bits $x_i = (s_i + 1)/2$. The condition is $\mathbf{c} = T$ for the Ising model and $\mathbf{c} = (g, T)$ for the J_1–J_2 model, each rescaled to numbers of order one, and the latent dimension is $d = 16$. The decoder returns one logit $\ell_i(\mathbf{z}, \mathbf{c})$ per site and treats the spins as independent Bernoulli variables given $\mathbf{z}$,

$$p_\theta(\mathbf{s} \,|\, \mathbf{z}, \mathbf{c}) = \prod_{i=1}^{N} \operatorname{sigm}(\ell_i)^{x_i} \big(1 - \operatorname{sigm}(\ell_i)\big)^{1-x_i}, \tag{10.1}$$

so that $\operatorname{sigm}(\ell_i)$ is the probability that spin i points up. The reconstruction term of the ELBO (8.2) is minus the BCE of the bits. It is zero if every spin is predicted with certainty and $N \ln 2 = 624$ nats for $N = 900$ if every spin is a fair coin.

Generating at temperature T uses the prior and the decoder only. We draw a latent vector, compute the logits and draw every spin independently, $s_i = +1$ with probability $\operatorname{sigm}(\ell_i)$. Alternatively we keep the most likely configuration, $\tilde{s}_i = \operatorname{sign} \ell_i$. At fixed $\mathbf{z}$ the decoder is a product over sites, the structure of a mean-field ansatz of statistical mechanics. The mean spin is $\langle s_i \rangle_{\mathbf{z}} = (+1)\operatorname{sigm}(\ell_i) + (-1)(1 - \operatorname{sigm}(\ell_i)) = 2\operatorname{sigm}(\ell_i) - 1 = \tanh(\ell_i/2)$. Two different spins are uncorrelated given $\mathbf{z}$, and every correlation between spins comes from the variation of $\mathbf{z}$,

$$\langle s_i s_j \rangle_{\mathbf{z}} = \tanh \tfrac{\ell_i}{2} \, \tanh \tfrac{\ell_j}{2} \quad (i \neq j), \qquad \operatorname{Cov}(s_i, s_j) = \operatorname{Cov}_{\mathbf{z}} \Big(\tanh \tfrac{\ell_i}{2}, \tanh \tfrac{\ell_j}{2} \Big). \tag{10.2}$$

All correlations must pass through 16 latent numbers. The magnetisation, one number per configuration, fits easily. The energy is a sum of nearest-neighbour products. Every spin whose probability lies away from 0 or 1 is drawn independently of its neighbours and breaks its bonds at random. The product structure itself, whatever the details of the training, limits what the synthetic configurations can carry.

10.2 Three variants

Three variants are compared, and each after the first adds one ingredient from physics. The plain cVAE has the prior $\mathcal{N}(0, I)$ and is trained on the raw data. The variant cVAE+Z2, named after the group $\mathbb{Z}_2$ of the global spin flip, uses this symmetry of H. Below T_c the data contain two mirror-image families with $m \approx \pm m_0(T)$, with $m_0(T)$ the spontaneous magnetisation. The plain model must place them separately in its latent space, and prior samples between them decode to mixtures of the two ordered states. In cVAE+Z2 every training configuration is replaced by its representative with $m \geq 0$, and every generated configuration is flipped with probability $\frac{1}{2}$. The relation $p(\mathbf{s}) = \frac{1}{2}[p_+(\mathbf{s}) + p_+(-\mathbf{s})]$ then holds for every configuration, with p_+ the distribution of the configurations before the flip. The variant cVAE+Z2+$p(\mathbf{z}|T)$ adds a conditional prior,

$$p_\psi(\mathbf{z} \,|\, \mathbf{c}) = \mathcal{N}\big(\boldsymbol{\mu}_\psi(\mathbf{c}), \operatorname{diag} \boldsymbol{\sigma}^2_\psi(\mathbf{c})\big), \tag{10.3}$$

computed by a small prior network with parameters ψ [7]. For digits the label and the style are nearly independent, since any digit can be written with any slant, and the latent vector of a cVAE lost the identity (Section 8). For spins the analogue of the style, the magnetisation and the size of the domains of one configuration, has a distribution that depends strongly on the temperature. A configuration thus reveals its own temperature, and the encoded configurations therefore of different temperatures occupy different regions of the latent space even when T is given to the networks. Generation from the fixed prior $\mathcal{N}(0, I)$ at temperature T draws from all these regions and produces a mixture of temperatures, while $p_\psi(\mathbf{z} \,|\, T)$ draws from the region of T. The KL term of the ELBO then compares the encoder cloud with p_ψ in place of $\mathcal{N}(0, I)$, and the conditional ELBO (8.2) becomes

$$\mathrm{ELBO}_\psi(\mathbf{x}, \mathbf{c}) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x},\mathbf{c})} \big[\ln p_\theta(\mathbf{x} \,|\, \mathbf{z}, \mathbf{c}) \big] - \mathrm{KL}\left(q_\phi(\mathbf{z} \,|\, \mathbf{x}, \mathbf{c}) \,\|\, p_\psi(\mathbf{z} \,|\, \mathbf{c}) \right), \tag{10.4}$$

maximised with respect to ϕ, θ and ψ together. Figure 32 draws this network.

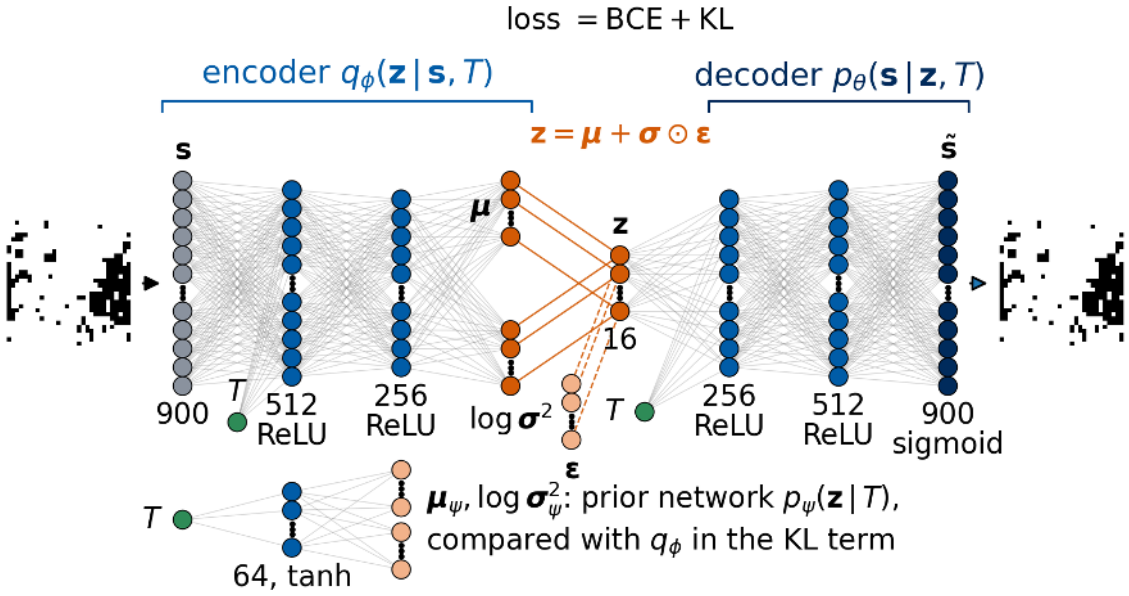


Figure 32: The spin cVAE with a conditional prior, $L = 30$. The encoder receives the 900 spins $\mathbf{s}$ and the temperature T (green), and its heads give $\boldsymbol{\mu}$ and $\ln \boldsymbol{\sigma}^2$. The sampling node forms $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}$ ($d = 16$), and the decoder receives $\mathbf{z}$ and T and returns 900 probabilities $\tilde{\mathbf{s}}$ that the spins point up. The prior network (bottom row) maps T to $\boldsymbol{\mu}_\psi$ and $\ln \boldsymbol{\sigma}^2_\psi$ of $p_\psi(\mathbf{z}\,|\,T)$, which enters the KL term in training and supplies $\mathbf{z}$ in generation. Layer widths and activations are written below the columns.

10.3 Training health

By the usual criteria the three variants train equally well (Figure 33). After 40 epochs their held-out $-$ ELBO is 391, 387 and 388 nats per configuration, against $N \ln 2 = 624$ nats for independent fair coins. All 16 latent coordinates are active, and the decoder recovers about 81 % of the spins from $\mathbf{z} = \boldsymbol{\mu}$ with $\tilde{s}_i = \operatorname{sign} \ell_i$. The training loss continues to fall while the held-out loss levels off, a sign that longer training would overfit the 128 configurations per temperature. The ELBO leaves open which variant generates the right physics.

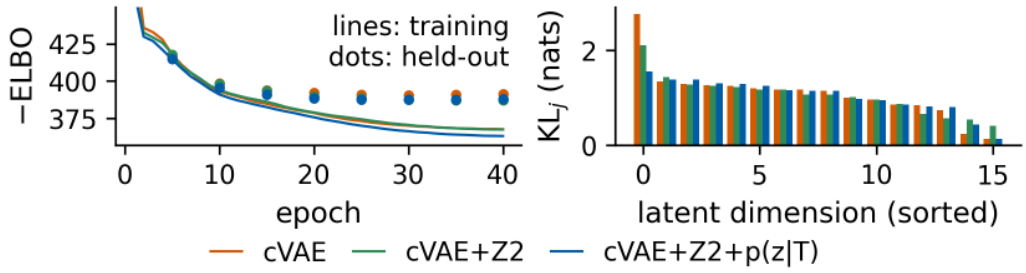


Figure 33: Training of the three Ising cVAEs. Left, $-$ ELBO per configuration in nats against the epoch, on the training data (lines) and on the held-out sets A and B (dots). Right, the KL term of every latent coordinate j on held-out data, sorted in decreasing order.

10.4 Drawing one configuration

Figure 34 follows one synthetic configuration of the best variant at $T = 1.8$ and near T_c. The decoder output is a smooth map of probabilities, close to 0 or 1 inside a domain and intermediate at domain walls and in disordered regions. A Bernoulli sample turns every intermediate probability into an independent coin flip, which produces isolated flipped spins inside the domains. Two samples of the

same $\mathbf{z}$ share the large-scale pattern and differ spin by spin. The most likely configuration removes the noise and keeps only smooth domains.

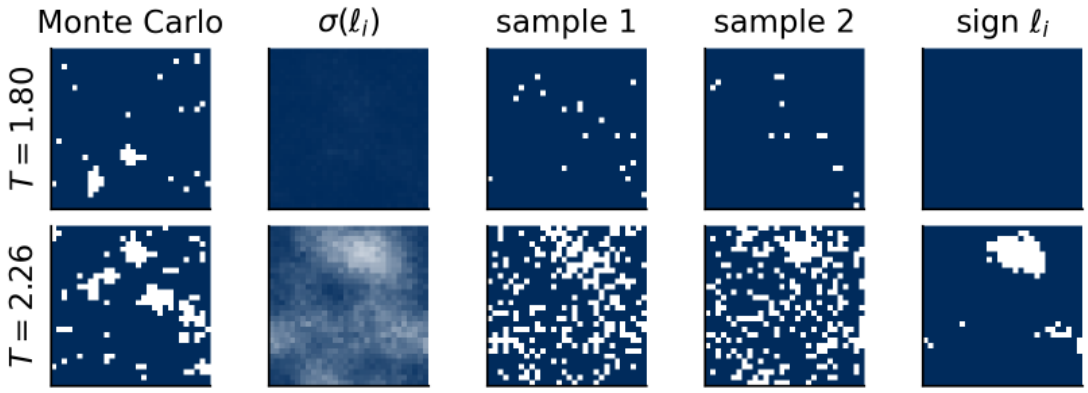


Figure 34: One synthetic configuration of cVAE+Z2+$p(\mathbf{z}|T)$, $L = 30$, at $T = 1.80$ (top) and $T = 2.26$ (bottom). From left, a Monte Carlo configuration of set A for comparison, the probabilities $\mathrm{sigm}(\ell_i)$ that the spins point up for one $\mathbf{z} \sim p_\psi(\mathbf{z}\,|\,T)$ (white 0, navy 1). Then follow two independent Bernoulli samples of these probabilities and the most likely configuration $\mathrm{sign}\,\ell_i$. Navy marks $s_i = +1$.

The numbers confirm the product structure (10.2). The nearest-neighbour correlation $G(1)$ measured on the Bernoulli samples is 0.894 at $T = 1.8$. The value predicted from the probabilities alone for independent spins, the average over $\mathbf{z}$ of $\tanh(\ell_{\mathbf{r}}/2)\tanh(\ell_{\mathbf{r}+\hat{\mathbf{e}}_\mu}/2)$, is 0.895. Near T_c the two are 0.520 and 0.521. The synthetic spins are independent given $\mathbf{z}$, as the decoder prescribes. Monte Carlo gives $G(1) = 0.930$ and 0.717 at the same temperatures, so the generator has fewer satisfied bonds, and hence a higher energy, than the Boltzmann distribution.

11 Physical properties of synthetic configurations

11.1 Snapshots

Each generator produces 64 configurations at each of the 37 temperatures, as many as each held-out set. Side by side (Figure 35) the plain cVAE and cVAE+Z2 barely follow the requested temperature, and their low-temperature configurations are speckled mixtures. With the conditional prior the configurations are ordered below T_c and disordered above, with isolated flipped spins inside the domains. The most likely configurations have no isolated spins and large smooth domains even at $T = 3.2$.

11.2 Tests by networks trained on Monte Carlo data

The first family of tests applies networks trained only on real configurations to the synthetic ones. Two small convolutional networks with periodic padding and global average pooling are trained on the Monte Carlo training set. A temperature regressor outputs an estimate $\tilde{T}$ and is trained with the loss $(\tilde{T} - T)^2$ at all temperatures. A phase classifier is trained only far from the transition, on $T \le 1.8$ (ordered) and $T \ge 2.8$ (disordered). Near the transition it becomes uncertain, and the minimum of its confidence $\max(p_0, p_1)$ against T marks the transition it sees. On the real held-out sets the regressor has a root-mean-square error (RMSE) of 0.12, and the confidence minimum lies at $T = 2.28$, next to T_c.

On synthetic data (Figure 36) the plain cVAE is read as far too hot at low temperature, with a mean $\tilde{T} = 2.77$ at $T = 1.5$. Its confidence minimum lies at $T = 1.70$. With the conditional prior the low-temperature end is restored, $\tilde{T} = 1.69$ at $T = 1.5$, and the confidence minimum moves to $T = 2.20$. The RMSE, 0.62, is still five times that of real data, because above T_c the synthetic configurations are read as too hot. The most likely configurations are read as far too cold. The networks thus recognise

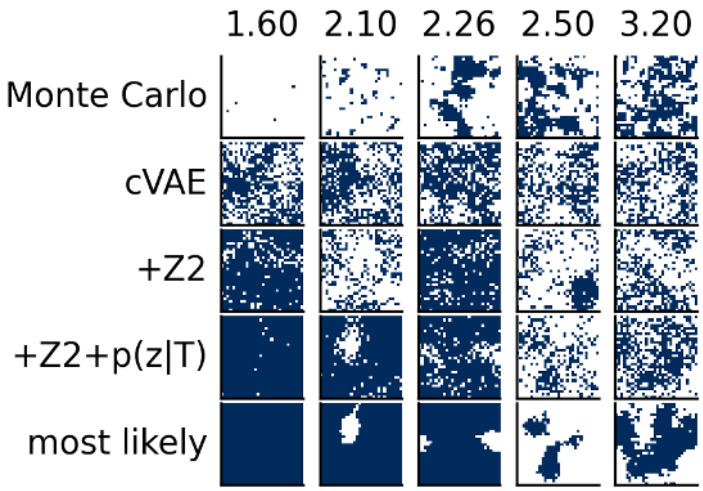


Figure 35: Monte Carlo and synthetic Ising configurations, $L = 30$, at the temperatures written above the columns. Rows, from top, Monte Carlo, cVAE, cVAE+Z2, cVAE+Z2+$p(\mathbf{z}|T)$ with Bernoulli sampling, and the most likely configuration sign ℓ_i for the same latent vectors. Navy marks $s_i = +1$.

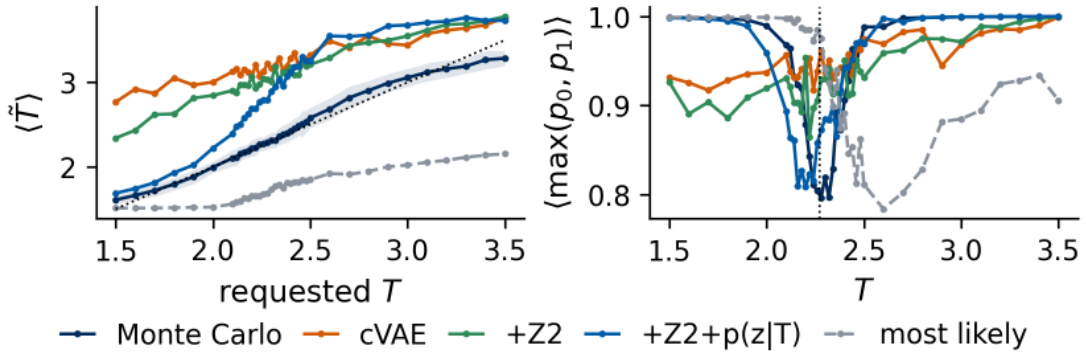


Figure 36: Networks trained on Monte Carlo data applied to Monte Carlo set A and to synthetic Ising configurations. Left, the mean regressor output $\langle \tilde{T} \rangle$ against the requested temperature, with the line $\tilde{T} = T$ (dotted) and the spread between sets A and B (grey band). Right, the mean classifier confidence $\langle \max(p_0, p_1) \rangle$ against T, with T_c dotted.

the phase of the synthetic configurations and, with the conditional prior, roughly their temperature below T_c.

11.3 Tests by the Hamiltonian

The second family of tests evaluates the estimators (9.4) on the synthetic configurations (Figure 37 and Table 7). The order parameter comes out right with the conditional prior, $R_{|m|} = 2.8$, $R_\chi = 2.0$ and $R_{\mathcal{W}_1(m)} = 1.3$, within a few noise units of Monte Carlo. Without it the plain cVAE is 13.3 noise units away in $\langle |m| \rangle$, too small below T_c and too large above. Below T_c we count the configurations with $|m|$ less than half the Monte Carlo value, mixtures of the two ordered states. Their fraction is 1 % in the Monte Carlo data, 45 % for the plain cVAE, 27 % with the imposed symmetry and 1 % with the conditional prior as well.

The energy fails in every variant. Even with the conditional prior $R_e = 34.8$, and near T_c the synthetic energy is $\langle e \rangle = -1.04$ per spin against -1.43 for Monte Carlo at $T = 2.26$. The specific heat is two to fifteen times the Monte Carlo value at six temperatures between 1.50 and 3.50, for example 12.54 against 1.94 at $T = 2.26$. The distribution $P(e)$ is far too broad (Figure 38). Isolated flipped spins inside the domains break four bonds each and raise the energy, and their number varies from configuration to configuration, which inflates its variance. The most likely configurations remove that noise and over-smooth instead. Their energy is too low, -1.86 at $T = 2.26$, their domains are large and compact, and $G(r)$ is too large at all distances.

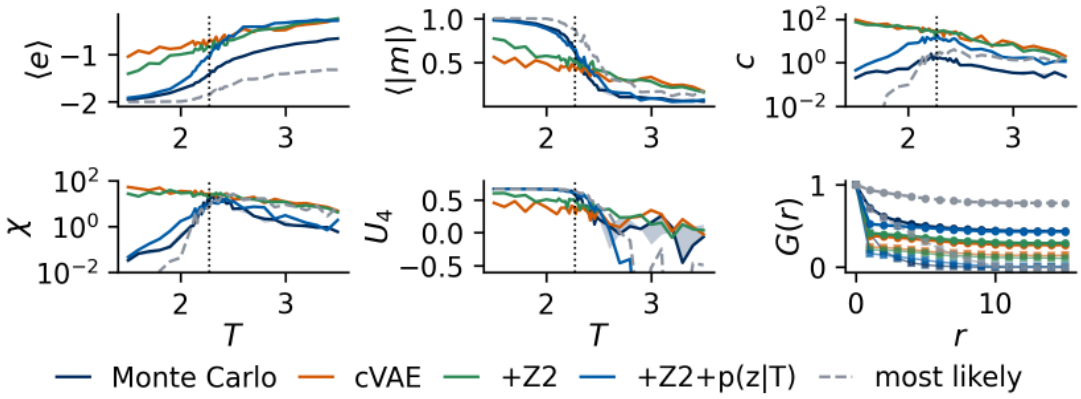


Figure 37: Hamiltonian observables of Monte Carlo set A and of the synthetic Ising configurations, $L = 30$. Top, from left, the energy per spin $\langle e \rangle$, the magnetisation $\langle |m| \rangle$ and the specific heat c (logarithmic axis) against T. Bottom, the susceptibility χ (logarithmic axis) and the Binder cumulant U_4 against T, and the correlation function $G(r)$ against the distance r at $T = 2.26$ (circles) and $T = 3.0$ (squares). The grey band is the spread between Monte Carlo sets A and B, and the dotted line marks T_c.

Table 7: Noise ratios R_O of equation (9.5) for the Ising model, $L = 30$, over the 37 temperatures. $R_O \approx 1$ means as close to Monte Carlo set A as the independent set B. The last column is $\mathcal{W}_1$ of the distribution of $|m|$, labelled $\mathcal{W}_1(m)$, and $G(L/2)$ is the correlation at half the lattice size.

	$\langle e \rangle$	c	$\mathcal{W}_1(e)$	$G(L/2)$	U_4	$\langle \lvert m \rvert \rangle$	χ	$\mathcal{W}_1(m)$
cVAE	60.0	155	38.1	15.0	2.1	13.3	11.8	3.3
cVAE+Z2	50.0	146	31.9	12.4	1.7	10.8	9.7	2.8
cVAE+Z2+$p(\mathbf{z}\|T)$	34.8	39.0	21.6	3.0	4.2	2.8	2.0	1.3
most likely	39.6	6.6	24.5	11.1	3.2	9.6	3.7	2.8

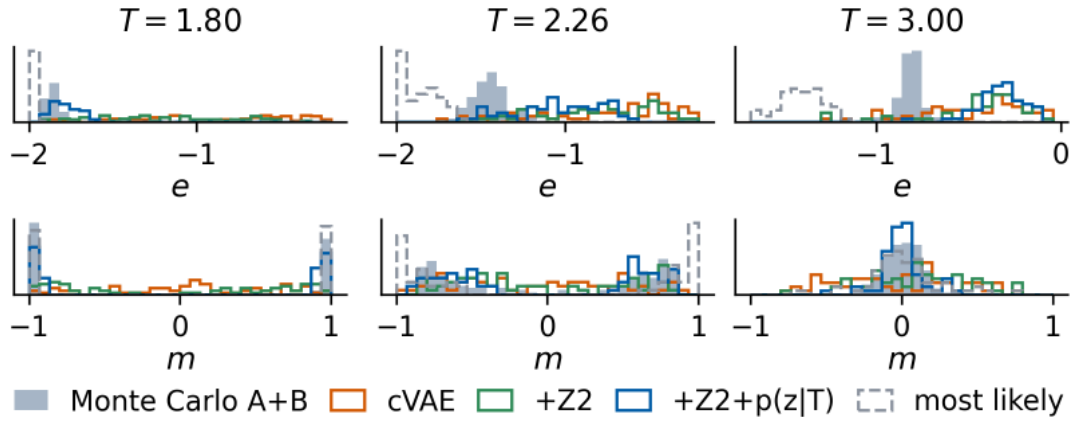


Figure 38: Distributions of the energy per spin e (top) and of the magnetisation m (bottom) at $T = 1.80$, 2.26 and 3.00. The grey histograms combine Monte Carlo sets A and B, and the lines show the synthetic sets. Histograms are normalised to the same area and share the vertical scale within a panel.

11.4 Two-sample test and memorisation

A classifier two-sample test (C2ST) [15] checks the distribution as a whole, beyond selected observables. If two sets come from one distribution, no classifier separates them better than chance. At three temperatures a fresh small convolutional network is trained to separate real from synthetic configurations, with balanced classes and a quarter of the data held out. We report its held-out accuracy. Real against real gives 0.50, 0.53 and 0.50 at $T = 1.8$, 2.26 and 3.0 (Figure 39). The best variant is separated with accuracies 0.73, 0.95 and 0.98, the other variants and the most likely configurations with 0.94 or more. Every synthetic set is distinguishable from Monte Carlo, most clearly near and above T_c.

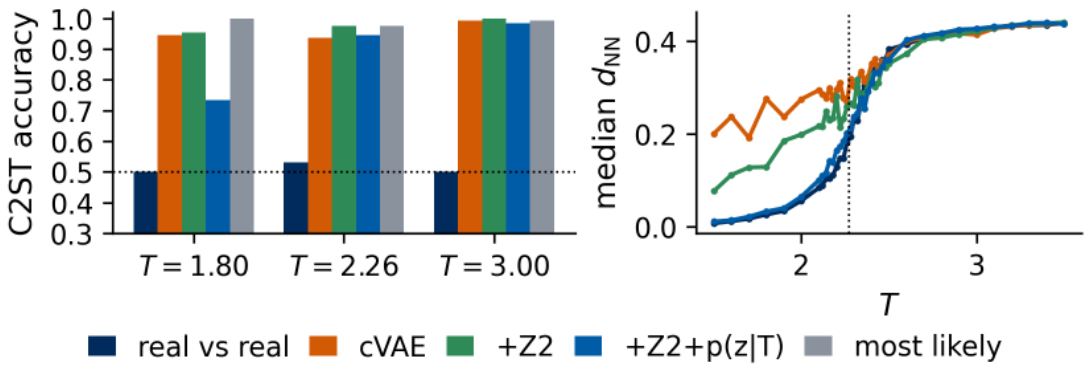


Figure 39: Left, held-out accuracy of the classifier two-sample test at three temperatures, 0.5 (dotted) meaning indistinguishable from Monte Carlo. Right, the median normalised Hamming distance d_{NN} of a configuration to the nearest training configuration at the same temperature, minimised over the global flip. It is plotted against T for held-out Monte Carlo configurations and synthetic ones, with T_c dotted.

A generator that copies its training set would pass every physics test. To exclude copying, for every configuration we compute the normalised Hamming distance to the nearest training configuration at the same temperature, minimised over the global flip, $d_{\mathrm{NN}}(\mathbf{s}) = \min_{\mathbf{t}} (1 - |\mathbf{s} \cdot \mathbf{t}|/N)/2$. Here $\mathbf{t}$ runs over these training configurations. Held-out Monte Carlo configurations, new draws from the Boltzmann distribution, give the reference. Their median d_{NN} is 0.03, 0.18 and 0.43 at $T = 1.8$, 2.26 and 3.0, and the best variant gives 0.03, 0.20 and 0.43. Its configurations lie as far from the training set as new Monte Carlo draws, which excludes copying.

11.5 Interpolating the J_1–J_2 phase diagram

The J_1–J_2 cVAE uses both ingredients of the best Ising model, the imposed spin-flip symmetry and a conditional prior $p_\psi(\mathbf{z} \,|\, g, T)$. It is trained only at every second g and every second T, a quarter of the grid called seen. It generates at every grid point, so three quarters of the conditions were never seen. Its held-out $-\mathrm{ELBO}$ is 476 nats at seen and 482 at unseen points, close to the training value 476. A three-class convolutional network trained on real configurations labels each configuration as ferromagnet, stripes or paramagnet, and the majority label at every grid point maps the phases.

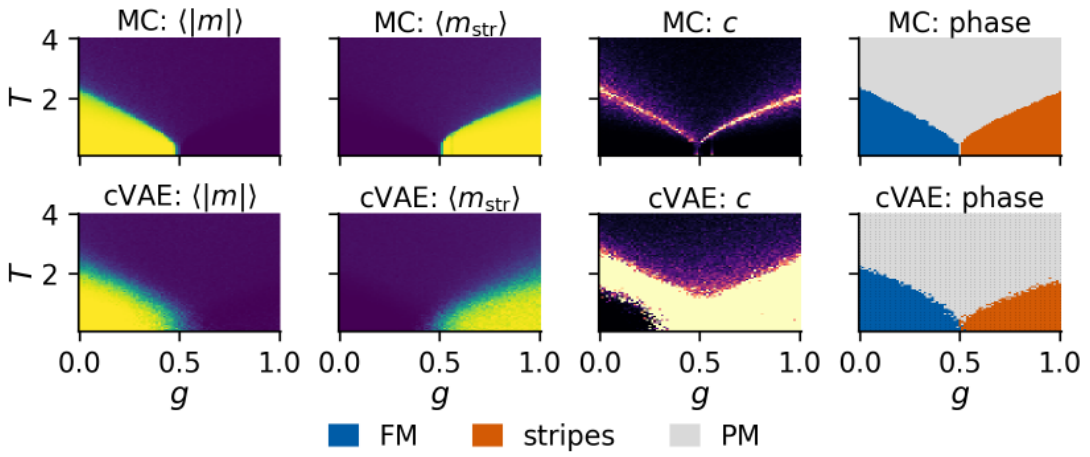


Figure 40: The J_1–J_2 phase diagram in the (g, T) plane from Monte Carlo set A (top) and from the cVAE (bottom). From left, $\langle|m|\rangle$ and $\langle m_{\mathrm{str}}\rangle$ (colour from 0, dark, to 1, yellow) are followed by the specific heat c (colour from 0 to 2.2, the 99th percentile of Monte Carlo). The last panel shows the majority phase assigned by the classifier trained on real data, ferromagnet (FM), stripes and paramagnet (PM). Dots in the bottom right panel mark the seen grid points.

The classifier assigns the synthetic configurations the same majority phase as Monte Carlo set A at 95.3 % of the seen and 95.4 % of the unseen grid points. Set B agrees at 99.7 % (Figure 40).

The generator interpolates the phase diagram as well at unseen as at seen conditions, with smeared boundaries and differences near $g = \frac{1}{2}$. The order parameters agree within a few noise units, $R_{|m|} = 4.8$ and $R_{m_{\mathrm{str}}} = 5.9$ at unseen points. Both stripe orientations are generated with nearly equal frequency, a fraction 0.48 alternating along x against 0.50 for Monte Carlo. The energy fails as for the Ising model, with $R_e = 63.9$. In the paramagnet the synthetic energy is close to zero, the value for independent spins, while the Monte Carlo energy carries the short-range correlations of the paramagnet (Figure 41). The specific heat deviates by $R_c = 1517$ noise units at unseen points. The most likely configurations are ordered almost everywhere and agree with the phase of set A at only 35.4 % of the grid points.

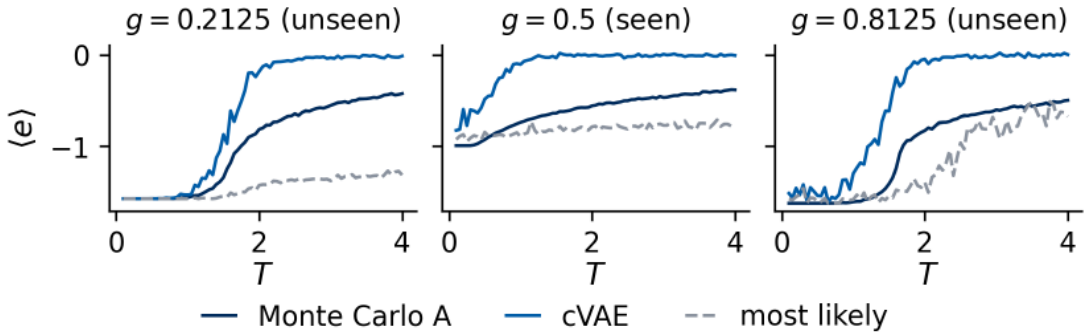


Figure 41: Energy per spin $\langle e \rangle$ of the J_1–J_2 model against T at three values of g, two unseen and one seen in training. The curves show Monte Carlo set A, the cVAE with Bernoulli sampling and the most likely configurations.

The phase, the order parameter and its distribution survive generation, also at conditions never seen, and so do the features that networks trained on real data respond to. The energy, its fluctuations and the short-range correlations are lost.

12 Further tests of synthetic configurations

12.1 Symmetry and correlation length

The Boltzmann distribution satisfies $P(m) = P(-m)$ at every temperature, which the two variants with the imposed symmetry satisfy by construction and the plain cVAE has to learn. Below T_c the fraction of synthetic configurations with $m > 0$ is 0.501 for the plain cVAE, 0.506 for cVAE+Z2 and 0.477 with the conditional prior. The two Monte Carlo sets give 0.515 and 0.500. We express the deviations from $\frac{1}{2}$ in binomial standard errors at each of the 15 temperatures below T_c and combine them in a chi-squared test. Its p-value, the probability of a deviation at least this large under exact symmetry, is 0.07 for the plain cVAE and at least 0.50 for the others. No variant deviates significantly from the symmetry at the 5% level (Figure 42).

Above T_c correlations decay exponentially, $G(r) \simeq Ae^{-r/\xi}$, with the correlation length ξ and an amplitude A, fitted here on $r = 1, \ldots, 4$. Monte Carlo gives $\xi = 4.01$, 2.48, 1.45 and 1.05 at $T = 2.4$, 2.6, 3.0 and 3.5, and the best generator gives 19.2, 9.62, 6.54 and 6.69. The amplitude averaged over $T \geq 2.4$ is 0.25 against 0.77. The synthetic correlations are too weak at short distances and too long-ranged, as the product structure implies. For $r \geq 1$ the synthetic $G(r)$ is the average over $\mathbf{z}$ of $m_{\mathbf{r}} m_{\mathbf{r}+r\hat{\mathbf{e}}_\mu}$, a product of mean spins from a smooth map that varies on the scale of the decoder's domains. Independent coin flips remove the strong correlations between neighbours. The spin-flip symmetry, a property of $P(m)$ as a whole, is preserved, and the correlation length, a property of neighbouring spins, is lost.

12.2 Monte Carlo started from generated configurations

A generated configuration carries the right order parameter and the wrong bonds. Local Monte Carlo updates repair bonds within a few sweeps, one attempted flip per spin. The slow modes of a

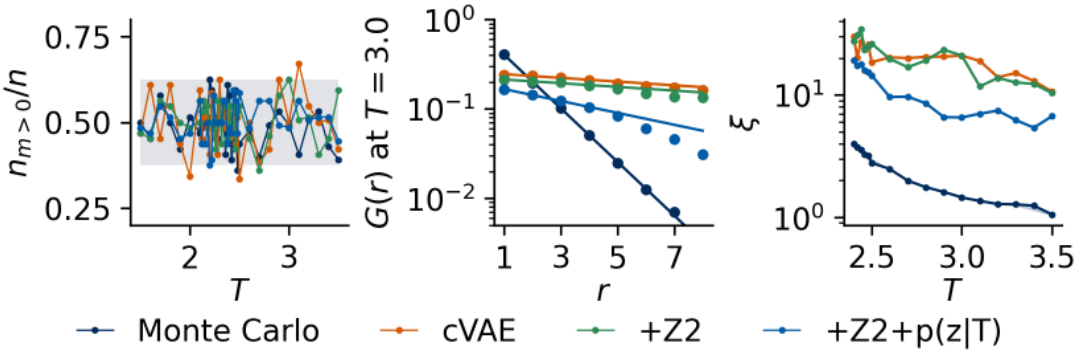


Figure 42: Spin-flip symmetry and correlation length, Ising model, $L = 30$. Left, the fraction $n_{m>0}/n$ of the 64 configurations at each T with positive magnetisation, with $\frac{1}{2} \pm 2$ binomial standard errors (grey band). Middle, $G(r)$ at $T = 3.0$ on a logarithmic axis (points) and the fits $Ae^{-r/\xi}$ on $r = 1, \ldots, 4$ (lines). Right, the fitted correlation length ξ against $T \geq 2.4$, logarithmic axis.

Metropolis chain are the large-scale ones, the growth of domains and the relaxation of $|m|$, which near T_c slow down with growing lattice size. Generated configurations can therefore start Metropolis chains. At $T = 1.8$, 2.26 and 3.0, 64 chains are run from each of three starts, generated configurations of the best variant, independent random spins (hot) and all spins up (cold). An observable counts as repaired when the chain average agrees with the mean of 128 held-out Monte Carlo configurations within three standard errors. This agreement must hold at a checkpoint and at every later one. The checkpoints are spaced by factors of two in the number of sweeps.

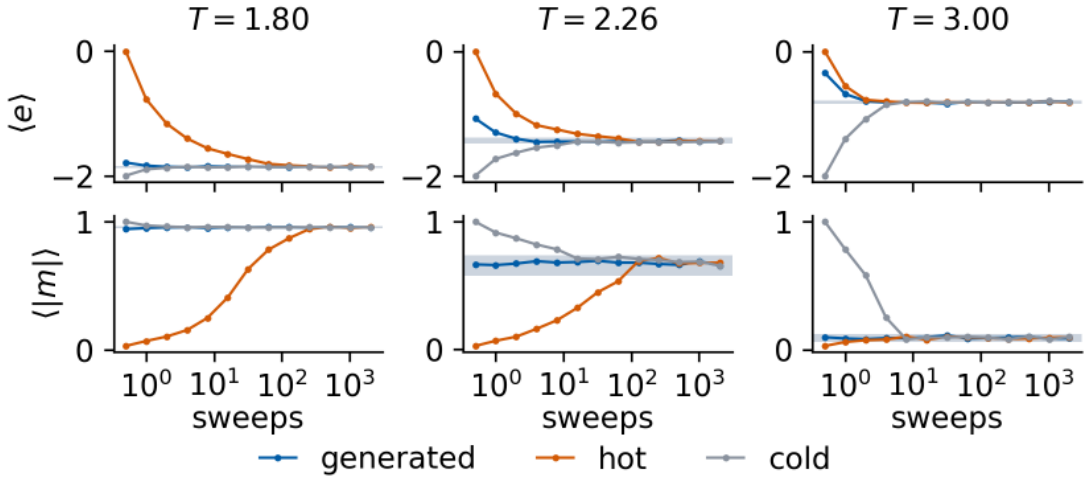


Figure 43: Metropolis chains started from generated (blue), hot (orange) and cold (grey) configurations, $L = 30$. Chain averages of $\langle e \rangle$ (top) and $\langle |m| \rangle$ (bottom) against the number of sweeps on a logarithmic axis (sweep 0 drawn at 0.5), at $T = 1.80$, 2.26 and 3.00. The grey band is the Monte Carlo mean with its tolerance.

Generated starts reach the Monte Carlo energy within 2 sweeps at all three temperatures (Figure 43), and their $\langle |m| \rangle$ is right from the start near and above T_c. Near T_c the generated start needs 2 sweeps for e, $|m|$ and $G(r)$ together, the cold start 16 and the hot start 128. Below T_c the generated start matches the cold start. The hot start needs 256 sweeps for the energy and 512 for $|m|$, because random spins first form domains that must coarsen. Above T_c hot and generated starts behave alike. The generator removes the slow part of the relaxation, which it captures, and leaves the bonds to the Markov chain. On this 30×30 lattice the gain over the best simple start is a factor of 8 near T_c and none away from it. The repaired configurations are Metropolis samples of $e^{-H/T}/Z$. Their correctness comes from the Markov chain, and the generator contributes only the starting point.

12.3 Finite-size behaviour

On a finite lattice the transition is rounded. The susceptibility peaks at a temperature $T_c(L)$ that approaches T_c as L grows, and its height grows as $L^{7/4}$ for the two-dimensional Ising universality class. The Binder cumulants of different sizes cross close to T_c, because at T_c the shape of $P(m)$ is the same for all large L [12]. The best variant is trained at $L = 20$, 30 and 50 with the same protocol and compared with 128 held-out Monte Carlo configurations per temperature (Figure 44).

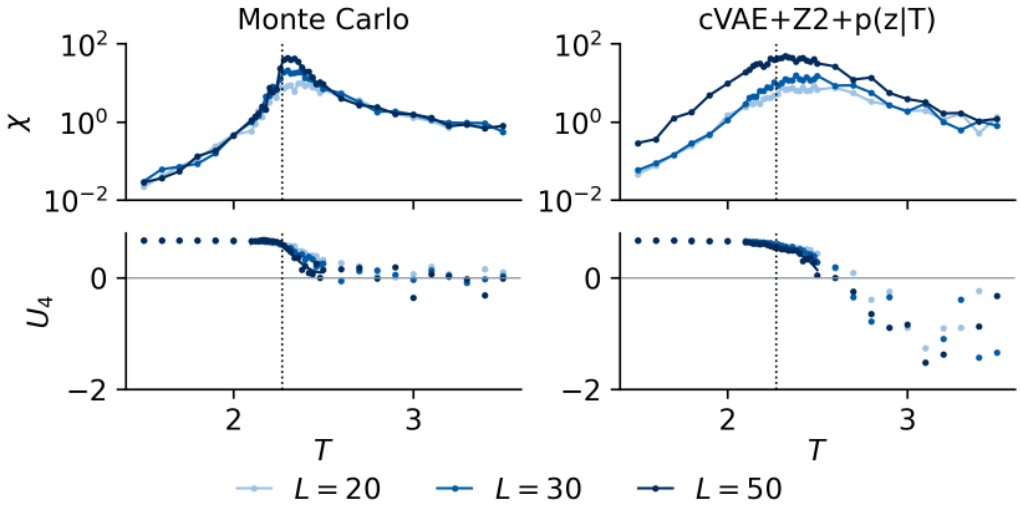


Figure 44: Finite-size behaviour of Monte Carlo (left) and cVAE+Z2+$p(\mathbf{z}|T)$ (right) for $L = 20$, 30 and 50. Top, the susceptibility χ against T on a logarithmic axis. Bottom, the Binder cumulant U_4 against T, with cubic fits near T_c (lines) used for the crossings. Dotted, T_c.

The growth of the susceptibility peak is preserved approximately. The Monte Carlo heights are 10.2, 21.1 and 43.6 for $L = 20$, 30, 50, with a slope of $\ln \chi_{\max}$ against $\ln L$ of 1.58. The synthetic heights 8.5, 15.7 and 48.8 give 1.92, with broad peaks. The Binder crossing is lost. The Monte Carlo curves cross at 2.231 ± 0.020 ($L = 20$ and 30) and 2.252 ± 0.007 (30 and 50), close to T_c. The synthetic curves of 20 and 30 cross at 2.347 ± 0.061. Those of 30 and 50 have no crossing near T_c, in the full data and in 65% of the bootstrap resamples. Above T_c the synthetic U_4 becomes strongly negative, $\langle m^4 \rangle > 3\langle m^2 \rangle^2$, which means a $P(m)$ with heavier tails than a Gaussian. Each synthetic configuration draws its own latent vector, and some vectors produce partially ordered maps. The peak of χ is a second moment of m and survives, while the crossing needs the tails of $P(m)$, which the generator misses.

12.4 The Ising lattice gauge theory

A model without a local order parameter tests the rule further. In the Ising lattice gauge theory [16] variables $\sigma_\mu(x) = \pm 1$, unrelated to the encoder widths σ_k, sit on the links of a periodic cubic lattice, here with $L = 6$ and $3L^3 = 648$ links. The product of the four links around an elementary square p, a plaquette, is $U_p = \pm 1$. The energy is $H = -\sum_p U_p$ with $3L^3$ plaquettes, at the coupling $\beta_g := 1/T$. Choosing a sign $\tau(x) = \pm 1$ at every site and replacing $\sigma_\mu(x) \to \tau(x)\sigma_\mu(x)\tau(x + \hat{\mathbf{e}}_\mu)$ leaves every plaquette unchanged, a local gauge symmetry (Figure 45). Elitzur's theorem states that a local symmetry cannot be broken spontaneously [17]. Every quantity that changes under gauge transformations therefore has zero average, in particular $\langle \sigma_\mu(x) \rangle = 0$ on every link at every coupling. Physical statements use gauge-invariant quantities only. These are the plaquette average $\langle U_p \rangle$, which is minus the energy per plaquette, and the specific heat c from its fluctuations. A further set are the Wilson loops $W(R, S)$, the averages of the product of links around $R \times S$ rectangles. The model has a continuous confinement–deconfinement transition at $\beta_g = 0.7614$. For smaller couplings the loops decay with their area, $W(R, S) \sim e^{-\sigma_s RS}$ with a string tension $\sigma_s > 0$, and for larger couplings with their perimeter. The Creutz ratio $\chi_C(2, 2) := -\ln[W(2, 2)W(1, 1)/W(2, 1)^2]$ removes the perimeter

term and estimates σ_s [18].

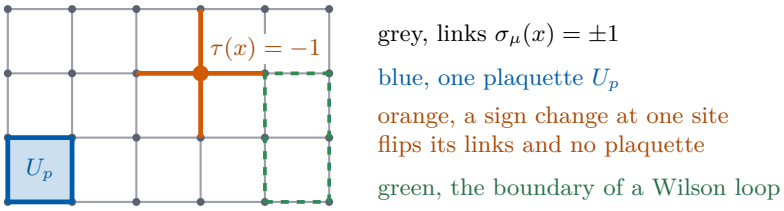


Figure 45: The Ising lattice gauge theory, one plane of the cubic lattice, sketch. Variables live on the links (grey). A plaquette U_p is the product of the four links around a square (blue). A gauge transformation at one site flips the links that touch it (orange) and leaves every plaquette unchanged. A Wilson loop is the product of the links along a closed path (green, dashed).

The Monte Carlo configurations were stored after an independent random gauge transformation each. Every link is therefore $+1$ or -1 with probability $\frac{1}{2}$ at every coupling, and the physics sits in products of four or more links. A cVAE with a conditional prior trained on these raw links learns nothing. Its $-$ ELBO of 449.1 nats equals $N \ln 2 = 449.2$ nats for $N = 648$, the cost of independent fair coins, so the decoder outputs probability $\frac{1}{2}$ for every link. Its gauge-invariant observables are those of random links, with $|\langle U_p \rangle| \leq 0.002$ at every tabulated coupling, and its link average agrees with Elitzur's theorem only trivially. The second variant is trained on gauge-fixed links. A maximal tree is a set of $L^3 - 1$ links that touches every site and contains no closed loop. A gauge transformation can set all its links to $+1$, every gauge orbit contains a single such configuration, and all gauge-invariant quantities are unchanged. Gauge fixing plays the role that the spin-flip representative with $m \geq 0$ plays for the Ising model.

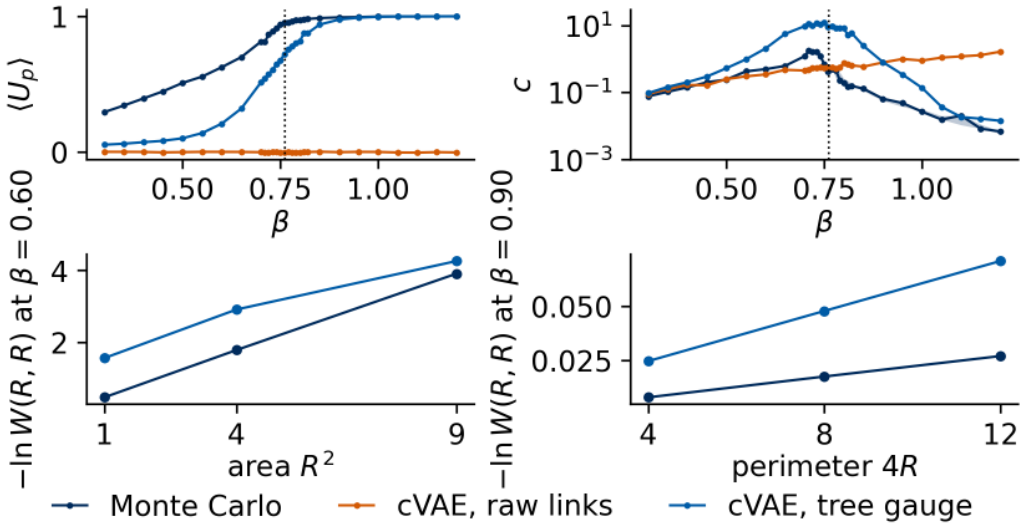


Figure 46: Three-dimensional Ising lattice gauge theory, $L = 6$, Monte Carlo and cVAEs trained on raw links and on tree-gauge links. Top, the plaquette average $\langle U_p \rangle$ and the specific heat c (logarithmic axis) against the coupling β_g, labelled β on the axes, with the transition dotted. Bottom, $-\ln W(R,R)$ for $R = 1, 2, 3$ against the area R^2 at $\beta_g = 0.60$ (confined) and against the perimeter $4R$ at $\beta_g = 0.90$ (deconfined).

After gauge fixing the generator learns (Figure 46). Deep in the deconfined phase it matches the plaquette average, 0.996 against 0.997 at $\beta_g = 1.00$, and the perimeter law of the Wilson loops. Around and below the transition the energy fails. The plaquette average is 0.575 against 0.865 at $\beta_g = 0.72$, and the specific heat peaks at about 9.732, against 1.615 for Monte Carlo. The noise ratios of $\langle U_p \rangle$ and c lie between 42.6 and 74.1 on the two sides of the transition. In the confined phase the Monte Carlo loops follow the area law, while the synthetic $-\ln W(R,R)$ grows linearly in R, a perimeter law. The Creutz ratio is 0.222 against 0.406 at $\beta_g = 0.60$. The generator underestimates the string tension and blurs the transition. In the tree gauge a free link is a product of many plaquettes. A decoder that draws links independently cannot represent the short-range correlations between

plaquettes of the confined phase. The link average of the tree-gauge samples, 0.752, reflects the gauge choice and has no physical meaning, so only gauge-invariant quantities enter the comparison.

12.5 Long-range interactions

The rule locates the energy failure in the connected correlations between interacting spins. If every spin interacts with many others, its local field is an average over many spins and fluctuates little. The energy is then close to its value for independent spins with equal mean spins, the regime in which mean-field theory works. The failure should therefore shrink as the interaction becomes longer-ranged. The long-range Ising model on a periodic $L \times L$ lattice,

$$H = -\frac{1}{2}\sum_{i \neq j} J_{ij} s_i s_j, \qquad J_{ij} = \frac{J_0}{r_{ij}^{\alpha}}, \tag{12.1}$$

tests this prediction, with r_{ij} the shortest distance on the torus and J_0 chosen so that $\sum_j J_{ij} = 4$ for every α, as for nearest neighbours. Renormalisation-group theory gives mean-field critical exponents in two dimensions for $\alpha < 3$ [19] and the short-range Ising exponents for $\alpha > 3.75$ [20]. Metropolis data for $L = 16$ at 13 temperatures and $\alpha = 2.5$, 3.5, 6 and ∞, the last being nearest neighbours only, train the best variant (Figure 47).

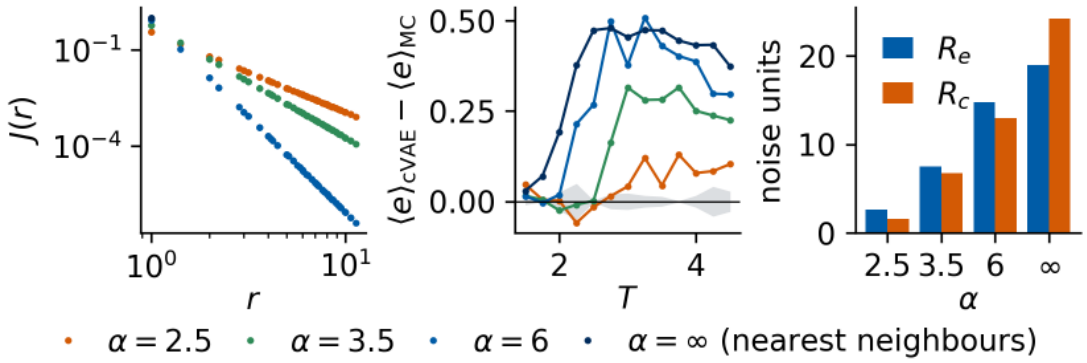


Figure 47: Long-range Ising model, $L = 16$. Left, the couplings $J(r)$ of one site against the distance r, logarithmic axes. Middle, the energy difference $\langle e\rangle_{\mathrm{cVAE}} - \langle e\rangle_{\mathrm{MC}}$ against T for each α, with the spread between Monte Carlo sets A and B for nearest neighbours (grey band). Right, the noise ratios R_e and R_c of the energy and the specific heat against α.

The energy failure shrinks monotonically as the range grows. Averaged over the temperatures the synthetic energy deviates from Monte Carlo by 0.362 per spin for nearest neighbours, 0.286 for $\alpha = 6$, 0.164 for $\alpha = 3.5$ and 0.057 for $\alpha = 2.5$. The real-versus-real noise is about 0.021, so R_e falls from 18.9 through 14.8 and 7.5 to 2.7. The specific heat follows, $R_c = 24.2$, 13.0, 6.8 and 1.6. The order parameter and the susceptibility agree for every α, with $R \leq 2.2$. In the mean-field regime the generator comes within a few noise units of every Hamiltonian observable tested. These results come from one lattice size, one training seed per α and 60 held-out configurations per temperature.

12.6 Uses and limits of synthetic configurations

Properties of the configuration as a whole survive generation (Table 8), also at conditions never seen. Properties fixed locally by the Hamiltonian are lost, together with the tails of $P(m)$, and the loss shrinks only where the interactions are long-ranged and mean-field theory applies. Networks trained on real data accept synthetic configurations with the right phase, while the two-sample test and the Hamiltonian reject them. Acceptance by a network is thus a weaker test than correct thermodynamics. Synthetic configurations answer questions about the phase and give useful starting points for Monte Carlo. Their distribution differs from the Boltzmann distribution $e^{-H/T}/Z$, so thermodynamic averages still require Monte Carlo sampling.

Table 8: What the spin cVAE preserves and loses, with the best variant and the key comparison. Noise ratios R_O are in units of the difference between two Monte Carlo sets.

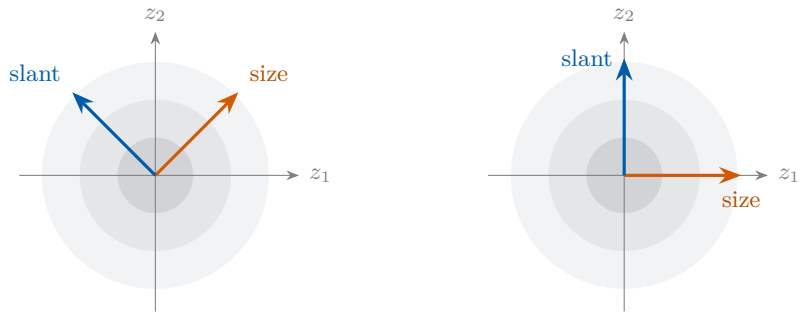
preserved	lost
$\langle\lvert m\rvert\rangle$, χ, $P(m)$, $R_{\lvert m\rvert} = 2.8$, $R_\chi = 2.0$ (Ising)	$\langle e\rangle$, c, $P(e)$, $R_e = 34.8$, $R_c = 39.0$ (Ising)
phase at unseen (g, T), 95.4 % of grid points (J_1–J_2)	energy at unseen (g, T), $R_e = 63.9$
spin-flip symmetry, p-values ≥ 0.07	correlation length, $\xi = 6.54$ against 1.45 at $T = 3.0$
growth of $\chi_{\max}$ with L	Binder crossing of $L = 30$ and 50
deconfined plaquettes and perimeter law (tree gauge)	confined plaquettes and string tension
energy at long range, $R_e = 2.7$ for $\alpha = 2.5$	energy at short range, $R_e = 18.9$ for nearest neighbours

13 Disentangled representations and the β-VAE

13.1 One coordinate per factor

The cVAE controls a factor that is known and labelled. Often the factors are unknown, and the task is to find what varies in the data. The process that produced a data set has generative factors, the few independent numbers from which a sample arises. A rendered sprite has a shape, a scale, an orientation and a position. A face has a pose, a hairstyle and an illumination, and a spin configuration in equilibrium has a magnetisation or a stripe order. A latent representation is called disentangled when each factor is carried by its own coordinate. Moving one z_k then changes one property of the decoded sample and changing one factor changes one z_k. Such a representation can be read and controlled. For a physical system it can identify the collective variables without being given them. We look for a coordinate that follows the magnetisation of Ising configurations and for one coordinate per order parameter of the J_1–J_2 model.

Figure 48 shows the difference on a sketch with two factors, the size and the slant of a digit. In both panels the latent vectors of the data fill the same round Gaussian. On the left the two factors change along diagonal directions, so moving z_1 alone changes size and slant together. On the right each factor has its own axis. The traversals of Figure 23 show that a plain VAE trained on digits is of the left kind, as the rotation symmetry at the end of Section 6 implies. The two panels share the prior and the distribution of latent vectors, and with a decoder that absorbs the rotation their likelihoods are equal. The ELBO cannot prefer the right panel, so the loss has to change.



entangled, every axis mixes size and slant

disentangled, one factor per axis

Figure 48: Two latent spaces of the same data, sketch. The grey disc is the distribution of the latent vectors, the same round Gaussian in both panels. Arrows show the directions in which the decoded image changes its size (orange) and its slant (blue). Left, the factors lie along diagonals, and moving z_1 alone changes both. Right, each factor has its own coordinate.

13.2 The cost of information

The β-VAE [21] changes one number of the loss, multiplying the KL term by a weight $\beta \geq 0$,

$$\mathcal{L}_\beta(\mathbf{x}) \coloneqq \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[-\ln p_\theta(\mathbf{x}\,|\,\mathbf{z})\big] + \beta\ \mathrm{KL}\left(q_\phi(\mathbf{z}\,|\,\mathbf{x})\,\|\,p(\mathbf{z})\right), \tag{13.1}$$

which equals $-\,\mathrm{ELBO}$ for $\beta = 1$. From here on β denotes this weight, and temperatures are written as T. For $\beta > 1$ every nat of information in the code costs more. The encoder keeps only what improves the reconstruction most, switches off coordinates that carry little and returns blurred reconstructions. For $\beta < 1$ information is cheap, and the code drifts away from the prior while the reconstructions sharpen. The original proposal motivates β as a Lagrange multiplier. The best reconstruction under a budget $\mathcal{R}_0$ for the averaged KL term has the Lagrangian $\mathbb{E}_q[\ln p_\theta(\mathbf{x}|\mathbf{z})] - \beta(\mathrm{KL} - \mathcal{R}_0)$, whose maximum is the minimum of $\mathcal{L}_\beta$. A smaller budget corresponds to a larger β.

Averaged over the data, the two terms have names from information theory [22]. They are the distortion $\mathcal{D} \coloneqq \mathbb{E}_{\mathbf{x}}\mathbb{E}_{q_\phi}[-\ln p_\theta(\mathbf{x}\,|\,\mathbf{z})]$ and the rate $\mathcal{R} \coloneqq \mathbb{E}_{\mathbf{x}}\,\mathrm{KL}(q_\phi(\mathbf{z}\,|\,\mathbf{x})\|p(\mathbf{z}))$, both in nats per sample, so that $\mathcal{L}_\beta = \mathcal{D} + \beta\mathcal{R}$ and $-\,\mathrm{ELBO} = \mathcal{D} + \mathcal{R}$. Every trained model is a point in the $(\mathcal{R}, \mathcal{D})$ plane, and the points reachable by an architecture lie above a decreasing curve $\mathcal{D}^*(\mathcal{R})$. For a decreasing convex curve, minimising $\mathcal{D} + \beta\mathcal{R}$ selects the point where $\mathrm{d}\mathcal{D}^*/\mathrm{d}\mathcal{R} = -\beta$. The weight β is thus minus the slope of the rate–distortion curve at the selected model. A large β selects the flat, low-rate end, a small β the steep, high-rate end.

13.3 The weight on digits

We train five VAEs with $d = 10$ and $\beta = 0.5$, 1, 2, 4 and 8 on MNIST from one initialisation. As β grows the rate falls from 22.9 to 4.5 nats per test image and the distortion rises from 85.2 to 138.5 nats, along a decreasing convex curve. Each model lies where a line of slope $-\beta$ touches it (Figure 49). The ELBO is best for $\beta = 1$, $-\,\mathrm{ELBO} = 107.1$ nats, as expected for the only model trained on $-\,\mathrm{ELBO}$ itself. The models with $\beta = 0.5$ and 2 are within 1.1 and 2.4 nats of it, while their codes carry 4.5 nats more and 4.9 nats less. The ELBO alone barely distinguishes models whose latent codes differ strongly. As β grows, reconstructions and samples blur. The number of active coordinates falls from 10 at $\beta \leq 2$ to 8 at $\beta = 4$ and 5 at $\beta = 8$, with the switched-off coordinates carrying about 10^{-3} nats.

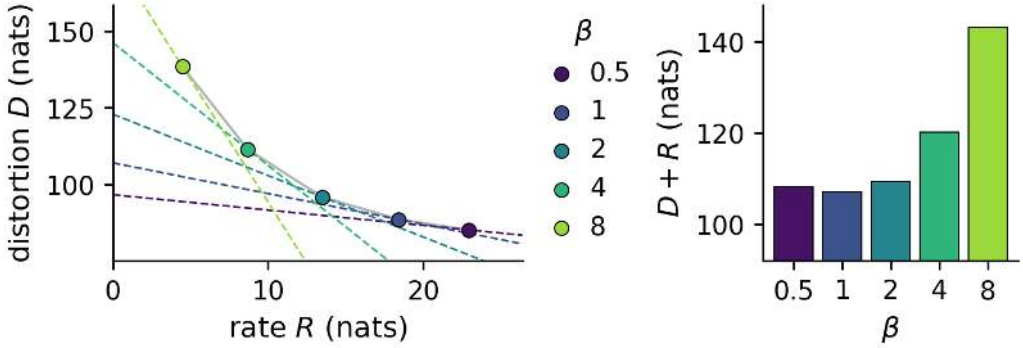


Figure 49: Rate and distortion of β-VAEs on MNIST, $d = 10$, measured on the test digits. Left, the distortion $\mathcal{D}$ against the rate $\mathcal{R}$, both in nats per image, for five values of β (dots), with the lines $\mathcal{D} + \beta\mathcal{R} = \text{const}$ through each model (dashed). Right, $-\,\mathrm{ELBO} = \mathcal{D} + \mathcal{R}$ for each β.

13.4 Three parts of the rate

The rate splits into three terms with separate meanings [23, 24], and one of them measures the independence that disentangling needs. We pick an image index i uniformly from the n images and then a latent vector from its cloud. This two-step draw defines the joint distribution $q(i, \mathbf{z}) \coloneqq \frac{1}{n}\, q_\phi(\mathbf{z}\,|\,\mathbf{x}_i)$, whose marginal over i is the aggregate posterior $q_\phi(\mathbf{z})$ of equation (5.8). The marginal of one coordinate is denoted $q_\phi(z_k)$. The mutual information of two random variables X and Y,

$$I(X;Y) \coloneqq \mathrm{KL}\left(p(X,Y)\,\|\,p(X)\,p(Y)\right), \tag{13.2}$$

is the divergence of their joint distribution from the product of their marginals. It vanishes only for independent variables and measures by how many nats knowing Y reduces the uncertainty about X. The rate is the average over i of $\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x}_i)}[\ln q_\phi(\mathbf{z}|\mathbf{x}_i) - \ln p(\mathbf{z})]$. Multiplying and dividing inside the logarithm by $q_\phi(\mathbf{z})$ and by $\prod_k q_\phi(z_k)$ gives

$$\ln \frac{q_\phi(\mathbf{z}|\mathbf{x}_i)}{p(\mathbf{z})} = \ln \frac{q_\phi(\mathbf{z}|\mathbf{x}_i)}{q_\phi(\mathbf{z})} + \ln \frac{q_\phi(\mathbf{z})}{\prod_k q_\phi(z_k)} + \sum_k \ln \frac{q_\phi(z_k)}{p(z_k)},$$

where the last step uses the factorisation of the prior, $p(\mathbf{z}) = \prod_k p(z_k)$. Each term is then averaged over $q(i, \mathbf{z})$. In the first, $q(i, \mathbf{z})/[q(i)\, q_\phi(\mathbf{z})] = q_\phi(\mathbf{z}|\mathbf{x}_i)/q_\phi(\mathbf{z})$ because $q(i) = 1/n$, so its average is the mutual information $I_q(i; \mathbf{z})$. The second depends on $\mathbf{z}$ only, so its average runs over $q_\phi(\mathbf{z})$ and gives a divergence. Each term of the sum depends on one coordinate only and is averaged over $q_\phi(z_k)$. Together,

$$\mathcal{R} = \underbrace{I_q(i; \mathbf{z})}_{\text{index–code information}} + \underbrace{\mathrm{KL}\left(q_\phi(\mathbf{z}) \,\middle\|\, \prod_k q_\phi(z_k)\right)}_{\text{total correlation TC}} + \underbrace{\sum_k \mathrm{KL}\left(q_\phi(z_k) \,\|\, p(z_k)\right)}_{\text{dimension-wise KL}}. \tag{13.3}$$

The first term is the mutual information between the index i of an image and its latent vector. It cannot exceed $\ln n$, and it is the information that lets the decoder tell the n images apart. The second and third terms are divergences and never negative, so the rate is an upper bound on the information that the latent vectors carry about the images. The total correlation (TC) vanishes only when the coordinates of the latent vectors are statistically independent over the data. It captures every kind of dependence, including nonlinear ones. The last term compares each coordinate alone with $\mathcal{N}(0, 1)$.

Estimated on a subset of 2000 test digits, the three terms add up to the rate of the same subset computed directly from the encoder clouds, within 0.01 nats. These rates, 22.47 nats for $\beta = 0.5$ and 18.02 nats for $\beta = 1$, lie slightly below the values of the full test set quoted above. The index–code information stays near its maximum $\ln 2000 = 7.60$ up to $\beta = 2$ (7.59, 7.55, 7.38 nats). It then falls to 3.86 at $\beta = 8$, where reconstructions deteriorate. The total correlation falls fastest, from 14.50 nats at $\beta = 0.5$ and 10.22 at $\beta = 1$ to 1.69 at $\beta = 4$ and 0.26 at $\beta = 8$. A larger β thus makes the coordinates nearly independent. The dimension-wise KL stays at or below 0.38 nats.

13.5 Alignment of coordinates with factors

A higher weight can align coordinates with factors in three ways. With $\beta > 1$ the encoder can afford only the few factors that improve the reconstruction most. Each costs nats that grow with the logarithm of the precision with which it is stored. Because the prior has independent coordinates, mixing factors that are non-Gaussian, as most physical factors are, makes the coordinates dependent and costs extra nats of total correlation, equation (13.3) and Figure 50. This pressure lasts only as long as the coordinates stay non-Gaussian, and the dimension-wise KL pushes them towards Gaussians. The diagonal covariance of the encoder clouds, mentioned in Section 6.3, breaks the rotation symmetry even for Gaussian factors. A posterior that is narrow along one factor and wide along another is approximated best by such a cloud when these directions are the coordinate axes [27]. The weight β raises the price of all information equally, the cost of dependence included. It forces the code to be economical, and an economical code is pushed towards the cheaper arrangement. Whether the coordinates end up disentangled depends on the data, and the cost is always a poorer reconstruction.

13.6 Independence and the choice of axes

Independent coordinates are necessary for disentangling, and they leave the axes undetermined in general. If two factors are Gaussian with equal variances, every rotation of them is again a pair of independent Gaussians, and no criterion of independence prefers the original axes. For non-Gaussian factors a rotation destroys the independence (Figure 50), and penalising the total correlation favours

the unrotated axes, the principle of independent component analysis [25]. Without assumptions on the model and the data a disentangled representation cannot be identified from unlabelled data [26]. Infinitely many transformations of the latent space that preserve the prior, combined with a matching change of the decoder, leave the model distribution of the data unchanged. A large study in the same work found that the random seed and the hyperparameters influence the result more than the choice of the loss. A coordinate found by a β-VAE is therefore a hypothesis to be checked against known or measured quantities.

The dimension-wise KL pushes every coordinate towards a Gaussian, while a physical factor may be uniform, like the position of a sprite, or bimodal, like the magnetisation below T_c. The two are compatible, because the decoder can apply a monotone map to each coordinate, the change of variables of Section 3.7. A map applied to one coordinate alone leaves the coordinates independent, so a factor stays in its own coordinate after such a map. Recovering a factor therefore means recovering it up to its sign, a monotone change of scale and the order of the coordinates. A latent coordinate is compared with a physical quantity by correlations and fits, and z_k is never expected to equal the magnetisation.

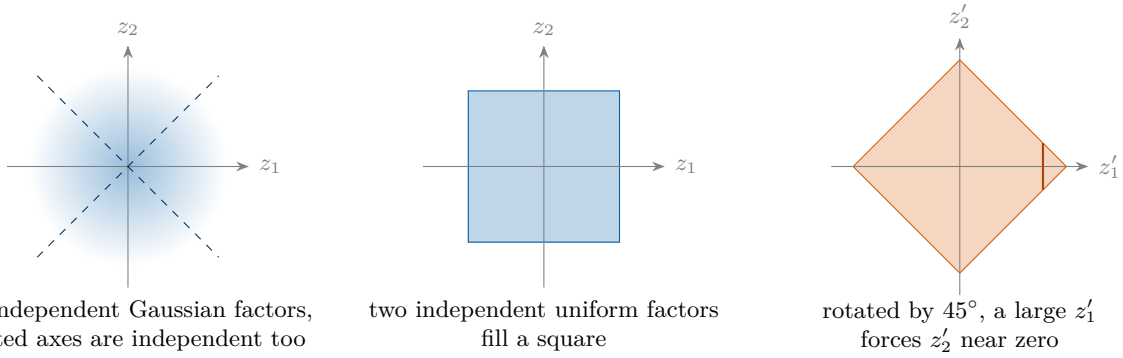


Figure 50: Independence and rotations, sketch. Left, the density of two independent standard Gaussian factors is round, and coordinates along rotated axes (dashed) are again independent. Middle, two independent uniform factors fill a square. Right, the same factors in axes rotated by 45° fill a diamond, and the allowed range of z_2' depends on z_1' (orange bar). The rotated coordinates are therefore dependent and have a positive total correlation.

On MNIST the coordinates of the $\beta = 4$ model are fewer and individually somewhat simpler than those of the plain VAE. Both models nevertheless change the identity of the digit along every axis. The identity is a discrete factor with ten values, and the latent space is a continuous Gaussian. No single coordinate can therefore turn a 3 into an 8 without passing through shapes in between. The cVAE avoids this by taking the identity as a label. The disentanglement of the remaining coordinates cannot be measured on MNIST, because nobody has listed the true factors of handwriting.

13.7 Measuring the alignment

The images of dSprites (disentanglement testing sprites) [28] are rendered from five known factors, shape, scale, orientation and two positions, so the alignment can be measured. The mutual information gap (MIG) [24] measures, for every factor v_j, how much more one latent coordinate knows about it than any other,

$$\mathrm{MIG} := \frac{1}{M} \sum_{j=1}^{M} \frac{I(z_{(1)}; v_j) - I(z_{(2)}; v_j)}{S(v_j)}, \tag{13.4}$$

where $z_{(1)}$ and $z_{(2)}$ are the coordinates with the largest and the second-largest mutual information with v_j. Here $S(v_j)$ is the entropy of the factor and M the number of factors. The mutual information is estimated from histograms of the encoder centres in 20 bins. Each term lies between 0 and 1. It is close to 1 when one coordinate determines the factor and no other carries information about it. It is close to 0 when the factor is shared by two coordinates or carried by none. With this estimator

the true factors used as a code give 0.90, and the same code with the two positions rotated by 45° gives 0.56. Trained models give 0.04 for $\beta = 1$, 0.10 for $\beta = 4$ and 0.22 for $\beta = 16$, whose three active coordinates follow the two positions and the scale. Raising β helps on dSprites, at the cost of a distortion that grows from 24.9 to 64.5 nats, and the values remain far from those of the true factors.

13.8 Faces

On faces the eye judges at once whether a traversal changes one property, such as the turn of the head, or several. A β-VAE trained on the celebrity photographs of CelebA (CelebFaces Attributes) [29] at 64×64 pixels has single coordinates that turn the head, change the smile or change the hair. In a plain VAE the coordinate that turns the head also changes expression and gender [21]. A smaller model trains on an ordinary processor. A convolutional β-VAE with $d = 32$ and $\beta = 3$ trained on 20 000 faces of 32×32 colour pixels uses 18 of its coordinates and has a rate of 15.8 nats per face. Its reconstructions keep pose, skin and hair colour, background and the shape of the face, and lose eyes, mouths and glasses (Figure 51). The most informative coordinates change global properties, the background, the direction of the light, the colour of skin and hair, the length of the hair. Their Pearson correlation with the 40 binary attributes of CelebA reaches at most $|\rho| = 0.35$, since the attributes describe details that a rate of 15.8 nats cannot afford.

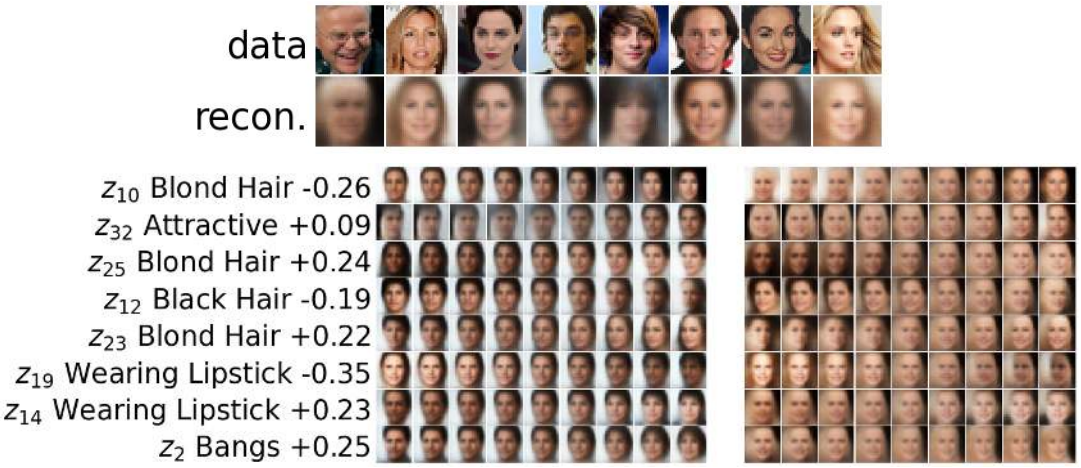


Figure 51: A β-VAE on 32×32 CelebA faces, $\beta = 3$, $d = 32$. Top, test faces (upper row) and the decoder means at their cloud centres (lower row). Bottom, traversals of two test faces (left and right blocks). Each row moves one of the eight coordinates with the largest $\overline{\mathrm{KL}}_k$ from -3 to 3. Its label names the attribute with the largest $|\rho|$ between the coordinate and the attribute over the training faces.

14 Order parameters from a β-VAE

14.1 The probing protocol

Besides generating, a VAE can be used to learn about the system that produced its data. A β-VAE receives only spin configurations, without temperature, phase or Hamiltonian, and compresses each into a few numbers that the weight β pushes towards independence. If the compression is good, these numbers describe the main ways in which configurations differ, and for a system in equilibrium these are macroscopic quantities. The first principal component of Ising configurations was found to be the magnetisation [30], and the latent variable of a VAE trained on Ising configurations correlates with it [31]. We train a β-VAE with $d = 8$ and a Bernoulli decoder on configurations only, with $\beta = 1$ and $\beta = 4$. We keep the active coordinates, $\overline{\mathrm{KL}}_k > 0.1$ nats on held-out configurations. For every held-out configuration we compute the latent mean $\mu_k(\mathbf{s})$ and its Pearson correlation with each observable

$O(\mathbf{s})$ computed from the spins,

$$\rho_{kO} \coloneqq \frac{\langle \mu_k O \rangle - \langle \mu_k \rangle \langle O \rangle}{\sqrt{\operatorname{Var} \mu_k \operatorname{Var} O}} \in [-1, 1], \tag{14.1}$$

with averages over the held-out configurations of all temperatures. We also inspect scatter plots of μ_k against the best-matching observable. The observables are m, $|m|$, the energy per spin e, the simulation temperature T, and for the J_1–J_2 model m_x, m_y, their moduli, m_{str} and g. Pearson's coefficient is ± 1 for a linear relation without scatter and detects linear relations only. A coordinate that follows m has $|\rho| \approx 1$ with m and $\rho \approx 0$ with $|m|$, because both signs of m occur.

14.2 The Ising model

The Ising configurations of the 30×30 lattice are split into 80% for training and 20% held out. The coordinate with the largest KL term, z_7, carries 3.04 nats for $\beta = 1$ and 2.23 for $\beta = 4$ and has $\rho = +0.99$ with m in both models. No other coordinate exceeds $|\rho| = 0.16$ with any observable. For $\beta = 1$ all eight coordinates are active. The other seven carry information about individual configurations that none of the macroscopic observables captures. For $\beta = 4$ only three of them remain, with about half a nat each. In the scatter plots (Figure 52) μ_7 is a smooth monotonic function of m. Against the energy it forms two branches, one for each sign of m, which meet at high temperature. The two ordered phases sit at the two ends of the μ_7 axis and the disordered phase at its centre. The energy is related to $|\mu_7|$, with $\rho(|\mu_7|, e) = -0.88$ for $\beta = 4$, a symmetric relation that $\rho(\mu_7, e)$ cannot detect.

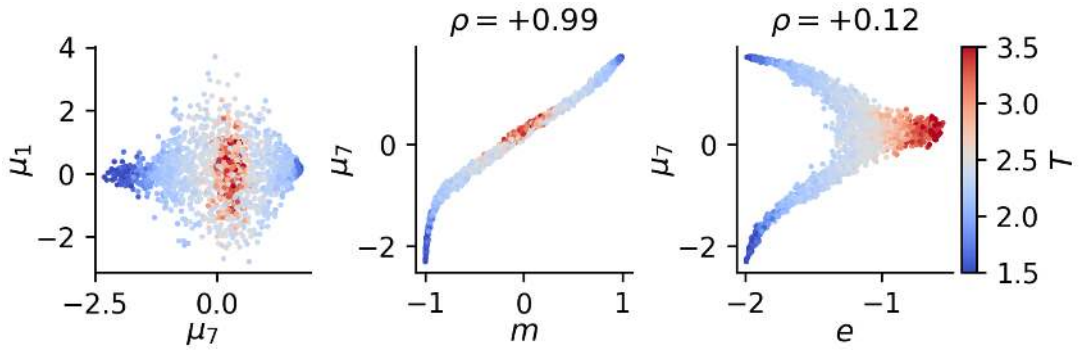


Figure 52: A β-VAE on Ising configurations, $L = 30$, $\beta = 4$, $d = 8$, held-out configurations coloured by their temperature T. Left, the latent means (μ_7, μ_1) of the two most active coordinates. Middle, μ_7 against the magnetisation m. Right, μ_7 against the energy per spin e. The titles give the Pearson correlation ρ.

14.3 The J_1–J_2 model

The J_1–J_2 model has three order parameters, m for the ferromagnet and m_x and m_y for the two stripe orientations. From the 81×79 grid, 12 configurations per point are used, 10 for training and 2 held out. The $\beta = 4$ model has three active coordinates, and each follows one order parameter (Figure 53), $\rho(\mu_7, m_x) = -0.97$, $\rho(\mu_5, m_y) = +0.98$ and $\rho(\mu_8, m) = +0.80$. The network never saw g, T or the Hamiltonian, and its coordinates carry the signs that distinguish the degenerate ground states. The $\beta = 1$ model also has three active coordinates, which mix the order parameters. One follows m with $\rho = -0.86$. The other two are combinations of m_x and m_y, for example $\rho = -0.58$ and $+0.67$ for μ_5, a rotated version of the stripe plane. The stronger pull towards independent coordinates has aligned the axes with independent order parameters.

Averaging $|\mu_k|$ over the configurations of every grid point maps where each coordinate is used (Figure 54). The magnetisation coordinate is large in the ferromagnet and moderate in part of the stripe phase, the small mixing seen in its correlation with m_{str}. The two stripe coordinates are non-zero only in the stripe phase, in complementary bands. At a given g the cooled Metropolis chains

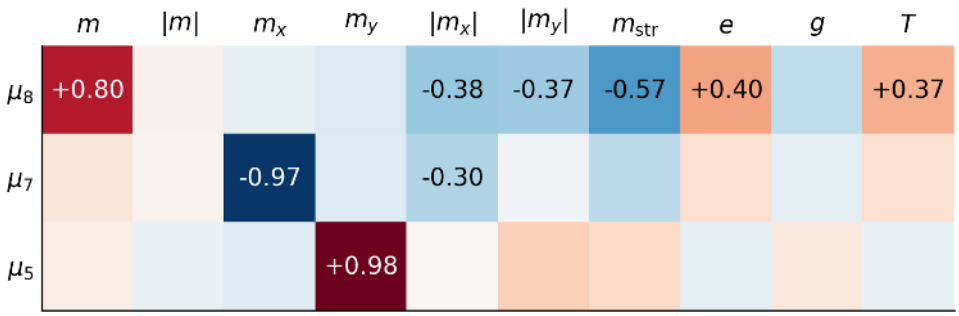


Figure 53: Pearson correlations $\rho(\mu_k, O)$ of the three active coordinates (rows) of the $\beta = 4$ model with observables of the J_1–J_2 configurations (columns), on held-out configurations, red positive and blue negative. Values with $|\rho| \geq 0.3$ are written in the cells.

keep the orientation chosen when the stripes formed. The bands are therefore a property of the data, while in equilibrium both orientations are equally likely. Together the maps trace the phase boundaries without any label.

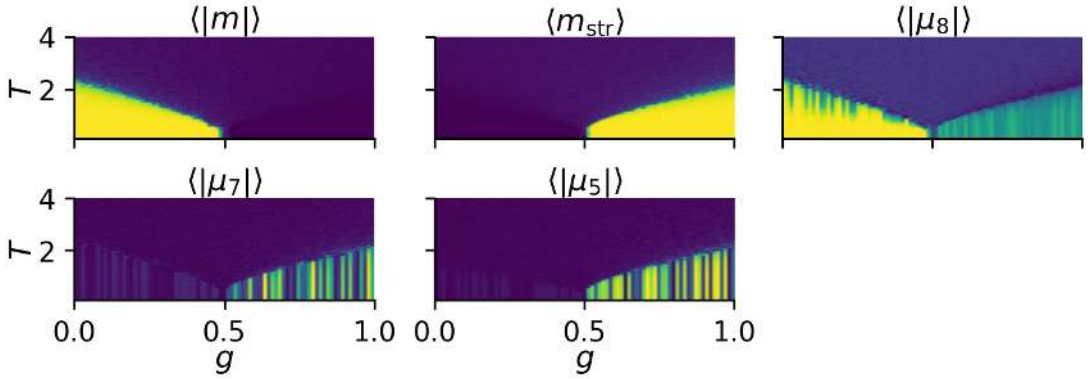


Figure 54: Maps over the (g, T) plane of $|m|$, m_{str} and $|\mu_k|$ for the three active coordinates of the $\beta = 4$ model, averaged over the configurations of each grid point. Each map runs from its minimum (dark) to its maximum (yellow).

For each observable we compare the largest $|\rho|$ over the active coordinates with the second largest. Their difference, the correlation gap, is large when one coordinate alone follows the observable. It transfers the idea of the MIG from mutual information to correlations and summarises the effect of β in one number. For the Ising magnetisation the gap is 0.90 for $\beta = 1$ and 0.95 for $\beta = 4$. For the J_1–J_2 model it grows with β from 0.09 to 0.85 for m_x, from 0.22 to 0.84 for m_y and from 0.46 to 0.68 for m.

14.4 Candidate order parameters

For these two models the order parameters were known and the probes served as a check. For a model whose phases are unknown, an active coordinate whose mean modulus changes abruptly across a range of parameters is a candidate order parameter. The next step expresses it through the spins. We fit μ_k by least squares with eight standardised features of a configuration, m, m_x, m_y, their moduli, and the nearest- and next-nearest-neighbour correlations. The fit explains a fraction $R^2 = 0.982$, 0.992 and 0.989 of the variance of the three coordinates, where $R^2 := 1 - \sum(\mu_k - \tilde{\mu}_k)^2 / \sum(\mu_k - \langle\mu_k\rangle)^2$ with $\tilde{\mu}_k$ the fit. The largest coefficients belong to m, m_x and m_y, and a low R^2 would point to a quantity outside the chosen features.

A large $|\rho|$ confirms that a coordinate follows a quantity and leaves the cause open, because quantities such as the magnetisation and the energy are themselves correlated across a transition. The sign and scale of a coordinate are arbitrary, and a nonlinear relation escapes a linear probe. The coordinates that appear depend on the model, the data, β and the seed. A probe finds only the quantities it tests, and a coordinate that matches none of them is either noise or a quantity not yet

considered.

15 Physical parameters from trajectories

15.1 The noisy damped oscillator

A particle of unit mass on a spring moves in a fluid. The spring pulls it back with the force $-\omega^2 x$, the fluid slows it with the friction force $-\gamma\dot{x}$, and the molecules of the fluid kick it at random. Newton's equation then reads

$$\ddot{x} = -\omega^2 x - \gamma\dot{x} + \sqrt{2D}\,\eta(t), \qquad \langle\eta(t)\rangle = 0, \quad \langle\eta(t)\eta(t')\rangle = \delta(t-t'), \tag{15.1}$$

where ω is the angular frequency without friction, γ the friction rate and D the noise strength. The Gaussian white noise $\eta(t)$ describes random kicks independent at different times, and the brackets average over many realisations of the noise. In the stationary state $\langle\dot{x}^2\rangle = D/\gamma$, and for a fluid at temperature T equipartition requires $\langle\dot{x}^2\rangle = T$. Hence $D = \gamma T$, and friction and noise have the same origin.

Without noise the ansatz $x = e^{\lambda t}$ gives $\lambda^2 + \gamma\lambda + \omega^2 = 0$ with roots $\lambda_\pm = -\gamma/2 \pm \sqrt{\gamma^2/4 - \omega^2}$. For $\gamma < 2\omega$, the underdamped regime, the particle oscillates at $\omega_d = \sqrt{\omega^2 - \gamma^2/4}$ inside an envelope $e^{-\gamma t/2}$. For $\gamma > 2\omega$, the overdamped regime, it creeps back without oscillating, dominated at long times by the slower rate $|\lambda_+|$. For $\omega \ll \gamma$ the expansion $|\lambda_+| \approx \omega^2/\gamma$ shows that an overdamped trajectory depends on the two parameters mainly through the combination ω^2/γ (Figure 55).

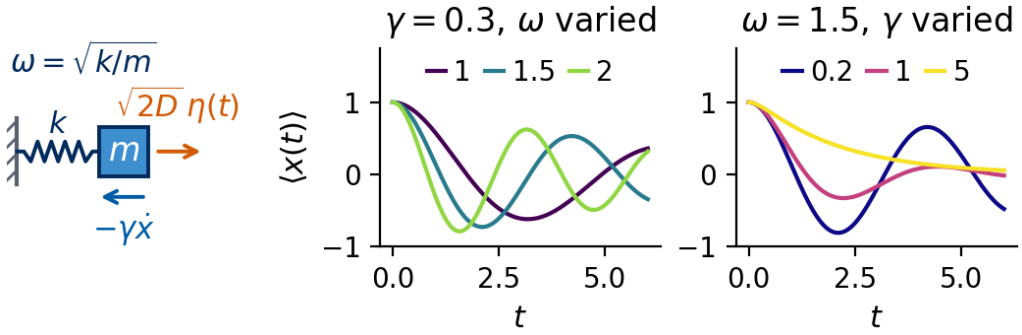


Figure 55: The noisy damped oscillator. Left, sketch of the mass m on a spring of constant k in a fluid, with $\omega = \sqrt{k/m}$, the friction $-\gamma\dot{x}$ and the random kicks $\sqrt{2D}\,\eta(t)$. Both are per unit mass as in the equation of motion. The letters m and k in the sketch denote the mass and the spring constant only here. Middle, the mean trajectory $x(t)$ for several ω at fixed γ. Right, the mean trajectory for several γ at fixed ω, from a slowly decaying oscillation to an overdamped creep.

We integrate the equation with small time steps Δt. Over one step the random kicks add up like a random walk, so their sum has variance $2D\,\Delta t$ and typical size $\sqrt{2D\Delta t}$. The update reads

$$x_{n+1} = x_n + v_n\Delta t, \qquad v_{n+1} = v_n + (-\omega^2 x_n - \gamma v_n)\Delta t + \sqrt{2D\Delta t}\,\eta_n, \qquad \eta_n \sim \mathcal{N}(0,1), \tag{15.2}$$

the Euler–Maruyama scheme [32]. The noise term must scale with $\sqrt{\Delta t}$, since a term proportional to Δt would make the particle stop diffusing as $\Delta t \to 0$. With $\Delta t = 0.01$ the stationary variance of the position agrees with the exact $D/(\gamma\omega^2)$ up to an error of order Δt. An ensemble gives 0.00185 against the exact 0.00178 for $\omega = 1.5$ and $\gamma = 0.5$.

Each trajectory of the data set has its own ω, uniform in $[1, 2]$, and its own γ, uniform in $\ln\gamma$ on $[0.2, 6]$. It also has its own initial position $x_0 \in [0.5, 1.5]$ and velocity $v_0 \in [-0.5, 0.5]$, with $D = 0.002$ for all. The positions are recorded at 100 times in $[0, 5.94]$, and 20 000 trajectories form the training set, 21% of them overdamped. The positions have a standard deviation of 0.43 over the data set, and the noise moves a trajectory away from its mean by 0.031, a small and visible perturbation. The four numbers $(\omega, \gamma, x_0, v_0)$ are the factors of a trajectory. Of these, x_0 and v_0 describe the start and the other two the oscillator.

15.2 A β-VAE for trajectories

Positions are real numbers, so the decoder is Gaussian. Given $\mathbf{z}$ every recorded position is Gaussian around the output $G_\theta(\mathbf{z})_i$ of a decoder network with fixed width $\sigma_x = 0.1$, about three times the noise. The reconstruction term is then the squared error (3.7), $\|\mathbf{x} - G_\theta(\mathbf{z})\|^2/2\sigma_x^2$, and $G_\theta(\mathbf{z})$ is a smooth mean trajectory. A dense network with $d = 8$, twice the number of factors, is trained with $\beta = 0.5$, 1, 4 and 8, three seeds each. A model that knew the four factors and nothing about the noise would reach the distortion of the exact mean trajectory, 4.93 nats, the noise floor. For $\beta \le 1$ the distortion lies below it, so the code stores part of the individual noise, and for $\beta = 4$ and 8 it lies above.

The number of active coordinates is $4, 4, 4$ for the three seeds at $\beta = 1$, $5, 4, 4$ at $\beta = 4$ and $3, 3, 3$ at $\beta = 8$. At $\beta = 1$, and for two of the three seeds at $\beta = 4$, it equals the number of factors, four, so the network finds how many numbers describe a trajectory. At $\beta = 8$ the network drops the initial velocity, whose best correlation with any coordinate falls to 0.23.

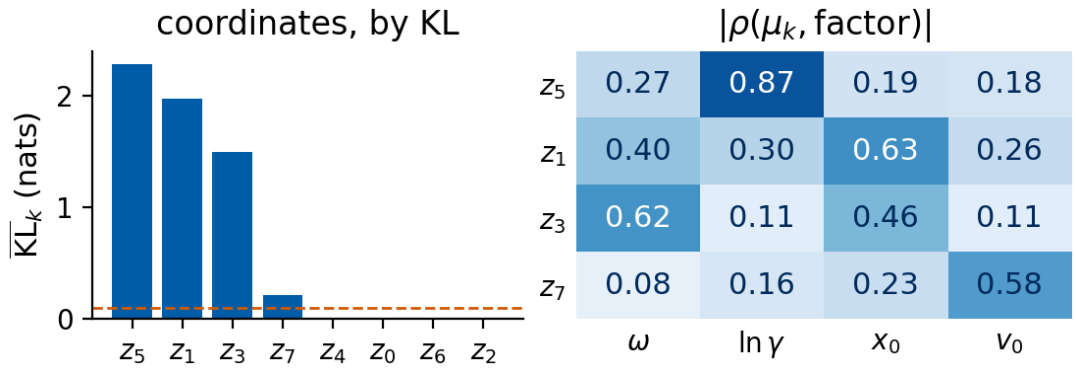


Figure 56: A β-VAE on oscillator trajectories, $\beta = 4$, seed with the largest MIG of three. Left, the mean KL term $\overline{\mathrm{KL}_k}$ of the eight coordinates, labelled z_0 to z_7, with the threshold 0.1 nats for an active coordinate (dashed). Right, $|\rho(\mu_k, \text{factor})|$ of the four active coordinates (rows) with ω, $\ln\gamma$, x_0 and v_0 (columns) over 5000 test trajectories.

15.3 Coordinates for damping and frequency

For the model of Figure 56 the most informative coordinate follows $\ln\gamma$ with $|\rho| = 0.87$ and nothing else above 0.3. Moving this damping coordinate turns a damped oscillation into an overdamped creep. The frequency ($|\rho| = 0.62$) and the initial position (0.63) are shared between two coordinates, and the initial velocity sits in the weakest one (0.58). Over all twelve models the damping is the factor found best, its best $|\rho|$ averaged over seeds reaching 0.82 at $\beta = 4$. The MIG (13.4), estimated from histograms, gives 0.92 for the true factors and 0.46 when ω and x_0 are rotated by 45°. The trained models stay between 0.03 and 0.07 for every β. No value of β gives a disentangled code here, and the spread between seeds is as large as the change with β.

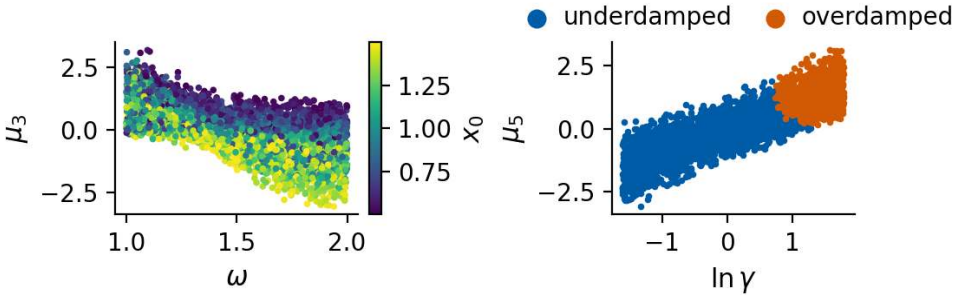


Figure 57: Left, the frequency coordinate μ_3 against ω, coloured by the initial position x_0. Right, the damping coordinate μ_5 against $\ln\gamma$, underdamped (blue) and overdamped (orange) trajectories.

The scatter plots show the structure that the coefficients compress (Figure 57). Among underdamped trajectories the damping coordinate has $|\rho| = 0.80$ with $\ln\gamma$, and among overdamped ones only 0.13. There it follows $\ln(\omega^2/\gamma)$ with $|\rho| = 0.69$. In that regime the network reports the slow rate, the combination that the data contain. The frequency coordinate changes with ω with a slope that grows with x_0. A least-squares fit by ω alone explains $R^2 = 0.38$ of its variance, and with x_0 and the product ωx_0 0.63. The frequency can be measured only as well as the oscillation is visible, and its visibility is set by the amplitude x_0.

The derivatives $\partial\langle x(t_i)\rangle/\partial p_a$ of the mean trajectory with respect to the factors p_a give the response of a trajectory to each factor. They also show whether two factors change the trajectory in similar ways. We change one factor by one standard deviation of its distribution and measure the change of the whole trajectory in units of σ_x. It is $17, 23, 12, 5$ for ω, $\ln\gamma$, x_0 and v_0. The initial velocity, the factor dropped first, is the least visible. The cosine of the angle between the derivative vectors of two factors is 0.79 for ω and v_0 and 0.69 for ω and $\ln\gamma$. Within the window these changes look alike, and the data give the network little reason to separate them. The most visible and least confusable parameter ends up in its own coordinate.

15.4 A network that predicts from latent neurons

SciNet [33] consists of an encoder that compresses an observation into a few latent neurons and a decoder that answers a question from them (Figure 58). Here the question is the position $x(t)$ at a requested time t, possibly beyond the observed window. Its cost is the squared error of the answer plus β times the KL term, with $\beta = 10^{-3}$. Trained on a damped pendulum without noise, $\ddot{x} = -\kappa x - b\dot{x}$ with fixed amplitude and phase and varying spring constant κ and damping factor b, SciNet stores κ and b in separate neurons [33]. Our reimplementation reproduces this separation after 1000 epochs, with $|\rho| = 1.00$ for κ and 0.98 for b and second-best values of 0.05 and 0.06. The prediction error is 0.027 over $t \in [0, 10]$. After 100 epochs both of its active neurons follow κ, and b reaches only $|\rho| = 0.50$. The β-VAE with $\beta = 4$ separates the two parameters on the same data in 60 epochs, $|\rho| = 0.99$ and 0.97. This speed is consistent with its much larger weight of the KL term, which pushes the code towards independent coordinates sooner. With random initial positions and velocities both networks store the most visible parameters first.

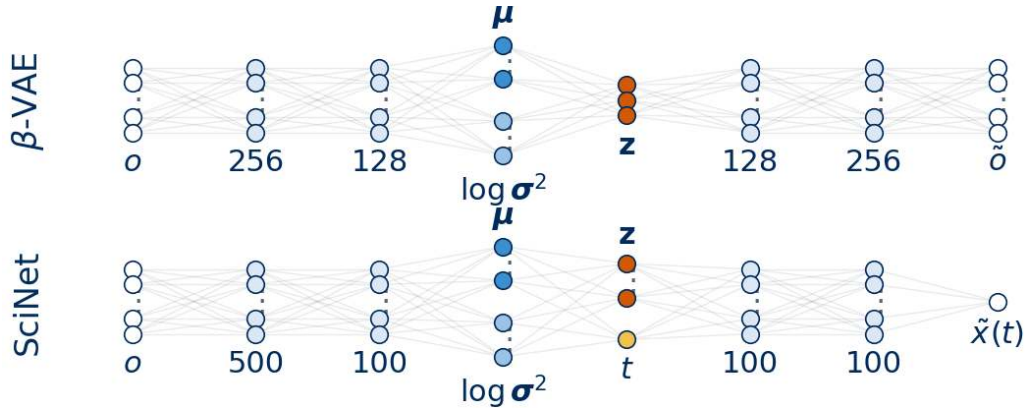


Figure 58: The β-VAE for trajectories (top) and SciNet (bottom), drawn as nodes and edges with the layer widths below the columns. Both encoders map the observed trajectory o to $\boldsymbol{\mu}$ and $\ln\boldsymbol{\sigma}^2$ and draw the latent neurons $\mathbf{z}$ (orange). The β-VAE decoder returns the whole trajectory, while the SciNet decoder receives $\mathbf{z}$ together with the question t (yellow) and returns the answer $\tilde{x}(t)$.

For two equal masses coupled by springs, the motion separates into two normal modes with frequencies ω_1 and ω_2, and the trajectories depend on the two springs only through them. A β-VAE with $\beta = 4$ finds one coordinate that follows the in-phase frequency, $|\rho| = 0.69$. A second coordinate follows the coupling spring (0.65) and the anti-phase frequency (0.63) about equally, a mixture of the two descriptions. A coordinate becomes interpretable when its parameter is visible in the data and no other parameter changes the data in a similar way. It has to be checked against known physics

before it is named.

16 Galaxy morphology

16.1 Images of galaxies

Real scientific images come without a list of factors. The Galaxy10 DECaLS data set [34] holds 17 736 colour images of galaxies from the Legacy Imaging Surveys of the Dark Energy Spectroscopic Instrument (DESI). Most of them were observed by the Dark Energy Camera Legacy Survey (DECaLS) [35]. We reduce them to 64×64 pixels of $1.05''$ and apply a nonlinear stretch that keeps faint regions visible. The near-infrared z band is shown as red, the red r band as green and the blue-green g band as blue. The colour carries physics, as old stellar populations and dust make a galaxy redder and young massive stars make it bluer. Each image carries one of ten morphology classes from the votes of Galaxy Zoo volunteers [36] (Figure 59). The classes are smooth galaxies from round to cigar-shaped, barred and unbarred spirals, edge-on disks with and without a bulge, and disturbed and merging systems. The β-VAE never sees the labels.

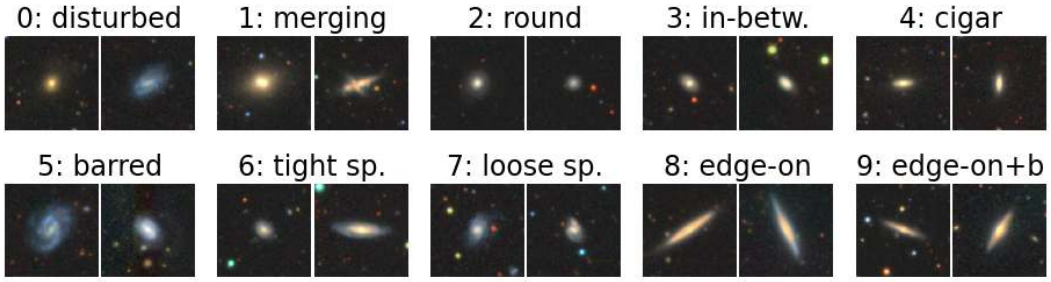


Figure 59: Two galaxies of each of the ten Galaxy10 DECaLS classes, 64×64 pixels, colours from the z, r and g bands.

16.2 Content of the code

Galaxy pixels are continuous and mostly sky, and we train a convolutional β-VAE with a Gaussian decoder, $d = 10$ and $\beta = 1$ or 4 on 15 000 images. The rate falls from 25.3 to 11.4 nats per image, while the mean squared error per pixel of the reconstruction rises only from 3.01×10^{-3} to 3.13×10^{-3}. The reconstructions keep size, brightness, elongation and orientation and lose arms, bars, companions and even the colour. Traversals of the $\beta = 4$ model suggest names for the coordinates (Figure 60). One grows a faint point into a large bright blob, others stretch the galaxy along a diagonal or along an image axis.

Naming a coordinate requires quantities measured from the pixels alone. We use a mask of pixels brighter than the sky by three noise levels. Inside it the size R_g is the radius of a disc of the same area, and the brightness F_g is the summed sky-subtracted intensity. The flux-weighted second moments I_{xx}, I_{yy}, I_{xy} give the ellipticity components $\epsilon_1 = (I_{xx} - I_{yy})/(I_{xx} + I_{yy})$ and $\epsilon_2 = 2I_{xy}/(I_{xx} + I_{yy})$. They also give the axis ratio q_ax, the square root of the ratio of the eigenvalues of the moment matrix. For a thin disk whose axis makes an angle ϑ with the line of sight, $q_\mathrm{ax} \approx \cos\vartheta$. The colour is the flux-weighted difference of the red and blue channels. Spearman's rank correlation is the Pearson correlation of the ranks and equals ± 1 for every monotonic relation. It compares each coordinate with each quantity over the 2 735 of the 2 736 test galaxies for which the quantities could be measured (Figure 61).

In the $\beta = 4$ model one coordinate, z_5, carries size and brightness together, with $|\rho| = 0.93$ and 0.97, and the orientation is stored in two pairs of coordinates following ϵ_1 and ϵ_2. The axis ratio has no coordinate, $|\rho| \leq 0.13$. Moving an ellipticity coordinate in either direction elongates the galaxy,

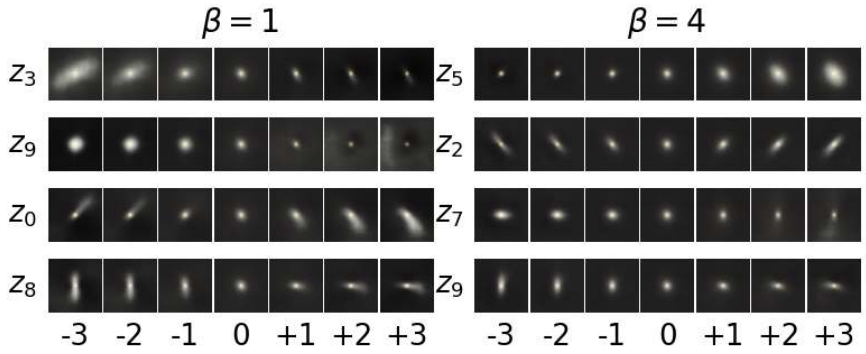


Figure 60: Galaxy traversals. Each row decodes the code of one round smooth test galaxy with one coordinate z_k shifted by -3 to 3 (columns). The rows show the four coordinates with the largest $\overline{\mathrm{KL}}_k$ of the models with $\beta = 1$ and $\beta = 4$. Coordinates are numbered from z_0.

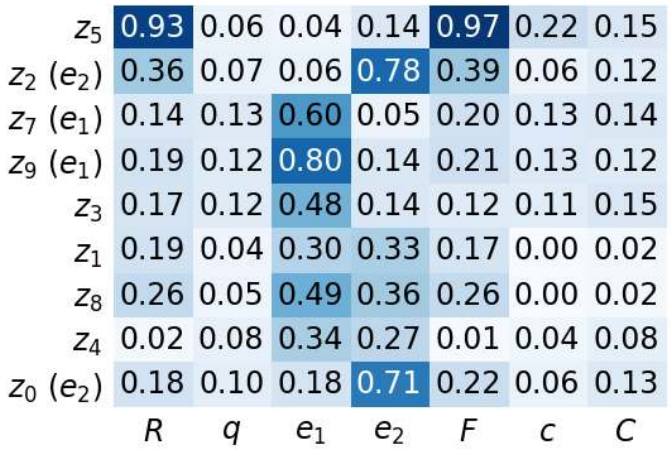


Figure 61: Absolute Spearman correlation $|\rho|$ between the latent mean μ_k of each active coordinate (rows) of the $\beta = 4$ model and quantities measured from the test images (columns). The quantities are size R (R_g in the text), axis ratio q (q_ax), ellipticity components e_1 and e_2 (ϵ_1, ϵ_2), brightness F (F_g), colour c and concentration C. The concentration is the ratio of the radii containing 80% and 20% of the light.

so q_ax is a symmetric function of it that a rank correlation cannot detect. The β-VAE stores the shape as the two components (ϵ_1, ϵ_2) that astronomers use, and the axis ratio, with it the inclination, follows from the length of that vector, $|\boldsymbol{\epsilon}| = (1 - q_\mathrm{ax}^2)/(1 + q_\mathrm{ax}^2)$. The colour reaches at most $|\rho| = 0.22$, consistent with the grey reconstructions. The mean correlation gap over the seven quantities rises from 0.12 for $\beta = 1$ to 0.22 for $\beta = 4$, almost entirely through size and brightness. A second seed at $\beta = 4$ gives 0.14. Two seeds leave open whether $\beta = 4$ disentangles better than $\beta = 1$, and they show that the seed matters as much as β.

16.3 Counterfactual galaxies

A convolutional classifier trained on the real images and their classes reaches a test accuracy of 0.640, and on the blurred reconstructions only 0.315. The β-VAE lets us test how the classifier would respond to a galaxy that is more inclined and otherwise unchanged. A test galaxy is encoded, one coordinate is shifted by δ, $\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}) + \delta\,\mathbf{e}_k$, and the decoded image is classified, with the unshifted reconstruction as reference. We select the 249 galaxies whose reconstruction the classifier calls a face-on spiral. For them a shift of $\delta = 3$ along the ϵ_1 coordinate most correlated with the axis ratio lowers the median measured q_ax of the decoded images from 0.65 to 0.37. It turns 12% of the decisions into edge-on disks, against 6% for the opposite shift (Figure 62). The classifier responds to elongation in a physically sensible way. The decoded images are, however, blurred grey blobs far from real galaxies, and since the model cannot add or remove arms or bars, such counterfactuals probe only the

coarse factors.

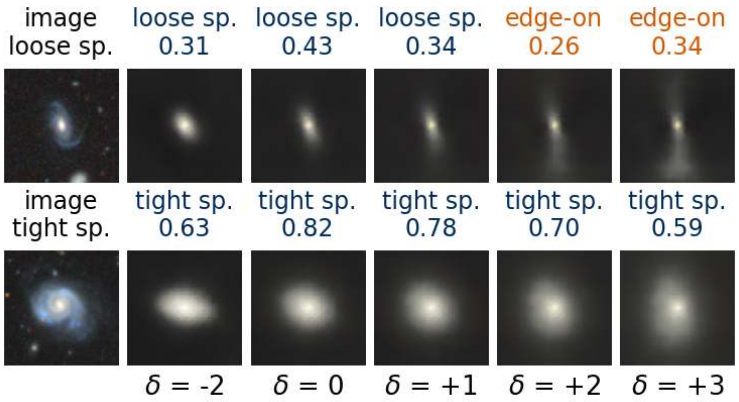


Figure 62: Counterfactual inclination. Two galaxies labelled face-on spirals (rows), the real image and decoded images with the inclination coordinate shifted by δ towards smaller axis ratio. Above each image, the class predicted by the classifier and its probability.

17 Conditioning and discovering

The two ways of controlling a variational autoencoder serve different purposes (Table 9). The cVAE applies when the factor is known and labelled, such as the digit, the temperature of a simulation or the point (g, T) of a phase diagram. It guarantees control of that factor and generates on demand, also between the conditions it was trained on. The β-VAE applies when the varying factors of the data are still unknown. It found the magnetisation and the two stripe amplitudes of the J_1–J_2 model, the damping and the slow rate of an oscillator, and the size and ellipticity of galaxies. It mixed the frequency with the initial position and dropped the colour. These findings are hypotheses to be checked against physics, after which they can become labels for a conditional model.

Table 9: The conditional VAE and the β-VAE.

	conditional VAE	β-VAE
training data	pairs $(\mathbf{x}, \mathbf{c})$	samples $\mathbf{x}$ only
loss	$\mathcal{D} + \mathcal{R}$, $\mathbf{c}$ given to encoder and decoder	$\mathcal{D} + \beta\mathcal{R}$
factor of interest	chosen and given as $\mathbf{c}$	found by the network, named afterwards
guarantee	$\mathbf{c}$ controls what it labels	none in general
latent vector carries	everything except $\mathbf{c}$	everything, ideally one factor per coordinate
use	generate on demand	explore the data, find collective variables

For physics the main result concerns the synthetic configurations. A generator whose output is a product of independent factors given a few latent numbers is a mean-field ansatz. It keeps what is collective, the phase, the order parameter and its distribution, and it loses what the Hamiltonian fixes locally, the energy, its fluctuations and the correlations between neighbours. The loss shrinks where interactions are long-ranged, and networks trained on real configurations do not detect it.

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